

United Nations Development Programme

Project title: Towards a Sustainable City through Energy Efficiency in Kairouan		
Country(ies): Tunisia	**Implementing Partner (GEF Executing Entity):** Ministry of Environment **Other Executing Entity:** UNDP	**Execution Modality**: Assisted National Implementation Modality (NIM)
Contributing Outcome (UNDAF): Strategic priorities 2: Take action to help reduce inequalities and improve Tunisia's resilience to climate crises and risks. Outcome 4: By 2025, all the actors involved ensure equitable, transparent and sustainable management of natural resources, ecosystems and territories, improving their resilience/adaptation as well as that of populations, especially the most vulnerable, in the face of crises and climate risks.		
UNDP Social and Environmental Screening Category: Moderate	**UNDP Gender Marker:** *2*	
Atlas Award ID: 01000110	**Atlas Project/Output ID:** *this should be created at PIF stage TBC*	
UNDP-GEF PIMS ID number: 6686	**GEF Project ID number:** 11040	
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Latest possible date to submit to GEF: *28.02.2023*		
Latest possible CEO endorsement date: *N/A*		
Planned start date: 1 April 2024	**Planned end date:** 31 March 2027	
Expected date of Mid-Term Review: *Not applicable*	**Expected date of Terminal evaluation:** 31 December 2026	
Brief project description: *The UNDP-GEF project is designed to promote low-carbon buildings and efficient public lighting in the Municipality of Kairouan. The results will eventually be scaled up to other municipalities and cities in Tunisia to support the National Programme for Sustainable Cities. The project will directly support the implementation of the Sustainable Development Strategy of Kairouan, the action plan for sustainable energy, and the updated Nationally Determined Contribution of Tunisia. This will be achieved through three outcomes: (i) institutional strengthening with enabling conditions, methodologies and tools for enforcing regulatory framework for new buildings; (ii) strengthening municipal institutions for the implementation of an efficient lighting system;*		

and (iii) contributing to the development of a municipal learning organisation capable to implement and manage sustainable city strategy.

The lifetime global environment benefits will accrue from application and enforcement of building energy codes and energy efficient public lighting through the proof-of-concept interventions in Kairouan and will be 20,219 tCO_{2e}. Post-project indirect emission reductions amounting to at least 31,064 tCO_{2e} are also expected. The project will produce co-benefits such as job creation in the energy efficient buildings value chain as well as the improved financial security of the Municipality of Kairouan due to savings on city-wide energy bills. The number of direct beneficiaries has been estimated to be around 187,000 persons (94,248 female; 92,752 male).

FINANCING PLAN	
GEF Trust Fund	USD 846,906
UNDP TRAC resources	USD 0
Confirmed cash co-financing to be administered by UNDP (Ministry of Environment co-financing)	USD 500,000
(1) Total Budget administered by UNDP	**USD 1,346,906**
(2) Total confirmed co-financing to this project not administered by UNDP)	USD **2,723,000**
(3) Grand-Total Project Financing (1) +(2)	**USD 4,069,906**

SIGNATURES:

NOTE: IF THE PROJECT DOCUMENT IS IN FRENCH OR SPANISH, THE FINAL PROJECT DOCUMENT MUST BE CLEARED BY THE RTA BEFORE SIGNATURE.

Signature: print name below	**Agreed by Government Development Coordination Authority**[1]	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*
Signature: print name below	**Agreed by Implementing Partner**[2]	**Date/Month/Year:** *within 6 months of GEF CEO endorsement*
Signature: print name below	**Agreed by UNDP**[3]	**Date/Month/Year:** *within 6 months of GEF*

[1] *Other evidence of government agreement may be accepted in lieu of a signature, unless the programme country government requires a signature.*

[2] *Not required when UNDP is the implementing partner (i.e. DIM implementation modality). If a UN Agency is the implementing partner, and has signed a SBEAA with UNDP, then the Government Development Coordination Authority, UNDP and UN Agency sign the project document. If an IGO is the implementing partner, and has signed a SBEAA with UNDP, then the Government Development Coordination Authority, UNDP and IGO sign the project document. If a CSO/NGO is the implementing partner, the Government Development Coordination Authority and UNDP sign the project document and attached it to the Project Cooperation Agreement to be signed by the CSO/NGO and UNDP.*

[3] *For NIM projects this is the Resident Representative. For DIM projects in a single country this is the Resident Representative. For global, regional DIM projects this is BPPS.*

		CEO endorsement

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ACTE	Alliance of Communes for Energy Transition (Alliance des Communes Pour la Transition Energétique)
ANME	Agence Nationale pour la Maîtrise de l'Energie (National Agency for Energy Efficiency)
CCM	Climate Change Mitigation
CTMCCV	Centre Technique des Matériaux de Construction et du Céramique et du Verre (Technical Centre for Construction Materials and for Ceramic and Glass)
DT	Dinar Tunisien (Tunisian Dinar)
EE	Energy Efficiency
ESMF	Environmental and Social Management Framework
ESMP	Environmental and Social Management Plan
FSP	Full Sized Project
FTE	Fonds de transition énergetique (Energy Transition Fund)
GDP	Gross Domestic product
GEF	Global Environment Facility
GEFSEC	Global Environment Facility Secretariat
GHG	Greenhouse gas
INNOPRI	Institut National de la Normalisation et de la Propriété Industrielle (National Institute of Standardization and Industrial Property)
IQMS	Inventory Quality Management System
ktCO2	Kilotonne of carbon dioxide
ktoe	Kilotonne of oil equivalent
LED	Light Emitting Diode
LMP	Labour Management Procedure
M&E	Monitoring and Evaluation
MSP	Medium Sized Project
MtCO2	Million tonne of carbon dioxide
PIF	Project Identification Form
PIR	Project Implementation Report
PAED	Sustainable Energy Action Plan (Plan d'action en faveur de l'énergie durable)
POPP	Programme and Operations Policies and Procedures
PPG	Project Preparation Grant
QCF	Quality Control Framework
QMP	Quality Management Protocol
QMS	Quality Management System
RTNB	Réglementation Thermique des Nouveaux Bâtiments (Energy Code for New Buildings)
SDG	Sustainable Development Goal
STAP	Scientific Technical Advisory Panel
TGBC	Tunisian Green Building Council
tCO2	Tonne of carbon dioxide
ToC	Theory of Change
UNDP-GEF	UNDP Global Environmental Finance Unit
WCCD	World Council on City Data

II. DEVELOPMENT CHALLENGE

1. In Tunisia, as in most countries of the world, rapid urbanization is often uncontrolled. Urban growth generates many dysfunctions in cities and a deterioration of the quality of life of populations. Tunisian cities today experience several failures due to degraded infrastructure, obsolete public transport, insufficiently controlled housing, and lack of both human and material resources to operate basic social services. Added to this is the lack of decent housing, and the increase in pollution. As a result, living conditions are declining for urban communities, a situation that has grown even more since the 2011 revolution. The national macroeconomic situation (Gross Domestic Product, GDP-1.1%; inflation rate-4.2%; unemployment rate-15.6%; poverty rate-15.2%)[4] is reflected at the sub-national level (e.g. governorates, cities, communes etc.).

2. COVID-19 has had a detrimental impact both as a health issue that has had economic spillover effects. On the health front, there have been 1,150,217 confirmed cases and 29,291 deaths as of 25 January 2023. Tunisia has experienced several peaks in infection cases and deaths since March 2020, with the highest being on 23 January 2022 with 9,341 infections (7-day average). The most recent peak was experienced on 12 July 2022 with 4,979 cases. Nevertheless, the most number of deaths (381) occurred on 1 August 2021, and the highest number of deaths related to the peak number of cases in July 2022 was 75 deaths. The trends show that the combination of sanitary measures, a successful vaccination campaign (13,241,985 doses administered as of 15 January 2023) and natural immunization have led to a decreasing severity of the virus in terms or mortality rate.[5] The number of new cases has been mostly zero in the month of January 2023, except for a spike of 2,488 on 17 January 2023. The same goes for deaths that showed a spike at 11 deaths on 24 January 2023. These figures, although not alarming, show that if not controlled, Covid-19 is still a risk. The stringent lockdown in 2020 coupled with trade openness (i.e. exposure to global value chains and tourism shocks) resulted in a GDP contraction of 8.8%. The financial impacts were more severely felt by households in the low-income quartiles, the self-employed and among informal wage workers. The coping strategies of households were to spend their savings and/or to rely on family support.[6] A gendered impact was also noted with women more adversely affected than men.[7]

3. The transition towards sustainable cities has to contend with the socioeconomic situation prevailing in Tunisia, and exacerbated by the COVID-19 crisis. However, we have to contend with other challenges. Despite the ecological potential of Tunisia, the transition towards "sustainable cities" is impeded by several environmental challenges, including: climate change, depletion of natural resources, drought, floods, waste management and industrial pollution. These environmental issues, including the increase of greenhouse gases emissions are aggravated by an exponential growth of urban populations coupled with a lack of integrated urban land use planning, among other barriers that are discussed below. To tackle these environmental challenges, including greenhouse gas emissions, the Tunisian Government is working on the National Programme for Sustainable Cities in Tunisia. This program has identified a cohort of 14 cities to implement this program, such as Zaghouan, Kasserine and Kairouan.

4. The city of Kairouan has been chosen based on the corresponding size of the city and demographic dynamics (Table 1)[8]. The criteria were aligned with those in the GEF7 CCM1-3 programming objective. Besides addressing climate change mitigation as discussed below, the proposed project will also have to deliver significant sustainable development co-benefits across several Sustainable Development Goals (SDGs), such as gender (high levels of illiteracy and unemployment among females) and job creation (high levels of unemployment).

[4] https://www.tn.undp.org/content/tunisia/fr/home/countryinfo/ - accessed 21 September 2021.

[5] https://covid19.who.int/region/emro/country/tn - accessed 25 January 2023.

[6] ILO & ERF (2022) Tunisia COVID-19 Country Case Study (https://www.ilo.org/wcmsp5/groups/public/---africa/---ro-abidjan/---sro-cairo/documents/publication/wcms_839018.pdf - accessed 26 January 2023).

[7] Ben Hassen, T, et al. (2022) Gendered Impacts of the COVID-19 Pandemic on Food Behaviors in North Africa: Cases of Egypt, Morocco, and Tunisia. Int. J. Environ. Res. Public Health 19, 2192. https://doi.org/10.3390/ijerph19042192

[8] UNDP (n.d.) Stratégie de développement Durable de la Ville de Kairouan 2030 ; Ministère des Affaires Locales et de l'Environnement (2019) Réalisation d'une étude diagnostique sur les villes tunisiennes et proposition d'une vision stratégique pour une ville durable.

This is particularly important since the COVID-19 pandemic has worsened the socioeconomic situation in a city like Kairouan,[9] including an amplification of gender disparities.

Table 1. Selected characteristics of the city of Kairouan

Geographic location	153 km from Tunis and 57 km from Sousse (Port); crossroads connecting north and south, and east and west; geographical coordinates (35.6712° N, 10.1005° E)
Area (km^2)	47
Population (2014)	187,000
Unemployment rate (%)	18.7 (exceeding the national average) [28.25% among female population versus 14.04% among male population]
Poverty rate (%)	34,5 (largely exceeding the national average)
Illiteracy rate (%)	17.64 (exceeding national average of 13%)
Demography	Population growth of 1.6% exceeding the national average of 1%; net emigration showing lack of attractiveness of the city
Gender disparities	- Illiteracy is more prominent among females - Unemployment among female population is higher despite the fact that women are more qualified
Climate[10]	- Daily mean temperature range: 11.5℃ – 29.5℃ - Average high temperatures: 17.2℃ – 37.7℃ - Record high temperatures: 30.0℃ – 48.1℃ - Average low temperatures: 6.9℃ – 22.9℃ - Record low temperatures: -4.5℃ – 12.0℃

5. Building energy use: There is a strong climate mitigation justification for investments in energy efficient or green buildings. The building sector is the main final energy consumer sector (37% in 2019, including biomass-energy). It is also related to the building materials industries which are among the most GHG-emitting sectors. From a value chain perspective, the building sector would represent – from upstream to downstream – around 50% of final energy demand in Tunisia, and at least 55% of GHG emissions attributable to energy in 2019.[11] Out of the 13.8 MtCO2 emission reductions to 2030 (relative to the business-as-usual scenario) expected from the energy sector, ~2MtCO2 are expected to be obtained from energy efficiency in buildings. The selection of Kairouan as pilot city also stems from the fact that it already implementing a sustainable energy plan (PAED), implying strong buy-in and ownership of the proposed project by the local government.[12] The emissions profile for 2010 for different economic sectors is shown in Table 2,[13] and based on projections the total emissions were expected to reach 402,762 tCO2 by 2020. As shown in Figure 1, electricity use produced around 50% of residential emissions, while buildings in the tertiary sector emitted around 20 ktCO2. In terms of primary energy consumption, the city consumed 68.5 ktoe, of which the municipality and residential sectors accounted for 0.65 ktoe and 16.76 ktoe, respectively.

Table 2. Emissions profile of Kairouan, 2010.

Economic Sector	**Emissions, tCO2**
Municipal	3,521
Residential	62,896

[9] GDP growth in 2020 hit a low of -8.8%.

[10] Institut National de la Météorologie (precipitation days/humidity/sun 1961–1990, extremes 1951–2017) ;

[11] Republic of Tunisia (2021) Updated Nationally Determined Contribution – Tunisia (draft).

[12] EU (2013) Commune de Kairouan Plan d'action en faveur de l'énergie durable (PAED).

[13] Updated values will be reconciled during the PIF formulation stage while engaging with stakeholders such as the National Energy Efficiency Agency (ANME) and the Governorate / Municipality of Kairouan.

Tertiary	18,923
Industrial	45,404
Transport	107,577
TOTAL	**238,321**

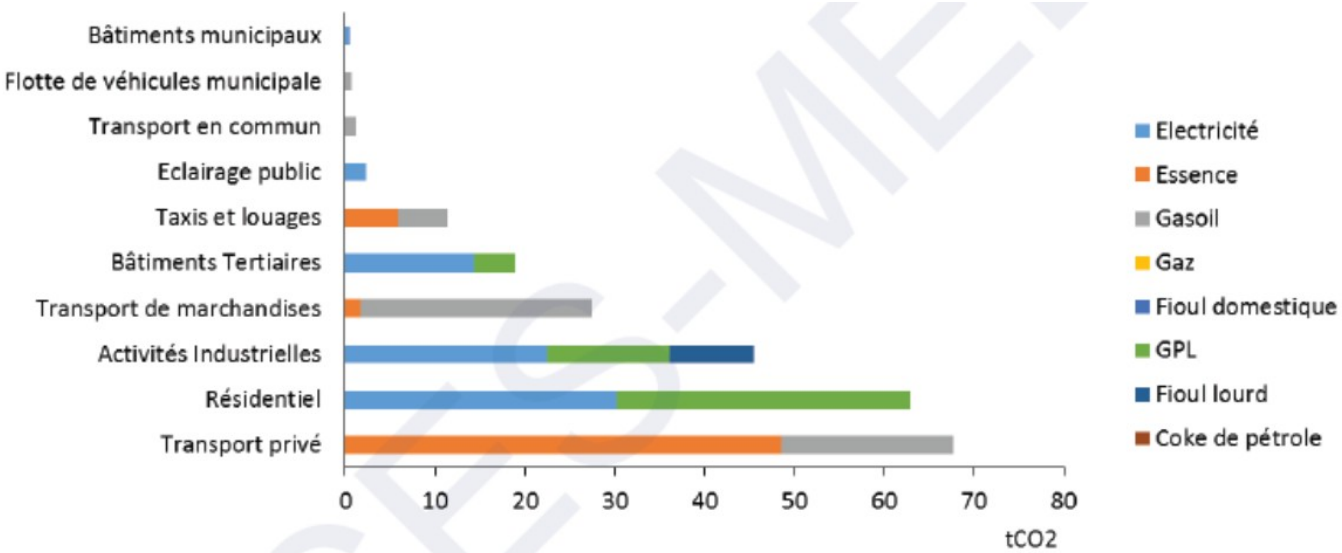


Figure 1. Sources of emissions in Kairouan by fuel type, 2010.
(Source: PAED)

6. These data demonstrate the large potential for emissions reductions through energy efficiency in buildings. In fact, the PAED proposes a number of energy efficiency interventions in buildings, including among others: (1) energy efficiency equipment in municipal buildings; (2) demonstration of a green building through energy efficient equipment and building envelope; (3) capacity development of stakeholders on energy efficiency in buildings. As discussed below, the project proposes to address the PAED interventions. This approach would be aligned with the objectives of the Sustainable Development Strategy of Kairouan,[14] and the National Programme for Sustainable Cities in Tunisia[15] to develop zero emissions buildings. The proposed project is also aligned with Tunisia's updated NDC as discussed in section III.

7. Public lighting: Another area that is of much concern for the municipality of Kairouan is the financial burden of public lighting. The municipality would like to decrease the operational costs of street lighting through energy efficiency. In this regard, an energy audit of public lighting was carried out by the national programme Alliance of Communes for Energy Transition (Alliance des Communes Pour la Transition Energétique, ACTE).[16] In 2019, 12.3 GWh of electricity was used for public lighting. The operational cost was 3 MDT representing 90% of the energy bill of the municipality. Using a grid emission factor of 447 tCO_2/GWh,[17] the corresponding public lighting emissions were ~55 $ktCO_2$.

8. Energy context: This has become an even pressing concern given the recent price inflation of imported fossil fuels, and, hence, of electricity prices, due to the global geopolitical energy conjecture. The exposure of the local economy to the price volatility is reflected in the high level of energy import dependency of Tunisia as

[14] UNDP (n.d.) Stratégie de développement Durable de la Ville de Kairouan 2030.

[15] Ministère des Affaires Locales et de l'Environnement (2019) Réalisation d'une étude diagnostique sur les villes tunisiennes et proposition d'une vision stratégique pour une ville durable.

[16] Perfect Engineering (n.d.) Rapport d'audit énergétique approfondi de l'éclairage public de la Commune de Kairouan.

[17] https://www.irena.org/IRENADocuments/Statistical_Profiles/Africa/Tunisia_Africa_RE_SP.pdf - accessed 19 May 2022.

shown in Figure 2. In 2021, natural gas and petroleum products represented 53.3% and 26.3% of primary energy demand, respectively, while renewable energy sources accounted for only 1.3%.

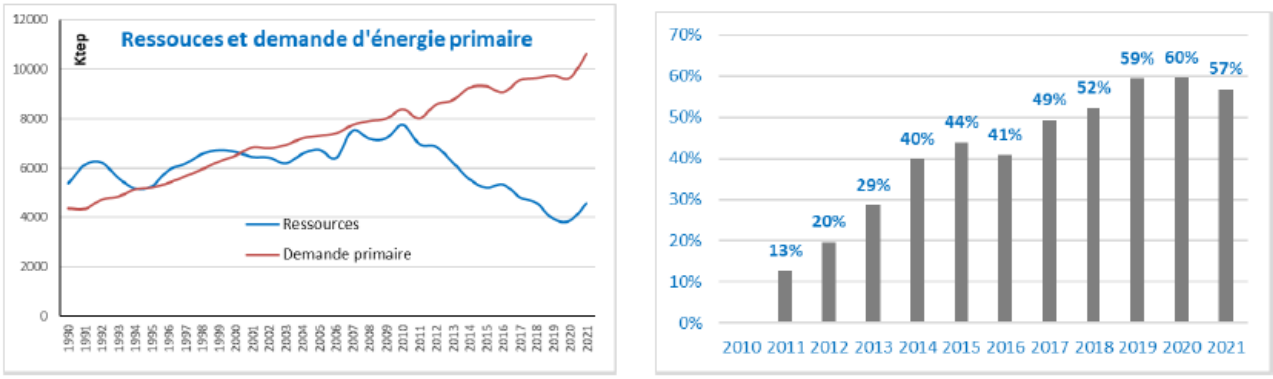


Figure 2. (left) Primary energy demand (red) and local production (blue); (right) dependency (%) on imported fossil fuels.
(Source: Authors' elaboration)

9. Regulatory frameworks: There is a number of regulations (Table 3) that are already in place to support energy efficiency in buildings and public lighting. The project will provide human and institutional strengthening so that the municipality of Kairouan is more capable to implement the legal provisions.

Table 3. Legal provisions for energy efficiency in buildings and public lighting.

Intervention	Regulation	Brief description
Energy auditing; stakeholder consultations and contractual agreements	Article 4, Law 2004-72	Mandatory energy audit for organisations with an energy use above a set threshold
	Article 5, Law 2004-72	New buildings (projects) or extensions with energy consumption forecasted to be above a threshold are verified by national agency for energy efficiency (ANME)
	Decree no. 2004-2144 and amended by decree no. 2009-2269	- Threshold for mandatory energy audits (industry – 800 toe/yr; transport, tertiary and residential – 500 toe/yr; frequency of energy audit – 5 years) - Threshold for extension projects (industry – 800 toe/yr; tertiary and residential – 200 toe/yr) - Auditors accredited by ANME - Energy audit carried out on project plan by 2 persons (engineer + architect) - Establishing the terms of reference of energy auditors
	Regulation from Ministry of Energy date 11 June 2007	Establishing the technical requirements for enery audits of development projects in tertiary and residential sectors
Building energy codes	Article 10, Law 2009-7	Projects for the construction of new buildings and projects for the extension of existing buildings are now obliged to meet minimum technical specifications for energy efficiency, which will be set by joint decree of the Minister for Housing and the Minister for Energy.
	Joint regulation of 23 June 2008 and amended regulation of 17 December 2010	- Lays down the minimum technical specifications to which projects for the construction of new buildings and projects for the extension of office buildings or similar purposes are subject (either performance-level or prescriptive-level approach) - Building thermal performance levels (public buildings: 1-3; private buildings: 1-5)

		- The prescriptive approach consists in setting the minimum technical specifications for the thermo-physical properties of the building envelope, depending on the climatic zone and the rate of the bay windows of the heated or cooled spaces, as well as their distribution over the different orientations - The building to be constructed or the parts of the building to be added are subject to a technical study by a consultancy firm or a specialized consulting engineer and verified and then approved by a technical controller authorized by the Ministry of Equipment, Housing and Spatial Planning - Any applicant of a building permit for the construction or extension of a building for office use or similar must attach to the permit application a technical sheet specifying the thermal performance of the building project according to the approach followed
	Joint regulation of 1 June 2009	Lays down the minimum technical specifications for the construction of new and extension of communal residential buildings
Energy efficient public lighting	Article 11, Law 2004-72	Obligation to comply with technical specifications relating to energy saving when installing public lighting networks
	Joint regulation of 9 February 2006	The decree sets out the technical specifications of the equipment used in public lighting networks for energy saving purposes

10. Financial derisking: To promote investment in the field of energy management and facilitate the implementation of programs contributing to energy management, the Tunisian State has created the Energy Transition Fund (Fonds de Transition Energétique, FTE, replacing the National Energy Management Fund created in 2005. The FTE provides financial assistance between 30,000 and 200,000 DT (municipal scale projects) for technical assistance (e.g. feasibility studies, energy audits) related to energy efficient buildings. It also provides financial support up to 400,000 DT for efficient building materials (e.g. demonstration projects, thermal insulation).

11. Barriers: In this contextual setting, a number of barriers need to be addressed in order to enhance the adoption of energy efficiency measures in buildings and public lighting. The main barriers are summarised in the Table 4. An issue that cuts across all the barriers is that women and young girls' voices and experiences of under-representation, often lead to a gender-blind designed project or a missing sense of ownership and sustainability component.

Table 4. Barriers preventing the adoption of energy efficiency in buildings and public lighting.

Barrier 1: Inadequate institutional capacity at the local level to implement legal frameworks
Buildings The Municipality of Kairouan does not have the institutional capacity to operationalise the legal framework for building energy efficiency in its permit and licensing process for the approval of construction projects (new and extension to existing buildings). This is because of low level of human capacity to apply the requirements of the legal framework described in Table 3; lack of procedures and operating manuals for integrating building energy codes in permits and licensing process; lack of surveillance / enforcement capacity of the municipality; and lack of technical capacity to use modeling tools such as CLIP developed by ANME. Public lighting The Municipality of Kairouan lacks the human and institutional capacity to carry out the following: (i) formulating a business plan for efficient public lighting; (ii) developing the terms of reference for shifting to efficient lighting, including technology assessments and identifying criteria for technology selection; and (iii) implement computer-assisted management of the maintenance of the lighting network.
Barrier 2: Absence of pilot to demonstrate the feasibility and cost-effectiveness of energy efficient building
As is typically the case for energy efficiency measures, the 'taste of the pudding is in the eating". Hence, social

acceptance relies heavily on the proof-of-concept. In the prevailing baseline, and with the traditional building materials and building skills having lost their social status in Kairouan, there are only few examples (e.g. Médina) of how an energy efficient building is compatible with a modern way of life. As discussed above, the proposed UNDP-GEF project could be developed around the proof-of-concept energy efficient building (building envelope using local materials, energy efficient equipment, solar thermal heating etc…)

Barrier 3: Lack of information on city sustainability

There is lack of a robust data management system to track the flows of materials in and out of the city. In the absence of verifiable data, it is difficult to support evidence-based policy-decision making. Data Standards for Sustainable Cities already exist such as the ISO 37120 Sustainable Development of Communities: Indicators for City Services and Quality of Life. The World Council on City Data (WCCD) has developed the first International Organization for Standardization (ISO) 37120 certification system and the Global Cities Registry™,[18] and the GEF-finance project could support Kairouan to become a member of the WCCD.

Barrier 4: Little use of ecological building materials (technology barrier)

Buildings

The National Programme for Sustainable Cities in Tunisia has noted in the case of Kairouan that there was little use of locally available ecological building materials. The low demand implies a shrinking of the local commercial chain to make such building materials available at competitive prices, and at scale. A compounding effect is the lack of national standards for building materials. The absence of such standards hinders the application of the thermal regulation for new buildings (Réglementation Thermique des Nouveaux Bâtiments, RTNB). There are also two associated causes for the non-adoption of energy efficiency in buildings, namely: (i) the lack of up-to-date buildings energy codes, and (ii) the limited scope of application of the existing legislation - i.e. the RTNB applies to a limited set of building types.

Barrier 5: Lack of skills to work with locally-available building materials

Buildings

The craft and workmanship with locally available construction materials is currently lacking, and it will be necessary to revitalize the traditional skills in order to support commercial value chains for producing the building materials locally and to build with the local materials.

Barrier 6: Perception barriers

Buildings

Lack of knowledge and negative perception of the benefits (relative to costs) of regulations and codes for energy efficiency in buildings still exist among stakeholders such as the construction industry players, the energy sector community and the general public. There is a lack of pedagogical and decision making tools widely available to the stakeholders based on lifecycle assessments. To date there has been no formal application of the ECOBat label and no capacity building and sensitization of stakeholders, including local authorities has been carried out.

Barrier 7: Financial barriers

Buildings and public lighting

There is concern among building developers regarding the higher upfront investment in energy efficient buildings. This situation is exacerbated by the split incentive paradox wherein the operating and maintenance costs of buildings are not borne by the developer but passed onto the owner or tenant. The Energy Transition Fund (Fonds de transition énergetique, FTE) managed by ANME is necessary but not sufficient for incentivizing investments in low-energy buildings. The prohibitive cost of shifting to energy efficient lighting is shown by example of a high pressure sodium lamp costing 250 DT, whereas a LED equivalent costing 800 DT. There is a need to make low-cost commercial capital available to developers while using FTE funds as a financial derisking instrument.

III. STRATEGY

[18] http://www.dataforcities.org/wccd/ - accessed 28 December 2016.

12. The Theory of Change (ToC) diagram for the GEF-financed and UNDP-implemented project shown in **Annex 1** captures the linkages between the developmental challenge and barriers discussed in Section II. It also shows how the project interventions at the level of the project outcomes address the root causes of the problems related to the application of building energy codes (efficient buildings), and in particular the RTNB (Réglementation Thermique des Nouveaux Bâtiments or Energy Code for New Buildings), and efficient public lighting at the municipal level. The project outputs are mentioned in the ToC diagram but they are not detailed. The detailed discussion of the project outcomes and outputs is the focus of Section IV.

13. The long-term impact of the project is to achieve greenhouse gas (GHG) emission reductions with strong sustainable development and adaptation co-benefits through implementation of building energy codes in and new buildings (intermediate goal), as well as the transition towards energy efficient public lighting. The long-term outcomes of the project relate to addressing the seven underlying problems that are given in Section II, namely: (1) low institutional capacity at the municipal level to implement legal frameworks; (2) the absence of pilot to demonstrate the feasibility and cost-effectiveness of energy efficient building; (3) lack of a robust data management system to track city sustainability through the flows of materials in and out of the city; (4) technological barriers arising from the low-level use of ecological building materials, especially those that are produced locally; (5) there is a lack of skills to work with locally-available building materials (related to the increasingly low-level use of local building materials); (6) lack of knowledge and negative perception of the benefits (relative to costs) of regulations and codes for energy efficiency in buildings among stakeholders; and (7) financial barriers arising from the higher upfront cost of energy efficiency measures. These barriers underpin the problem statement: 'significant barriers hamper the adoption of energy efficiency in buildings and public lighting thereby impeding the transition of Kairouan into a sustainable city'.

14. The intervention logic starting from the problem statement in Annex 1 is that IF the activities and outputs of the project are successfully implemented (given assumptions) THEN the three project outcomes will be achieved. In turn, the long-term project outcomes will THEN be achieved in the form of (i) enhanced institutional and human capacity for implementing energy efficiency measures at the city level; (ii) better engagement and coordination among stakeholders with positive gendered impacts; (iii) increasing the economies of scale for locally-manufactured building materials with a positive incidence on job creation and local wealth generation and retention; (iv) improved financial viability of municipal finances through reduced public lighting energy bills; and (v) scaled up action across other cities in Tunisia through sound knowledge management. The successful implementation of the long-term goals will THEN give the intermediate result of large-scale adoption of the RTNB and the ECOBat label for new buildings, as well as the city-wide adoption of efficient public lighting. With planned scaling up across sustainable cities in Tunisia, the impact will be reductions in the emission of greenhouse gases with sustainable development co-benefits.

15. There are different drivers of change and assumptions that come into play at different levels in the ToC. The external drivers are variables that fall outside the control of the project but which exert pressure in the relevant direction that support project interventions. When it comes to addressing the problem statement, there are a number of key enablers, including: (i) the need to improve the financial viability of the municipality and one lever is to reduce the energy bill through energy efficiency measures; (ii) a high level of political commitment (including the Mayor) for integrating a climate response in city planning; and (iii) the existing institutional knowledge regarding the requirements of the RTNB and knowledge with efficient public lighting. The key external drivers that will have a positive influence on the project logic are: (1) the high levels of fuel costs that exacerbate the energy bill of the municipality; (2) high level of unemployment requiring the need to develop job creation at the local level; and (3) the need for the municipality to deliver climate benefits to its citizens.

16. The technical assistance of the project will create enablers that are referred to as internal drivers of the change. The main internal drivers are: (1) the enhanced capacity to carry out enforcement of the RTNB (new buildings) supported by incentives and awareness-raising. This is the most critical element of the project since no global environmental (and other socio-economic) benefits can accrue without enforcement of Codes; (2) a performance data system is established that will ensure the tracking of the city metabolism, including energy use and GHG emission reductions, which will provide tangible evidence for investing in EE measures; and (3)

strengthened local administration with institutional and technical capacity for ensuring that EE measures are included in municipal buildings and public lighting. It is expected that the technical assistance, incentives and enabling framework that will be put in place by the GEF-financed project will lower the risk profile of investments in EE interventions so that the intermediate goal in the ToC is achieved.

17. The ToC contains a number of assumptions (shown in red boxes in **Annex 1**). At the level of project interventions, the main assumptions are:
    - Political and institutional support received at all levels (national, Kairouan, other municipalities)
    - Public and Private sector engagement in implementing RTNB;
    - Reliable and accurate data is available for monitoring and evaluating the project;
    - Successful enforcement of the RTNB.

18. At the level of the long-term outcomes, the assumptions are:
    - Project trained human resources are retained in the municipality of Kairouan, and tools and operational manuals are implemented;
    - Regulatory and institutional frameworks and economic environment are conducive for investments in RTNB & efficient lighting;
    - Knowledge management used for scaling up mitigation actions and deliver sustainable development co-benefits.

19. The assumptions relating to the intermediate goal are as follows:
    - Building developers and owners are sufficiently stimulated by enabling environment and incentives to implement RTNB;
    - Sustainability is ensured beyond the project lifetime, including the mitigation of project risks;
    - Stakeholders are well duly coordinated to ensure enforcement of RTNB;
    - Performance of electricity network is adequate for capitalising on the benefits of efficient public lighting.

20. Response to COVID-19: The transition to a sustainable city, with its focus on enhancing the overall quality of life of citizens, is expected to improve socioeconomic conditions, including addressing the negative impacts of COID-19, especially on the most vulnerable sections of society. The project will directly support recovery from COVID-19 in the city of Kairouan through (i) savings for the municipality from the adoption of energy efficiency measures in municipal buildings and efficient lighting (investments under Outputs 1.6 and 2.3) and scaling up efficiency measures (Output 3.4) that then provides financial space for the municipality to invest in other social and infrastructure services; and (ii) job creation by supporting value chains for energy efficient building materials and lighting equipment (Output 1.2, Output 1.4, Output 1.5, Output 1.6, Output 2.2 and Output 2.3), as well as by developing a strategy for revitalizing the value chain for locally-fabricated bricks (Output 1.3).

21. Multi-stakeholder process. A challenge in the enforcement of the RTNB is to ensure that there is proper engagement and coordination of all key stakeholders within and outside the municipality. This is even more pressing in the UNDP-GEF project because interventions occur at distinct geographical and political levels - i.e. implementation at the municipal level while regulatory framework, financial incentives and standards are designed and implemented at the national level). The two geographical levels relate to the physical location where project beneficiaries are found, namely: national (e.g. national institutions such as ANME and INNORPI), and regional (e.g. municipalities). The political levels are loci where policy decision-making (e.g. development of Building Codes and accompanying standards, design and administration of financial incentives, and issuance of Building or Occupancy Permits) takes place in Tunisia. The engagement of all project stakeholders (Section IV) at different geographical and political levels has been ensured through a rigorous stakeholder engagement process.

22. Geographical coverage: The project interventions will take place in the city of Kairouan[19] that is located in the Governorate of Kairouan as shown in Figure 3.[20] The baseline new building that will be enhanced using GEF investments will be in the municipal district of Keblia. Details of the geographic location of the baseline building are given in Annex 2. The locations for retrofitting public lighting are listed in Table 5, and the corresponding geographical locations within the city are given in Annex 2.

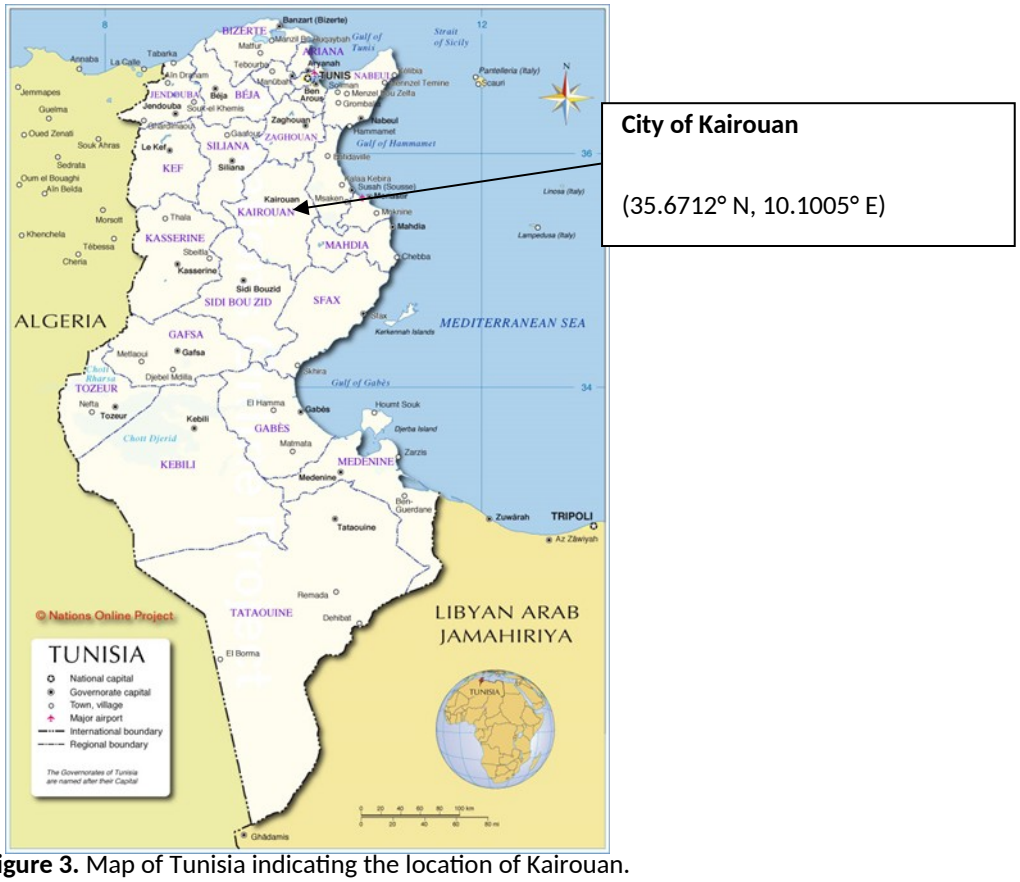


Figure 3. Map of Tunisia indicating the location of Kairouan.

Table 5. List of locations chosen for retrofitting street lights in the city of Kairouan.

Location	Number of lighting points	Existing Power rating (W)	Power rating with GEF investments (W)
Avenue de l'environnement et giratoire 'Zarbiya'	206	250	100
Place Sadeka	58	150	80
Cité commerciale et cité des martyrs	60	100	80
Avenue Ibn Aghlab	70	250	100
Avenue Touhami Negra	94	250	100
Avenue Sadek Lemkaddem	24	250	100
Avenue Ibn Jazzar	34	250	100
Avenue Moez Ibn Badis	28	250	100

[19] Project interventions will be in the City of Kairouan over which the Municipality has authority, and not over the entire Governorate of Kairouan. This said, the results of the project will be applicable more broadly to the Governorate and other cities in Tunisia.

[20] https://www.nationsonline.org/oneworld/map/tunisia-administrative-map.htm - accessed 17 May 2022.

Total	**574**	**128,700**	**55,040**

23. Transformation through enforcement and incentives. As discussed above, one of the problems confronting the implementation of energy efficiency designs in buildings is the lack of enforcement of the RTNB and the lack of incentives (monetary and non-monetary) to drive changes in the behaviour of the market for new buildings. These barriers persist despite the pressure exerted by external drivers for adopting energy efficiency in buildings, as well as the cost effectiveness of energy efficiency measures in general (as discussed in Section IV). This situation can be changed using a combination of enforcement of the existing RTNB by putting in place the appropriate institutional structures for linking third party verifiers, a certification system (ECOBat designed by ANME) and the issuance of Building and Occupancy Permits by the municipality. To change market behavior towards adopting the RTNB, a combination of monetary and non-monetary incentives will be provided. The monetary incentives will take the form of grants from the Energy Transition Fund (FTE) and fiscal incentives on municipal tax on housing and land, while the non-monetary incentives will take the form of increased floor area ratio (FAR) and building energy performance certificates. It is pointed out that the non-monetary incentives will translate into direct financial benefits in the form of higher real estate value of the energy-labelled buildings. The application of the RTNB and the ECOBat label will be applied to the baseline municipal buildings in the District of Keblia as proposed under Output 1.6 (Section IV). The UNDP-GEF project will also carry out advocacy for increasing the scope of application of the RTNB to non-collective residential buildings.

24. Gender mainstreaming: The participation of men, women, and youth in energy efficiency value chains is not equitable, and the challenges faced by women and men are discussed in Annex 12. Consequently, a gender responsive strategy, which is reflected in the ToC, is applied across all project components as follows:
    - There is low level participation of women across the entire energy efficiency value chain for buildings and public lighting. For GEF investments under Components 1 and 2, the gender impacts of investments in energy efficiency measures will be demonstrated;
    - Lessons learned reports (Component 3) will highlight the women and youth empowerment linkages with energy efficiency interventions. The project will optimize the use of gender-responsive training materials to project stakeholders and project interventions will ensure that the respective roles, and needs and interests of men, women and youth are taken into consideration;
    - The project will adopt the following principles in the day to day management: (i) gender stereotypes will not be perpetuated; (ii) women and other vulnerable groups will be actively and demonstrably included in project activities and management whenever possible, and (iii) derogatory language or behaviour will not be tolerated;
    - The Project Manager (PM) will be the designated focal point for gender issues to support development, implementation, monitoring and strategy on gender mainstreaming internally and externally. The PM will be supported by UNDP experts (Country Office and Regional Hub) on gender transformation issues; and
    - For regular monitoring and evaluation of the project progress and reporting, the project makes use of gender-disaggregated indicators (Gender Action Plan and Project Results Framework), and will facilitate involvement of women in the M&E and the implementation of Grievance Redress Mechanisms.

25. Knowledge management. Knowledge management is a cornerstone component in the project design since the impacts of the project will accrue from a scaling up of its results. Output 3.3 seeks to capture and disseminate lessons learned and best practices within Tunisia and other MENA countries. For instance, the development of an online tool under Component 1 (Output 1.4) for carrying out comparative socio-economic and environmental benefits of buildings with or without EE measures will be disseminated broadly to be used as a decision-making and pedagogical tool for promoting low-carbon buildings. Similarly, in Component 3, a replication and investment plan for scaling up the lessons learned on city-wide approaches for promoting energy efficient buildings and public lighting will be formulated. In addition, the UNDP-GEF project seeks to transform the municipality into a learning organisation that is characterised by evidence-based decision making using a data management system that will allow city sustainability to be tracked. For this, the project will assist the municipality of Kairouan to adopt Data Standards for Sustainable Cities, namely ISO 37120. Also, the project will support the city of Kairouan to become a member of the World Council on City Data (WCCD). The adoption

of ISO 37120 will be a precursor for the city of Kairouan to develop its inventory of greenhouse gases on a regular basis.

IV. RESULTS AND PARTNERSHIPS

26. Expected Results: The proposed project builds on UNDP's Urban Offer. UNDP's integrated support in this Offer coalesces around 5 areas: (1) building the capacities of cities, including for global engagement; (2) supporting the transformation of urban societies and economies to benefit all; (3) facilitating exchange of knowledge and resources (including technological) for best-fit solutions; (4) facilitating partnerships to build investment base and local digital ecosystems to support smart growth; and (5) ensuring urban developments as integral to Sustainable Development Goal (SDG) achievement by supporting local authorities to localize the 2030 Agenda, achieve the Paris Climate commitments, and the New Urban Agenda.

    UNDP adopts an integrated and multidimensional systems approach to achieve urban sustainability, inclusiveness, and resilience – including the articulation of urban poverty into programming using metrics and tools (as framed in UNDP's sustainable Urbanization strategy):

    - Assessments: Support cities to enhance coherence across national and local planning and budgeting instruments and processes (leveraging integrated development planning and inclusive growth, local governance and local development). Enhance monitoring and reporting of urban development progress and of achieving the SDG targets.
    - Thematic areas: Building on existing UNDP urban development programmes, facilitate access to lessons learned and best fit solutions on key areas for sustainable development areas, ensuring progress throughout the rural-urban continuum. This includes informal economy, circular economy, resilience-building and risk-informed development and digital transformation.
    - Building capacity of municipal authorities to address targeted urban development issues using tools such as poverty data analysis, vulnerability assessments, futures analysis and scenario-building, integrated urban planning. This can relate to local management of migration and displacement, urban and community risk management, and climate change adaptation).
    - Financing: Support municipal authorities to cost urban development interventions and manage finance, especially climate finance, from different local, national, and international sources, including public, private, and blended resources and innovative financing.

27. The total cost of energy efficiency interventions pledged in the updated Tunisia NDC is estimated at USD 5,755 million representing 49% of all funding required to implement measures in the energy sector. Several mitigation measures planned at the municipal level have not been implemented according to plans because of the prevailing barriers (Table 4), and particularly low levels of low-cost financing. Access to climate finance is one way of circumventing shortages of local funding that has plagued measures to reduce GHG emissions in Tunisia, and the proposed project will serve as catalyst for leveraging further climate finance. Since the GEF grant is 846,906 USD and cash co-financing of USD 0.5 million are relatively small compared to investments required to transform Kairouan into a sustainable city, the proof-of-concept of a low-carbon building, while addressing barriers discussed below, will serve as template for developing a larger project intervention under the next GEF cycle or targeting the Green Climate Fund. This would be in line with government's emphasis for a significant share of international climate finance to support conditional emission reductions in the NDC.

28. Using funding from the GEF and co-financing, the project will support baseline activities that face the barriers listed in Table 4. By reducing these barriers, and using the Theory of Change discussed in section III, the project will substantially enhance the prospects of both the baseline projects (Components 1 and 2), and follow-on projects catalyzed as a result (Component 3). By the end of the project, it is expected that:
    - Energy efficient street lighting is implemented in the City of Kairouan;
    - Proof-of-concept demonstrated for the application of the thermal energy code for new buildings, as well as the national label ECOBat for new buildings;

• Institutional mechanisms in the form of streamlined process with operational guidelines for mainstreaming the legal requirements of the RTNB in the new buildings permits and licensing process;
• National standards will be developed for constriction/building materials and outdoor lighting technologies that will provide better quality assurance to end-users;
• Replication plan and bankable Concept Note to attract additional financing will be developed for scaling up of emission reductions in the building sector and street lighting in Tunisia in support of the National Programme for Sustainable Cities;
• A local value chain for the production of ecological building materials will be developed that will have strong gender-differentiated socio-economic benefits (e.g. gender-differentiated jobs);
• Lessons learned and project experience will be shared nationally and regionally to increase the impacts of the project results;
• Tools will have been developed to provide stakeholders about information about the benefits of investing in Building Energy Code; and
• The enabling conditions created by the project will have the long-term impact of catalysing national and international investments to implement the RTNB and ECOBat label, and efficient street lighting that will lead to direct and indirect emission reductions as discussed below.

29. **Project Objective**: To promote green buildings and efficient lighting in alignment with the objectives of the enhanced Nationally Determined Contribution, the Sustainable Development Strategy of Kairouan and the National Programme for Sustainable Cities in Tunisia.

30. **Changes made in project design**: Baseline analyses were updated during PPG stage and changes were made in the project design (Table 6). Prior to describing the project components, outcomes and outputs, the changes brought to the project design are discussed. Changes have been brought at the output level only, without changing the project components and corresponding outcomes. While relatively few outputs have been removed compared to the aforementioned discrepancy between scope and budget at PIF stage, the updated project design has also introduced either new outputs or increased the scope of others.

Table 6. Changes brought in project design at PPG stage.

Output in PIF (or *new output*)	Changes made	Reasons for change
Component 1 / Outcome 1		
Output 1.6. Proof-of-concept new municipal building adopting RTNB and ECOBat label demonstrated	Two changes have been made: (i) the title has been changed to "Output 1.6. Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label" (ii) Baseline project changed from municipal Headquarters for the district of Metbasta to the municipal Headquarters for	As discussed under Output 1.2, the project will use the enforcement mechanism that will be put in place to also coordinate national and municipal level stakeholders for the review and update of the RTNB for instance to include a self-racheting mechanism for its regular update in order to increase its level of ambition as far as best available technology and scope of application are concerned. Given the short time duration of the project, there is no need to update the RTNB before carrying out investments under Output 1.6. This is because the ECObat label provides for building energy labelling that outperforms the minimum requirements of the RTNB through the use of best available technologies. As explained below, the baseline municipal building has been changed and this will constrain the actions that can be implemented. By applying the Class 3 level of building energy efficiency, the GEF investments - accompanied by enforcement modalities that will be put in place by

	the district of Keblia	the project – demonstrate the path towards zero-emission buildings, while noting that investments in roof-top production of renewable energy will remain outside the scope of the project. At PIF stage, the proposition was to use the municipal Headquarters for the district of Metbasta that is proposed to start building in 2023 as the baseline project to enhance using GEF investments. However, discussions with the Municipality of Kairouan, and, in particular, the Mayor, have revealed significant administrative and financial constraints that will delay the construction of the new municipal building in Metbasta. The Municipality of Kairouan, therefore, proposed to instead use the extension (phase 2) of the municipal headquarters for the district of Keblia as baseline project. Phase 1 of new building, which consists of Ground + 2 levels (G+2), will be completed around mid-2023. Phase 2 constructions will consist of extending the new building with 2 additional storeys. As discussed in more details for Output 1.6 below, the GEF investments will be used to improve the thermal properties of (i) the 2 storeys comprising Phase 2 extensions; and (ii) the outer façade (envelope) of the existing G+2 structure.
Component 2 / Outcome 2		
Output 2.3: A smart usage system for municipal lighting demonstrated	(i) Number of lighting points to be changed has been revised from 1,000 units to 574 units (ii) The PIF proposed to set up an intelligent public lighting system for the remote management of public lighting. This element has been removed from the project because of concerns of cost and also because of parallel initiatives	(i) In the PIF, it was proposed that 1,000 units of high pressure sodium lamps with 150 W power rating would be replaced with LEDs (80 W). Discussions with the municipality has shown that it would be more energy efficient (and with higher financial savings for the municipality) to replace SHP lamps of 250 W power rating (as shown in Table 5 above). Replacement of a 250 W SHP with LED results in a power saving of 150 W in comparison to 70 W savings when replacing a 150 W SHP with LED. For increased public visibility (part of strategy for choice of locations – Annex 2), few 100 W SHPs have been identified for replacement as well. Overall, since the cost of 100 W LEDs that are used to replace 250 W SHPs is higher, the number of lamps has been reduced. However, with fewer SHP lamp replacements in the ProDoc, the power savings are higher at 73.66 kW compared to 70 kW as proposed in the PIF. (ii) Since global environmental benefits in the form of greenhouse gas emission reductions accrue mainly from the replacement of lamps, and given that investments in an intelligent system for managing the public lighting

		network remotely would be prohibitively (relative to the small project funds), it was decided not to invest in the intelligent remote sensing system. This aspect of public lighting management can be covered by parallel initiatives. However, the 574 LED lamps that will be purchased will be compatible with an intelligent system of remote sensing. Consequently, this output has been reformulated as 'Output 2.3: Energy efficient municipal lighting demonstrated'.

31. **Component 1: Institutional strengthening for regulating the market for new efficient buildings**: In support of focusing on energy efficiency in buildings, the updated NDC notes that Tunisia intends to focus on the choice of the best energy technologies and practices for the building sector, and would therefore need to forge international partnerships in order to develop and generalize sustainable construction techniques, materials and building materials industries. The construction sector is therefore one of the sectors on which Tunisia intends to focus on in terms of technological innovation. As discussed above, this intention is already captured in the Sustainable Development Strategy of Kairouan that has a strong cultural heritage in building construction using indigenous materials. Given climate changes (Table 1) will place more stress on thermal cooling and heating needs in buildings, the indigenous building materials and techniques could prove to be effective and efficient in the passive design of buildings. In addition to providing thermal comfort, scaling up passive design will bring strong socioeconomic benefits by reigniting a local construction industry based on locally-sourced materials, and by developing traditional building skills. Such an approach will squarely support implementation of SDG11. For this to happen, the GEF-financed project will demonstrate the proof-of-concept of an efficient building by applying the thermal regulation for new buildings (RTNB) and the EcoBAT energy label using a public building listed in Table 5. It is pointed out that the EcoBAT label has been designed to incentivize building developers to adopt energy efficiency practices that go beyond the minimum requirements of the RTNB. While acknowledging that building energy codes need to be updated on a regular basis, Output 1.2 provides for a mechanism for the review and update of the existing legislation can be carried out. As discussed below (Output 1.6), the application of GEF investments will seek to adopt the possibility of using locally sourced materials and traditional skills will be explored. It will also strengthen the institutional capacity of the Municipality of Kairouan to issue permits and licensing process for energy efficient buildings, and, hence, to regulate the market for new efficient buildings.

32. Given the low application of the RTNB, and given the central role that the Municipality plays in the issuance of building and occupancy permits, it has been deemed suitable to use a municipal building as baseline project for energy efficiency enhancements using GEF financing. There are several reasons for this, namely: (i) interventions in a municipal building will be advantageous for the parallel capacity strengthening of the enforcement capacities of municipal staff because of easy access to the building compared to the case of a non-municipal building wherein permissions would be required (Output 1.1); (ii) better capacity to plan capacity building of municipal staff on operational procedures for enforcing building energy codes in accordance with the timeline for project implementation and construction of municipal building (since all three are managed by the municipality) (Output 1.2); (iii) the municipal building that is visited by a large segment for the population for municipal services is good for the project visibility; (iv) the public sector taking a lead on energy efficiency can be used as a strong message for private developers; and importantly (v) the selection of a municipal building for GEF investments does not exclude the participation of a private sector project ('the Proportion of Gold' – Activity 1.6.3) for the adoption of the ECOBat label. Finally, the municipality may also be more open for using locally-fabricated bricks in the construction of the municipal building, albeit on a limited scale, as part of the government responsibility for protecting local heritage. The scaling up of the results of the project will include a wide range of building types, over and above municipal/governmental buildings (Output 3.4).

33. The expected outcome from outputs proposed in Component 1 is: “institutional strengthening with enabling conditions, methodologies and tools for enforcing regulatory framework for new buildings”. The outputs proposed to achieve this outcome within the broader ambit of city sustainability are designed to overcome the barriers given in Table 4. The first five outputs will be through Technical Assistance, and Output 1.6 will be of Investment-type. Outputs 1.1, 1.2, 1.3 and 1.6 will be implemented in the City of Kairouan. Outputs 1.4 and 1.5 will be applicable at the national levels and supportive of local level interventions. The outputs are:
    - → Output 1.1: An operational manual for new building licensing and permits process in the Municipality of Kairouan to operationalise RTNB (Réglementation Thermique des Nouveaux Bâtiments) is designed and adopted;
    - → Output 1.2: Enforcement capabilities of municipality strengthened for ensuring new building compliance with RTNB;
    - → Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials;
    - → Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated;
    - → Output 1.5: National standards for building materials are developed and adopted; and
    - → Output 1.6: Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label.

34. *Output 1.1: An operational manual for new building licensing and permits process to operationalise RTNB is designed and adopted* – The Municipality is responsible for issuing building permits and licenses for new constructions. The existing procedures do not allow the municipality to effectively integrate operationalization of the RTNB in the permits and licensing process. This output will review the exiting process and develop an operational manual to ensure the application of the RTNB. The staff of the municipality will be trained to apply the new process. It is pointed out that the process will be applicable to the implementation of the RTNB in its current form as well as to future changes the building energy codes (Output 1.2). The following activities will be carried out:
    - Activity 1.1.1: Review of existing licensing and permits process. The existing licensing process will be reviewed and a gaps analysis will be carried out to understand the areas of improvements. The Municipality holds a key position in the issuance of Building and Occupancy Permits, and, hence, should play a central role in the application of the RTNB. The process for the issuance of these permits is illustrated in Figure 4. Whereas the onus is placed on the Developer to hire a Consultancy Firm to assess the thermal characteristics of a proposed building (excluding standalone residential buildings / houses), and to issue a Certificate of Conformity in relation to the RTNB), there is no mechanism in place for the verification of the conformity to the RTNB at the time of construction. At the step in the process related to the issuance of the Permit of Occupancy, the final inspection carried out by the Municipality does not allow for the verification of the thermal components of the building. A verification step under the oversight of the Municipality is missing during the period of construction as illustrated by the broken arrow in Figure 4. Technical assistance will be provided so that the deficiencies of the existing licensing and permits process are well understood, which will be used to propose a process that allows for verification of the application of the RTNB (Output 1.2). The focus here will be mainly on the institutional deficiencies of the municipality of Kairouan.

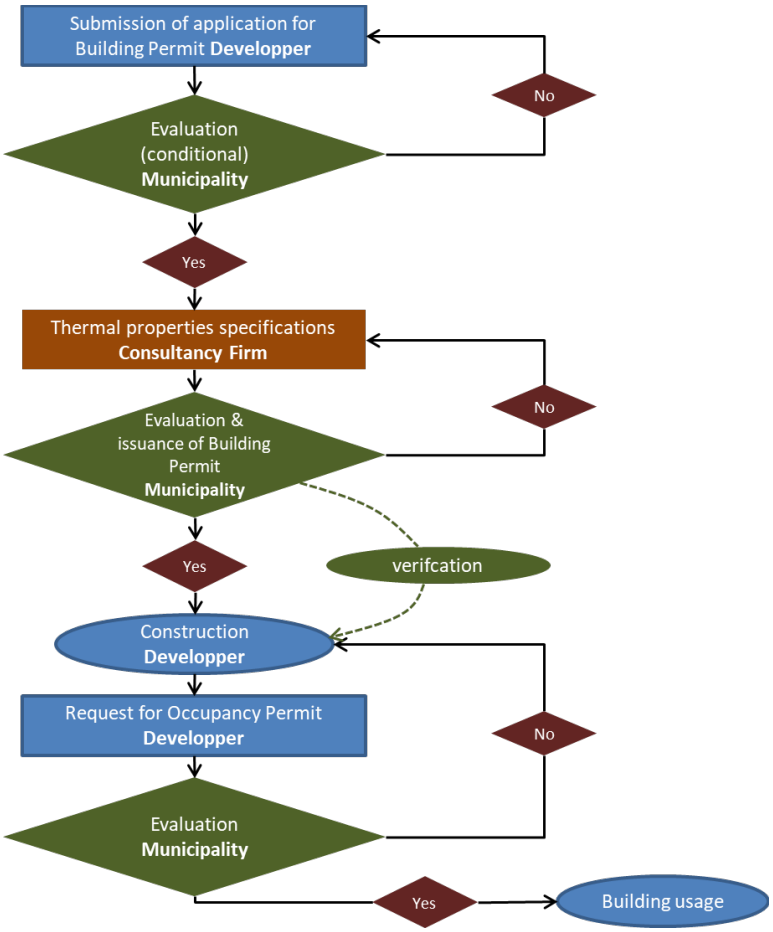


Figure 4. Mechanism presently used for the issuance of Building and Occupancy Permits.

- Activity 1.1.2: Operational manual formulated. An operational manual will be developed so that the permitting process designed under Activity 1.1.1 can be implemented.
- Activity 1.1.3: Training of staff. The staff of the municipality will be trained for the application of the operational manual for ensuring the application of the RTNB, and the enhancement of the municipal building at Keblia will be used as example.

35. *Output 1.2: Enforcement capabilities of municipality strengthened for ensuring new building compliance with RTNB* – This output is directly linked with the previous one. As mentioned above, the prevailing practice is that the RTNB is not enforced. For achieving the outcome of Component 1, the staff of the Municipality will be provided with the necessary capacity building to support enforcement of the RTNB through the licensing and permits process. However, as shown in Figure 4, the process involves several stakeholders. This output will propose the most effective and efficient institutional mechanism for the enforcement of the RTNB with clearly defined roles and responsibilities of all actors involved;
    - Activity 1.2.1. Cross-institutional enforcement mechanism established: There is recognition of the institutional weakness for ensuring compliance with the RTNB, especially through the enforcement during the construction phase. ANME is planning to develop a process to develop a multi-institutional mechanism with clear roles and responsibilities that will ensure enforcement of the energy building code. The UNDP-GEF project will provide technical assistance to this effect, and the mechanism will be operationalized through Activity 1.2.2.
    - Activity 1.2.2. Cross-institutional enforcement mechanism operationalised. The proposed institutional mechanism will be tested and operationalised using the enhancement of the municipal building in Keblia as proof-of-concept (Output 1.6), as well as the private building that will be developed by 'the Proportion of Gold' property developer (Output 1.6.3).

- Activity 1.2.3. Review and update of the RTNB. The RTNB was last updated in 2010 (Table 3) and it is known that building codes need to evolve with time and with best available technology. The institutional mechanism that will be set up under Activity 1.2.1 will be used to coordinate stakeholders (national and local) in order to update the RTNB; not just regarding making it more stringent in terms of energy efficiency exigencies but also in terms increasing the scope of types of buildings covered. It is expected that the review and update of the RTNB will reflect the use of BAT and to incorporate a self-ratcheting update mechanisms (e.g. every 5-7 years) to ensure it remains actual and its ambition is increased overtime.

36. *Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials* – The City of Kairouan has a strong cultural heritage. It hosts several heritage buildings and sites such as the Oqba Ibn Nafi mosque (also known as the Great mosque); the Bir Barrouta monument; the Sidi Abid mausoleum; the ponds of Aghlabides (made of locally-available stones); and Medina, which is a UNESCO World Heritage Site, among others (Annex 15). Of particular interest for the UNDP-GEF project is the Medina that is mainly constructed using locally-fabricated bricks. However, the local value chain has waned over the years with the introduction of mass-produced modern bricks that are relatively cheaper but without good thermal insulation properties and also not compatible with the ancestral architectural characteristics. The value chain for local production of bricks relies on demand from the National Institute for Heritage for the maintenance and renovation of cultural buildings, such as the Medina. This output will develop a strategy and action plan to support re-invigorating the local production of sustainable and low-carbon building materials in support of the Kairouan Sustainable Development Strategy. The emphasis will have to be on the stimulation of demand for locally-produced bricks, as well as ensuring quality standards and working conditions for what remains a highly labour intensive sector. The potential for new jobs creation will be a strong focus of the strategic plan given the relatively high rates of unemployment (Table 1);
    - Activity 1.3.1. Strategy and action plan. Tunisia has two good examples of how value chains for local construction materials can be developed. One is the locally-manufactured brick in Tozeur, and the other is the value chain for gypsum plaster (Annex 15). As is the case with the clay needed for the fabrication of bricks in Kairouan, Tunisia has important mining resources that constitute the raw materials for the manufacture of the Tozeur brick and plaster. The UNDP-GEF project will draw from lessons learned from those two experiences to propose a strategy and action plan for the enhancement of the local value chain for bricks produced in Kairouan.[21] One approach that has worked well for the development of a local value chain for the production of gypsum plaster in Tunisia is the application of the ValueLinks[22] that is comprised of the four components (and 11 modules) as shown in Figure 6. Since local stakeholders are already conversant with this methodology, it is proposed that the same be applied in the case of the value chain development for locally-fabricated bricks in Kairouan. During preparation of the Strategy and Action

[21] There are two types of natural resources that are used for the fabrication of construction materials. One is the 'white stone' that is mined from two quarries in Kairouan (Rouissat and Jloulla) by two companies employing a total of around 45 persons. The value chain for 'white stone' is healthy as there is good demand for it. The other is clay that is used for making bricks that are manufactured using indigenous techniques, such as manual preparation of the clay mixture, moulding of bricks and firing of bricks in traditional furnaces. The value chain for the brick made of clay is of interest in Output 1.3. The size of activities can be gauged by the number of furnaces that are operational. There are presently four traditional furnaces that are operated manually by around 10 persons. Because of dwindling demand for these traditional bricks, the livelihood condition of the workers is at best precarious. Discussions with a few workers have revealed that they have to supplement their income through other part-time income generating activities. The demand for the traditional clay bricks are mainly from the Institut National du Patrimoine that uses the bricks for the restoration of heritage buildings in the Medina; a UNESCO Cultural Heritage Site. There are visual signs that the restoration of buildings in the Medina is not always carried out using the traditional bricks, especially for residential buildings that are privately owned. The Association de Sauvegarde de la Medina has pointed out two main barriers that reduce demand the locally-fabricated brick. First, there is lack of clarity or a loophole in the existing legislation (Loi no 94-35 1994 – Code du patrimoine archéologique, historique et des arts traditionnels). For instance, the law does not stipulate the use of traditional building materials for the restoration of the Medina. Instead, it makes provision for a modicum fine of 250 DT for infringements that makes the economic calculus for using the cheaper modern brick more viable. Second, the Municiaplity of Kairouan does not have architects with the necessary knowledge for the protection of cultural heritage, which is a specialised area.

[22] Andreas Springer-Heinze (2018) ValueLinks 2.0 – Manual on Sustainable Value Chain Development, GIZ Eschborn.

Plan, a targeted and strategic assessment will be undertaken to assess potential associated environmental and social risks and take them into consideration in the final draft.

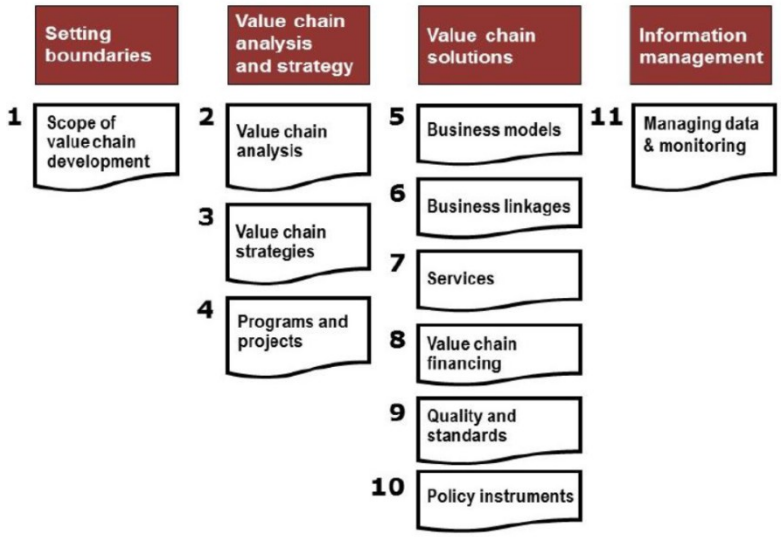


Figure 6. Components and sub-components of ValueLinks for the development of sustainable value chains.

- Activity 1.3.2. Characterisation of the properties of bricks. One important set of parameters that is needed to support the adoption of locally-fabricated bricks are their thermal and physical (mechanical) properties. To date, such properties are not available. The project will collaborate with the Technical Centre for Construction Materials and for Ceramic and Glass, CTMCCV (Centre Technique des Matériaux de Construction et du Céramique et du Verre) to measure the thermal and physical of locally-manufactured bricks.

37. *Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated* - In order to overcome perception and awareness barriers, an online tool will be developed to justify the socio-economic and sustainable development benefits of green buildings using life-cycle analysis. One of the persistent perception barriers is that the cost of investing in EE in buildings is prohibitive. The tool will be used for appraisal by investors; for green marketing by developers and professionals (to complement CLIP used in the process of obtaining building licenses and permits); and as a decision-making tool by end-users. The following activities will be carried out:
    - Activity 1.4.1. Assess the functional needs of end users. The design of the tool will be based on the needs of end-users. Hence, it is important to first assess the needs of a wide array of end-users such as investors, building developers, professionals in the construction industry, public officials, and end-users (mainly householders and private commercial business owners).
    - Activity 1.4.2. Design and validate the tool. The online tool will be designed under the oversight of the ANME and Tunisian Green Building Council (TGBC). A user-friendly interface will be designed to maximise the use of the online modeling tool by 'hiding' all the technical details of the calculations. With this interface, the tool can also be used as a pedagogical tool for teaching and training purposes.
    - Activity 1.4.3. Dissemination of tool. The tool will be made accessible on the websites of the ANME and TGBC. A methodological note will accompany the tool. The functionality of the tool will be ensured by ANME as the lead national agency mandated to promote energy efficiency in Tunisia. ANME will use its own resources to ensure the continued use and functionality of the tool, such as is the case today for the CLIP Tunisia modeling tool that is managed by ANME once it was developed under a previous UNDP-GEF project (Output 1.6 below).

38. *Output 1.5: National standards for building materials are developed and adopted* – Technical assistance will be provided to the National Institute of Standardization and Industrial Property (Institut National de la Normalisation et de la Propriété Industrielle, INNOPRI) for developing Tunisian quality standards for building materials. The following activities will be carried out.
    - Activity 1.5.1. Standard for bricks used for building envelope. There is currently a deficiency in the national quality standards for the bricks that are used for building envelope. Technical assistance will be provided to INNOPRI for the development of such standards. This will be supported using the results of Activity 1.3.2. Adoption of the standard can take place a multiple levels such as technical committee of INNOPRI, on the one hand, and by Cabinet of Ministers, on the other. Given the short duration of the project and the administrative and bureaucratic delays that are related with receiving Cabinet approval, the project will primarily seek approval of the standard by the technical committee of INNOPRI. Nevertheless, the process of seeking adoption by Cabinet will be supported by the UNDP-GEF project.

39. *Output 1.6: Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label* – All of the above outputs will be contextualised and institutionalised around the proof-of-concept for a new municipal building. In anticipation of the start of GEF project implementation in the second half of 2023, the baseline project 'municipal headquarters of the district of Keblia' is proposed to be enhanced to be in accordance with the RTNB. Further, the proof-of-concept will also make an application of the ECOBat label. The ECOBat energy label is a voluntary label designed to promote the adoption of building energy codes beyond the minimum energy performance standards prescribed in the RTNB. In its inception form, the ECOBat label is applicable to the following building types (same as those to which the RTNB is applicable: hotels, health institutions, commercial (office) buildings, and residential apartments with each type of building scored across three indicators, namely: (i) building envelope; (ii) electrical equipment; and (iii) management of resources (e.g. water, waste, health and project management).[23] The application and demonstration of the ECO-Bat label was initiated in 2022 through a pilot project involving two publicly-owned buildings, namely: (a) the Méditerranéenne-Soukra health clinic, and (b) the municipal building in Médenine. The two buildings are expected to be accredited at ECOBat label level 3 in order to demonstrate the application of the scoring and labeling system proposed in ECOBat.
    - Activity 1.6.1. Targeted assessment/ESMP for the building extension. As per the SESP screening, a number of risks have been identified that relate to the construction of the proof-of-concept and the scaling-up of energy efficiency in new buildings (see Annex 10). The Municipality of Kairouan will carry out a targeted assessment for the extension[24] of the municipal building in Keblia. In order to address the risks, the Environmental and Social Management Framework (ESMF) given in Annex 11 will be used to guide the formulation of the assessment, which will result in a site-specific Environmental and Social Management Plan (ESMP).
    - Activity 1.6.2. Enhancing energy efficiency of municipal building. The municipal headquarters for the district of Keblia has a surface area of 207 m^2. The CLIP Tunisie[25] software proposed by ANME has been used to model the thermal performance of the G+2 building as shown in Figure 5.Using the current building design and materials, the performance will place the building at Level 5 (Classe 5) on the ECOBat label (see Figure 7 below). This is the minimum performance level (i.e. score 0) admitted for a new building that is in conformity with the RTNB.

[23] ANME (2022) PROJET DE CAHIER DES CHARGES RELATIF AU LABEL DE PERFORMANCE ENERGETIQUE ECO-BAT ; Hassen Ben Hassine, Presentation entitled 'Label ECO-Bat – Choix fondamentaux et mise en place, Tunis, 24 February 2021.

[24] It is pointed out that the G+2 municipal building in Keblia did not require an Environment Impact Assessment (EIA). The justification from the Municipality is as follows: the building site was occupied by the Headquarters of the municipal police since the 1980s. Since the new building has maintained its public administration uses, an EIA was not required. The Commune mandated that the demolition wastes (of police headquarters) should be sent to controlled landfill sites.

[25] The CLIP model was developed under a previous UNDP-GEF project (GEF-2 PIMS 407/GEF ID 520 - Energy Efficiency In new Buildings | Tunisia).

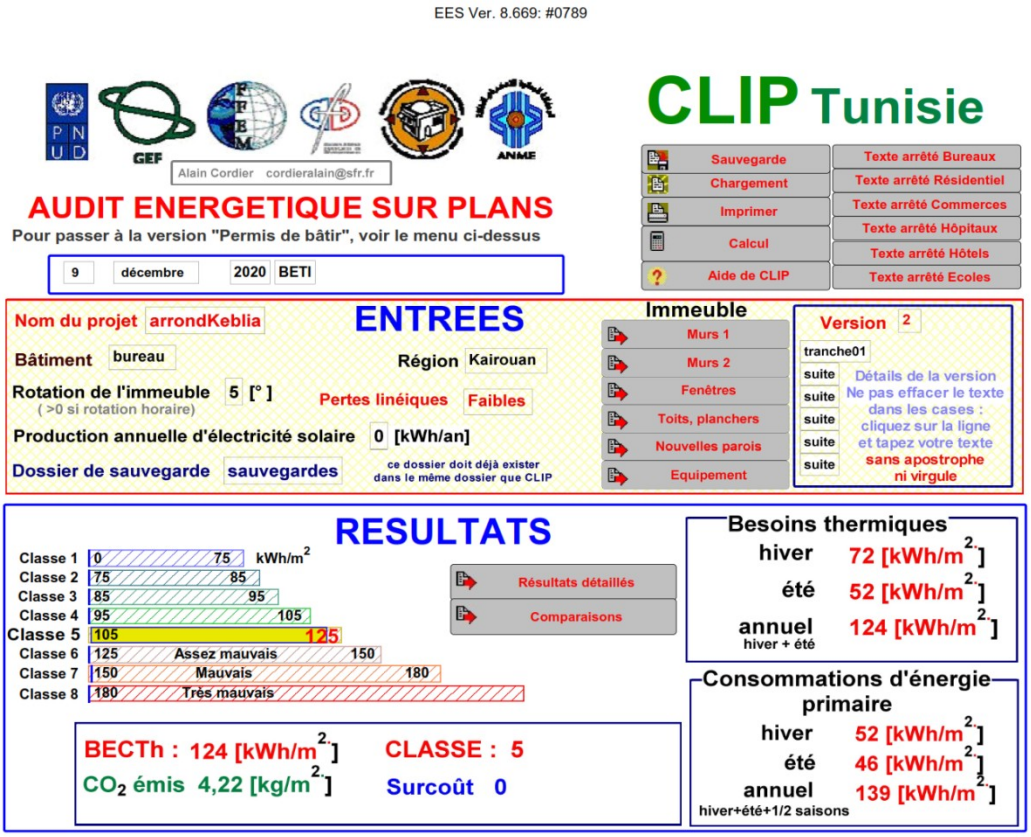


Figure 5. Modelled thermal performance of G+2 municipal building in Keblia.
(*Source: CLIP Tunisie modeling; courtesy - Municipality of Kairouan*)

Since the G+2 is yet to be commissioned, no physical interventions to improve its energy efficiency can be carried out. However, once it has been commissioned, then there will be the possibility to increase its thermal insulation properties by adding an external cladding of insulating material to the existing building envelope. This intervention can only take place one year after the warranty period following commissioning has expired. However, such constraints do not pose for the G3 and G4 extensions, as these will be subject to a separate building contract.

In the absence of the GEF project, the thermal performance of G3 and G4 will be at Level 5. Using Level 5 as the baseline (design related to Figure 3), the GEF investments will be used to enhance the thermal performance of G3 and G4 to Level 3 (Classe 3) as shown by the results of modeling in Figure 6. Hence, there will be an energy efficiency gain of 33 kWh/m2 from GEF investments, and the technical specifications for achieving this gain will be included in the design as summarized in Table 7 and Table 8.

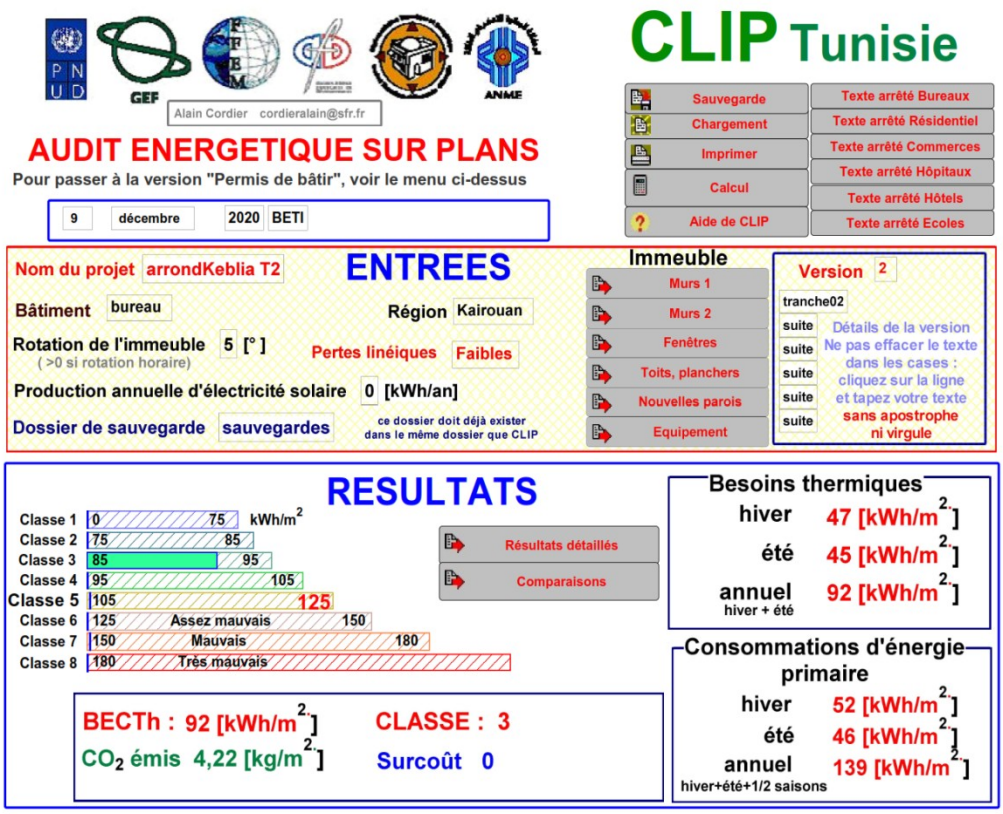


Figure 6. Modelled performance of G3 and G4 for municipal building in Keblia.
(*Source: CLIP Tunisie modeling; courtesy – Municipality of Kairouan*)

Table 7. Specifications of building envelope.

Floor	Surface insulated (m^2)	Façade	Surface area (m2)	Walls	Glass panes/windows, door	Roof
G3	156	North	258	* double-walled	**tinted glass	Concrete slab 16+5
G4	131	East	268	* double-walled	** tinted glass	***Concrete
		South	258	* double-walled	** tinted glass	
Total	287	West	268	*double-walled	** tinted glass	
		Total	1052			

Source: Municipality of Kairouan

Notes:
*double wall (35cm) with a layer of polyester (4cm)
**bronze-tinted glass (6mm)
*** Water-proofing material (0.5cm) + Cement mortar (3cm) + Cellular concrete (10cm) + Reinforced concrete (5cm) + Slabs (16cm) + Cement mortar rendering (2cm)

Table 8. Thermal specifications of openings.

N°	Type	L (m)	H (m)	Area (m^2)	East		South		West		North		TOTAL U
					U	Surf	U	Surf	U	Surf	U	Surf	
3	FA6	0.50	0.70	0.35	0	0.00	2	0.70	2	0.70	0	0.00	4

5	FA2	1.00	2.00	2.00	4	8.00	4	8.00	4	8.00	6	12.00	18
	TOTAL					8.00	6	8.70		8.70		12.00	22

Source: Municipality of Kairouan

The building envelope energy efficiency gains that will ensure Level 3 classification for the G3 and G4 will be obtained through the interventions listed in Table 7 and Table 8. An estimation of the cost of the energy efficiency interventions has been carried out and the results are shown in Annex 14 (Part D). The results are summarised in Table 9. The envelope insulation for G3/G4 will be inside (interior insulation), whereas for G/G1/G2 there will be retrofitting through exterior façade insulation that is more costly. The total cost of interventions for envelope insulation is estimated at USD 293,587. Although the GEF investments will not be used to install rooftop solar PV, a budget of USD 20,869 has been earmarked for (i) preparation of the insulated roof for roof-mounting structure; and (ii) designing the PV systems for achieving electricity autonomy of the municipal building at Keblia.

Table 9. Estimation of incremental costs of building envelope energy efficiency

	Walls (USD)	**Openings (USD)**	**Roof (USD)**
G3 + G4	20,857	15,258	16,698
G + G1 + G2 (retrofit)	217,889	22,887	Not applicable
Total (USD)	**238,746**	**38,145**	**16,698**

Given the relatively high unemployment level in Kairouan, all jobs created from this output will favour local labor and expertise. As far as practicable, the building materials, including inputs to improve energy efficiency will be sourced locally. Hence, there will be no influx of workers from outside Kairouan due to project activities, and local wealth creation will be privileged.

- Activity 1.6.3. Process to get ECOBat label. The ECOBat provides for three energy labels, namely Bronze, Gold or Platinum. The labelling system offers a progressively higher score for new buildings that rate higher than the minimum requirements of the RTNB (Level 5 or Classe 5) as illustrated in Figure 7 below. The UNDP-GEF project will support the Municipality of Kairouan to get a Level 2 (Classe 2) labelling for the municipal building at Keblia. It will also support the building that 'the Proportion of Gold' will construct to get ECOBat labelling. The liaison with ANME will be supported by the Project Manager and the Project Coordinator. No project funding will be used to finance the private building energy efficiency enhancements.

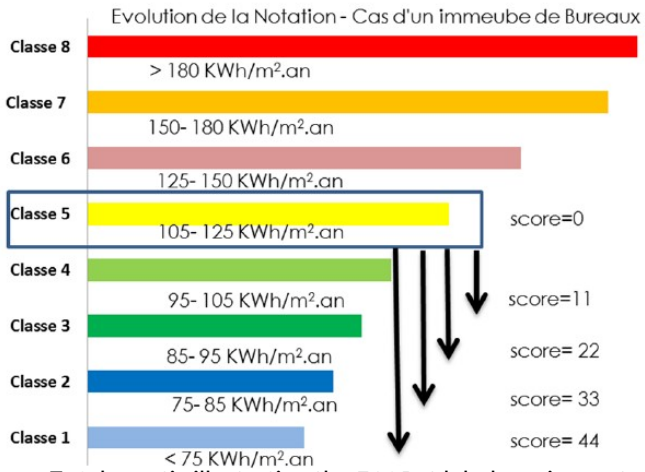


Figure 7. Schematic illustrating the ECOBat lebel scoring system.

40. **Component 2: Energy efficient public lighting:** This component will deal with the barriers that prevent the adoption and scaling up of efficient technologies for public lighting. This component will deliver reductions in the emission of greenhouse gases as calculated in Annex 14. A co-benefit will be a reduction in the high energy bill of the Municipality. The expected outcome is: "institutional strengthening for the implementation of an efficient municipal lighting system", and it will be achieved through the following outputs:
    - → Output 2.1: Capacity of municipality staff to implement efficient lighting projects strengthened;
    - → Output 2.2: National standards for public lighting technologies developed and adopted; and
    - → Output 2.3: Energy efficient municipal lighting demonstrated.

41. *Output 2.1: Capacity of municipality staff to implement efficient lighting projects strengthened* – As indicated in Table 4, the Municipality does not have the technical capacity to make the business case, including technical specifications for efficient public lighting. Also, a procedure for the safe storage and disposal in an environmentally-sound way will be developed for lamps used for public lighting at the end of their lifetime.
    - Activity 2.1.1. Training provided to municipality technicians for the adoption of efficient public lighting. Training will be provided to municipal staff for: formulating a business plan for efficient public lighting; and (ii) developing the terms of reference for shifting to efficient lighting, including technology assessments and identifying criteria for technology selection. A strategic plan will also be developed for replacing all public lighting in collaboration with ACTE project. This activity will also support the municipality to defined the technical specifications for the procurement of LED lamps in Output 2.3.
    - Activity 2.1.2. Mechanism for the safe disposal of lamps. There is currently no procedure or mechanism for the environmentally-sound disposal of failed lamps used in public lighting (both conventional technologies and LEDs). In collaboration with the Ministry of Environment (Directorate of Sustainable Development), the UNDP-GEF project will propose a procedure for the safe collection, storage and environmentally-sound disposal of lamps at the end of their lifetime. This mechanism will be applicable country-wide across all municipalities. The procedure will be guided by the ESMF (Annex 11).

42. *Output 2.2: National standards for public lighting technologies developed and adopted* - Technical assistance will be provided to Institut National de la Normalisation et de la Propriété Industrielle (INNOPRI) for developing Tunisian standards for efficient public lighting technologies.
    - Activity 2.2.1. Standards for LED lamps. There are currently no national standards for LED lamps in Tunisia. With the increasing adoption of LED lighting, it is important to support the market development of efficient lighting through quality standards. To keep with the scope of the UNDP-GEF project, INNOPRI will be supported to domesticate international standards for LEDs used in public lighting. Adoption of the standard can take place a multiple levels such as technical committee of INNOPRI, on the one hand, and by Cabinet of Ministers, on the other. Given the short duration of the project and the administrative and bureaucratic delays that are related with receiving Cabinet approval, the project will primarily seek approval of the standard by the technical committee of INNOPRI. Nevertheless, the process of seeking adoption by Cabinet will be supported by the UNDP-GEF project.

43. *Output 2.3: Energy efficient municipal lighting demonstrated* – GEF funding will be used to implement energy efficient and site-appropriate lighting technology implemented (e.g. LEDs with capacity for upgrade to intelligent system). The site-specific interventions for implementing efficient lighting using LED lamps are given in Table 5 and the locations shown in Annex 2.
    - Activity 2.3.1. Environmental and Social Management Plan (ESMP). This activity is subject to a targeted assessment that will result in a site-specific Environmental and Social Management Plan (ESMP) that includes a Waste Management Plan and an Occupational Health and Safety Plan.
    - Activity 2.3.2. Procurement of LED lamps. The technical staff of the municipality will develop the technical specifications for LED lamps with the technical assistance of the ANME (through project ACTE). The municipality will use its public procurement process to purchase the lamps, with oversight and support from the UNDP Country Office.

- Activity 2.3.3. Replacement of SHP lamps with LEDs. A total of 574 light points using high pressure sodium (SHP) lamps of between 100 W and 250 W capacities will be replaced with LEDs of 80W or 100 W capacities. It is expected that the GEF investments will reduce power usage in public lighting equal to 73.66 kW. The LED lamps will have technical specifications that make them compatible for integration in a smart public lighting management system that the municipality will eventually put in place through parallel initiatives.

44. **Component 3: Knowledge management, monitoring and evaluation, and scale-up strategy:** The third component addresses outreach, and capturing and dissemination of results for scaling up the results of the proposed project. It also seeks to ensure adequate monitoring and evaluation (M&E) of the project to support adaptive project management. Emphasis is placed on the use of enablers of institutional learning on all aspects of operationalisation of the building energy code for new buildings (RTNB) and the ECOBat energy label for buildings, and outdoor efficient lighting in the city of Kairouan to support knowledge management and approaches for scaling up sustainable cities in Tunisia. Since the Sustainable Development Strategy of Kairouan goes well beyond issues related to energy efficiency, the GEF project will also put in place a framework for sustainable city data management system to support evidence-based decision making, and for monitoring & evaluation purposes. The outcome is: "Municipal learning organisation capable to implement and manage sustainable city strategy". The following four outputs will be implemented in order to achieve this outcome.
    - → Output 3.1 Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, (iii) Terminal Evaluation carried out;
    - → Output 3.2: Data management system for benchmarking Kairouan City sustainability established and operational;
    - → Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries; and
    - → Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed.

45. *Output 3.1: Output 3.1 Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Terminal Evaluation carried out* – This output covers the monitoring and evaluation (M&E) of the project using standard activities required by both the GEF and UNDP. M&E is an essential part of adaptive project management in so far as measurement of indicators proposed in the project design (e.g. Project Results Framework) is carried out for tracking implementation performance. For this output, the following activities will be implemented:
    - Activity 3.1.1. Inception workshop. An inception workshop will be planned within 60 days of project CEO endorsement.
    - Activity 3.1.2. Monitoring and Evaluation. The results framework and GEF core indicators; project plans (e.g. Stakeholder Engagement Plan, Gender Action Plan and Risk Register), and Social and Environmental Safeguards will be monitored. Also, an independent evaluation will be carried out at the end of the project as per standard UNDP-GEF procedures. The financials of the project will also be verified by an independent accredited auditor on an annual basis.
    - Activity 3.1.3. Terminal Evaluation. An independent evaluation will be carried out by the UNDP within 6 months of project closure as per standard requirements described in Section VI.

46. *Output 3.2: Data management system for benchmarking Kairouan City sustainability established and operational* – The project will support the City of Kairouan to become a member of the World Council on City Data (WCCD), and to adopt ISO 37120 (Sustainable Development of Communities: Indicators for City Services and Quality of Life) to track progress in the implementation of the Sustainable Development Strategy. The data management system will integrate the indicators contained in the monitoring and evaluation framework for the National Programme for Sustainable Cities in Tunisia.
    - Activity 3.2.1. Establish a coordinating structure for data collection. The ACTE project has already established a cross-functional institutional arrangement for the management of issues related to energy use and energy efficiency at the municipal level. This structure is depicted in Figure 8 (except for the

function related to waste management); it combines high level political support with the Mayor acting as the Chairperson and the participation of two elected councillors (members of the municipal council), and high level technical functions through the participation of directors, deputy directors, officers-in-charge and engineers from different divisions (Annex 15 for more details). Since the scope of the city sustainable development performance is broader than the requirements of the ACTE project (that focuses primarily on energy efficiency), the existing structure can be enlarged to include other functional areas such with waste management given as an illustrative example.

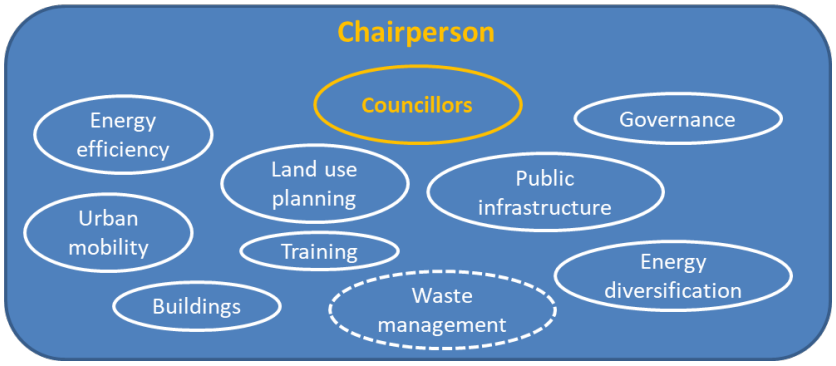


Figure 8. Configuration of cross-functional political and technical arrangement for the adoption of ISO 37120. (*Source: Based on the structure used to implement the ACTE project*)

- Activity 3.2.2. Establish data management system. Currently, there is no data Inventory Quality Management System (IQMS) for collating, analysing, reporting and archiving city data in formats that are useful to analyse and plan for city-wide sustainability performance. Such an IQMS (Figure 9) will be put in place so that the City of Kairouan is in a position to carry out sustainability reporting against the ISO 37120 using the coordinating structure established under Activity 3.2.1.

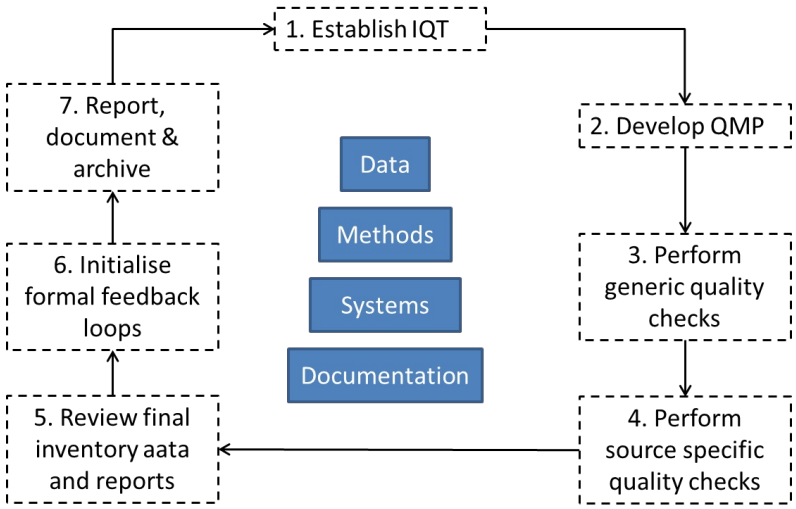


Figure 9. Inventory Quality Management System.

The IQMS will include a Quality Control Framework (QCF), which will be applied to the elements listed in Table 10.

Table 10. Elements of Quality Control Framework and their attributes.

Elements of QCF	Attributes of elements of QCF
Data	• Raw data (e.g. activity level, emission factors etc.) • Secondary data (e.g. sources and quality assurance systems used to generate data) • Quality control procedures to carry out high quality data collection • Improvements in data collection procedures
Methods	• Technical aspects of calculations and tools used for calculations (e.g. grid emission factor for national electricity system) • Selection, application, and updating of inventory methodologies as identified by ISO 37120 and new research to improve data quality
Systems	• Institutional and managerial procedures for developing data inventories
Documentation	• Records of data, methods, systems, processes, assumptions and estimates

The Quality Management System (QMS) will be managed using seven steps as summarised in Table 11.

Table 11. Description of the seven steps to implement the QMS.

Step	Description
1. Establish Inventory Quality Team (IQT)	• Responsible for implementing the QMS and to continually improve inventory quality • Coordination between relevant units in the Municipality of Kairouan, external entities providing secondary data (e.g. waste collection, emission factor of the electricity system in Kairouan), and independent verifiers • The IQT will be the coordinating structure that will be established under Activity 3.2.1
2. Develop Quality Management Plan (QMP)	• Outlining the steps taken by GAM to implement its QMS
3. Perform generic checks	• Developing appropriate quality checks for all data and calculations to generate indicators
4. Perform source-specific checks	• Developing guidelines for recalculation of indicators • Establishing criteria / circumstances for data / indicators restatements • Means of addressing uncertainty in data
5. Review final inventory data and reports	• Establishing a process of internal technical review of data inventory, followed by a process of internal managerial review for institutional approval of inventory • Establishing a process for external auditing of inventory data
6. Institutionalise formal feedback loops	• Establishing a process of feeding the results of reviews in 5 to the IQT for the continuous updating of the QMP
7. Report, document & archive	• Developing record keeping procedures, including how data is archived • Establishing which information is kept for internal use and which information is reported to external parties • Developing procedures for formal feedback

- Activity 3.2.3. Supporting process of membership to international organisations. The UNDP-GEF project will establish link with the WCCD Secretariat and it will facilitate the process of membership admission of the City of Kairouan to the WCCD. The knowledge and lessons learned from the UNDP-GEF funded project in the City of Amman, Jordan entitled "A systemic approach to sustainable urbanization and resource efficiency in Greater Amman Municipality (GAM)" will be sought as the City of Amman is already a member of the WCCD. The project will also support the City of Kairouan to seek membership with the C40 Cities and/or ICLEI. Regarding C40 Cities, experience sharing with the City of Amman and

the City of Marrakech will be sought. For the latter, collaboration will be sought with the Marrakech Sustainable Cities Child Project "Strengthening Marrakech's sustainable development through innovative planning and financing". The C40 Cities has very stringent requirements for membership. In order to pave the way for the City of Kairouan to become a member of C40 Cities, the capabilities of the City of Kairouan for mainstreaming climate change across municipal operations and services will be assessed against the requirements of C40's Climate Action Planning (CAP) Framework.[26] The result of this baseline assessment will be used to develop a plan that will enable the City of Kairouan to join the C40 Cities. The assessment will be carried out in collaboration with C40 Cities.

47. *Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries* – The results (lessons learnt and experiences) of the project will be captured and packaged into a knowledge product that will be disseminated inside Tunisia and more broadly in the MENA region. The PMU will also ensure adherence of the Municipality of Kairouan to the PANORAMA partnership (South-South Cooperation at paragraph 54 below) for enhancing knowledge sharing.
    - Activity 3.3.1. Knowledge product on lessons learned. A knowledge product will be developed to capture project best practices and pitfalls to avoid for dissemination (website, publications, manuals, participation in national, regional and international conferences and fora etc.), and to demonstrate the process for leveraging investments in EE buildings and public lighting. Since there will be no mid-term evaluation, lessons learned will be compiled on an annual basis. The contents of these lessons learned reports will then be used to develop the final knowledge product.
    - Activity 3.3.2. Adhesion to PANORAMA. The Project Manager will take the necessary steps for the Municipality of Kairouan to join the PANORAMA partnership so that it is able to share lessons learned from the UNDP-GEF project to a wider community of practitioners.

48. *Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed* – In support of the National Programme for Sustainable Cities in Tunisia, a replication plan will be developed based on lessons learnt to scale up the project results to other cities in Tunisia In designing this output, due consideration will be given at PPG stage for ensuring that all the concerns and guidance provided in the ESMF to manage project risks that can be amplified through scaling-up are taken into consideration.
    - Activity 3.4.1. Replication plan. Based on the lessons learned (Output 3.3), a Replication Plan will be developed for scaling up city-wide energy efficiency interventions in buildings (through the enforcement of the RTNB and adoption of the ECOBat label), and efficient public lighting. The replication plan will cover at least the 14 cities that are included in the National Programme for Sustainable Cities in Tunisia, and it will be accompanied by an investment plan. During preparation of the Replication Plan, a targeted and strategic assessment will be undertaken to assess potential associated environmental and social risks and take them into consideration in the final draft.
    - Activity 3.4.2. Bankable concept note for climate finance. The UNDP-GEF project and its results provide an opportunity for Tunisia to leverage international climate finance in order to promote the implementation of the National Programme for Sustainable Cities in Tunisia. Based on the results of the project (Output 3.3 and Activity 3.4.1), a concept note will be formulated using a programmatic approach to access climate finance from the GEF, the Green Climate Fund (GCF) and other funding sources for scaling up emissions reductions across Tunisian cities.

49. Global environment benefits: The calculations for GHG reductions emissions are detailed in Annex 14. The GEF methodology and tool for EE in buildings (file named 'GEF_EE_GHG_Tool_v1.0_Kairouan.xlxs') was used and is provided separately as part of Annex 14. Direct emission reductions accrue from three sources: (1) enforcement of the energy building code (Output 1.1 and 1.2); (2) investments in efficient building (Output 1.6); and (3) investments in efficient public lighting (Output 2.3). Discussions with stakeholders have revealed that apart from the results of 2010 shown in Figure 1 and Table 2 given above, there are no up-to-date baselines for the building sector based on which the global environmental benefits (direct and indirect) of the proposed project could be calculated. The CLIP model was used for estimating energy efficiency gains in buildings as shown in Figures 5, 6

[26] https://www.c40knowledgehub.org/s/article/Climate-Action-Planning-Framework?language=en_US – accessed 8 May 2023.

and 7. Regarding public lighting a baseline scenario up to 2040 was developed for the business-as-usual case with inputs from the Municipality of Kairoua. A decarbonization scenario was developed reflecting the technical and financial support from external sources, such as the UNDP-GEF project. Modeling to 2045 provided an indication of post-project market opportunity for decarbonization from adoption of LED lighting in Kairouan. The output of the model (file named 'PIMS 6686 – Public light (Annex 14)') was used as input in the GEF Tool to estimate indirect top-down emission reductions. The results of the modeling are summarised in Table 12.

Table 12. Summary of project-related direct and indirect emission reductions.

All Components	Cumulative		
	Total	*2023-2026*	*2027-2046*
Direct Electricity Savings (MWh)	45,232	4,142	41,090
Direct Natural Gas Savings (GJ)	0	0	0
Direct (District) Heat Savings (GJ)	0	0	0
N/A	0	0	0
Direct Total Energy Savings (GJ)	**162,836**	14,911	147,925
Direct GHG Emission Savings (tCO2)	**20,219**	1,851	18,367
Direct Post-project GHG Emission Savings (tCO2)	**36,796**		36,796
Indirect Bottom-up Emission Savings (tCO2)	**14,549**		14,549
Indirect Top-down Emission Savings (tCO2)	**31,064**		31,064

48. Partnerships: For the project to achieve its objectives, a number of project partners have been identified using the multi-stakeholder approach described in the previous section. Since the project operates at two geographical and political levels – i.e. national and local levels of governance - the importance of an inclusive approach to cater for the interests of key project partners cannot be overstated. The issue of partnership becomes even more prominent when considering gender-inclusivity as is discussed below. The project will be implemented in a context where there are parallel initiatives with which synergies should be formed to provide maximum benefits to the beneficiaries. The parallel and complementary initiatives are described in Annex 14 (Part B), and are reflected in the letters of co-financing that are given in Annex 15.

49. As further described in Table 13 below, these co-financed activities correspond to funds not flowing through UNDP accounts and whose results are not included in the project results framework. In this case, UNDP is accountable to monitor the risk to realization of co-financing amounts and realization amounts annually in the GEF PIR and at terminal evaluation. Specifically, potential risks associated with co-financing that may affect the project, including safeguards related risks that fall within the project context or area of influence will be considered in safeguards due diligence and the project risk register and monitored accordingly. Risk management measures identified will be only those within the control of the UNDP project (e.g. managing reputational risk). See the ESMF (Annex 11) for more details on the management of risks related to the different types of co-financed activities in this project.

Table 13. List of co-financed activities not included as project results.

Co-financing source	**Co-financing type**	**Co-financing amount**	**Included in project results?**	**If yes, list the relevant outputs**
Government (Ministry of Environment)	*Cash*	*500,000*	*Yes*	*Outputs 1.3, 1.4, 1.5, 1.6, 2.2, 2.3, 3.1, 3.2 & 3.4*

Government (Ministry of Environment)	*In-kind*	*150,000*	*No*	*n/a*
Government (Municipality of Kairouan)	*Cash*	*825,000*	*No*	*n/a*
Government (Municipality of Kairouan)	*In-kind*	*350,000*	*No*	*n/a*
Private Sector (the Proportion of Gold)	*Cash*	*133,000*	*No*	*n/a*
GEF Agency (UNDP)	*Cash*	*415,000*	*No*	*n/a*
Private Sector (the Proportion of Gold)	*Cash*	*850,000*	*No*	*n/a*

50. Risks: The project risks are detailed in Annex 8, and have been informed by the social and environmental screening contained in Annex 7 and the ESMF given in Annex 11. The SESP has rated the project as 'Moderate' risk. The main risks are summarised in Table 14 below. It is pointed out that a two-tiered Grievance Redress Mechanism is provided in the Stakeholder Engagement Plan (Annex 10) to support management of project risks. As per standard UNDP requirements, these risks will be monitored quarterly by the Project Manager. The Project Manager will report on the status of the risks to the UNDP Country Office, which will record progress in the UNDP ATLAS risk register. Management responses to critical risks will also be reported to the GEF in the annual PIR. Implementation Partner risks identified through HACT and PCAT are also covered.

51. COVID-19 risks and mitigation measures: COVID-19 poses a risk to project implementation both as a national and operational risk. The national level COVID-19 situation is given in paragraph 2 and the project's contribution to post-COVID-19 recovery articulated in paragraph 20. The several waves of infections since March 2020 and incidence of new infections, albeit relatively low, in January 2023 shows that the country is not immune from an ongoing health risk. This translates into an operational risk for the project. The COVID-19 Pandemic may slow down project implementation. In order to mitigate COVID-19 risks, the project design has incorporated a number of mitigation measures as summarized in Table 14.

Table 14. Assessment of key project risks and mitigation measures. See the complete list – in particular of social and environmental risks - in Annex 7 (SESP) and Annex 8 (Risk Register).

Risk	Rating	Mitigation
National and Operational risk COVID-19 pandemic is a national issue arising from a sanitary and health crisis that has negative socio-economic impacts. This translates into an operational risk for the project. The COVID-19 Pandemic may slow down	Moderate L = 2 I = 3	As discussed in paragraph 2, there has been a significant drop in new COVID-19 cases except for isolated spikes that are well contained. Tunisia has experienced a very strong uptake in COVID-19 vaccination and sanitary measures (especially for foreign visitors) are functioning well to prevent new cases. However, it is pointed out that Kairouan was one of the hardest hit cities at the peak of the pandemic in July 2021 because of its weaker health infrastructure.[27] Hence, there is an onus on the project to put in place measures to minimise potential infections arising from movement of project personnel (staff and consultants) and gatherings (workshops, trainings). The following project design provide mitigation actions that reduce the need for physical travel and meetings: • The biggest threat relates to the movement of international consultants. The project has been designed to make space for the use of local expertise as far as practicable by using the bundled approach for the recruitment of consultants that would favour local firms. • Data and information sharing will be shared via the website of the municipality and using its social media platforms (Annex 10). The use of virtual meetings to carry out stakeholder meetings and consultations will be used whenever practicable and desirable. For instance, the Inception Workshop for the PPG stage was carried out in mixed mode participation (virtual and physical), and this may be replicated during implementation.

[27] https://www.rfi.fr/en/tunisia-hospitals-struggle-with-covid-tsunami - accessed 26 January 2023.

project implementation. As discussed in section II, the response to COVID-19 in Tunisia has been. However, the pandemic is still evolving and it poses a tangible threat to the continued constrained mobility of persons that can slow down project implementation.		• Regarding in-person workshops and technical working group meetings, the project will adopt precautionary measures such as social distancing and mandatory sanitary protocols, including wearing of face masks and hand-sanitizing among others.
Political Risk	Moderate L= 3 I = 3	Since the Arab Spring, the country's socioeconomic situation has deteriorated. The discontent is especially high amongst the Tunisian youths as they resent the lack of economic opportunities. The situation has been aggravated by the Covid-19 pandemic. The difficult socioeconomic situation translates into a volatile political environment. The positive side is that the socioeconomic situation has been stabilised through intervention by the International Monetary Fund (IMF). While country-wide sovereign risks are beyond the control of the project, the proposed GEF project will have socio-economic benefits at the local level in the form of job creation and catalysing investments in energy efficiency value chains at the local level. In the past few years, there has also been a trend for more centralization of decision-making, implying the decreasing autonomy of regional and local governments. The situation is dynamic and will require close monitoring. Changes in the governance arrangements can be subsumed in the overall sovereign risk that is beyond the control of the project.
Organizational / Institutional Risk	Moderate L = 4 I = 3	The Ministry of Environment (Directorate of Sustainable Development) is the project Implementing Partner. The capacity assessment of the IP has revealed the IP to be 'moderate risk'. In order to mitigate this risk, the Support to NIM implementation modality will be used with project administrative support provided by the UNDP CO. Further, the Ministry of Environment has requested the UNDP CO to host the Project Management Unit. As the project was initiated by the Directorate of Sustainable Development (Ministry of Environment) and the Municipality of Kairouan it is expected that they will be willing to build the capacity for project implementation. The institutional (municipality) appropriation for implementation of the proposed project is high.
Climate Change Risks (SESP – Risk 5)	Moderate L = 2 I = 2	Measures that have been incorporated into this project include; 1) introduction of climate-responsive building techniques and elements to reduce the effect of heat and reduce demand on energy for cooling (under Output 1.3 that will support the uptake of locally-fabricated bricks, as well as characterize the thermos-physical properties of these bricks that are typically used in the maintenance and renovation of buildings of cultural heritage value, such as the Medina); 2) raising awareness on the long-term benefits of energy efficiency (Output 1.4 will develop an interactive tool to quantify the benefits of investments in energy efficiency in buildings); 3)

		application of building energy codes and support to choose building insulation to reduce energy demand (Output 1.1 and Output 1.2 with demonstration under Output 1.6). The ESIA that will be undertaken for the retrofitting of the municipality building (Output 1.6) will assess this risk and recommend measures that ensure that the construction works do not result in maladaptive practices.
Environmental Risks (SESP – Risk 4, Risk 8, Risk 9) (Construction and Electrical Wastes from investments and scaling up – Outputs 1.6, 2.3 and 3.4) (SESP – Risk 7) (Hazardous emissions – Output 1.3) (SESP – Risk 10) Damage to biodiversity	Moderate L = 3 I = 3 Moderate L = 4 I = 3 Moderate L = 2 I = 3	Risk 4 - A targeted assessment will be undertaken for the rehabilitation works of the municipal building (Output 1.6) to address this risk. The assessment will result in an Environmental and Social Management Plan (ESMP) that will include a Waste Management Plan that adopts international best practices. In Tunisia, the collection, transport, treatment and dumping of hazardous and special waste is regulated. The list of companies authorized by the Minister of the Environment for the management of hazardous waste is available on the ANGed website[28] and will be used to contract the company if hazardous waste needs to be disposed. Risk 8 - A targeted assessment will be undertaken prior to commencement of the replacement activities (Output 2.3) to analyze this risk and develop a Waste Management Plan (as part of the ESMP) that will ensure safe storage or disposal of these lamps. This may include storing them at public storage facilities to be reused elsewhere. Risk 9 - As outlined in the ESMF, a targeted assessment will be developed for the replication plan for scaling up energy efficient buildings and public lighting (Output 3.4) that will consider environmentally feasible alternatives to the disposal of obsolete lamps and construction waste, including potentially exporting them in line with the Basel Convention. Risk 7 - As outlined in the ESMF, a targeted and strategic assessment to consider potential social and environmental impacts will be undertaken for the strategic plan for development of a value chain for the local production of traditional construction materials (Output 1.3) that will address this issue to ensure proposed solutions do not lead to air pollution. Risk 10 - As outlined in the ESMF, a targeted and strategic assessment will be conducted to ensure potential downstream social and environmental impacts are considered in the development of the strategic plan for developing a value chain for the local production of traditional construction materials (Output 1.3)..
Social Risks (SESP – Risk 2) – Limiting women's opportunities	Moderate L = 3 I = 3	Risk 2 - The project Gender Action Plan contains actions to contribute towards overcoming gender-related disparities through four considerations, namely: (i) facilitating equitable participation, (ii) ensuring male engagement to prevent backlash against women, (iii) leveraging existing initiatives, and (iv) differing levels of literacy and education. The project will make use of gender-disaggregated indicators (Gender Action Plan

[28] ANGed (2022). November 2022: List of authorized hazardous waste companies. Retrieved from http://www.anged.nat.tn/Novembre_2022Liste_societes_autorisees_dechets_dangereux.html (Accessed 4 January 2023)

(SESP – Risk 3) – Discrimination against vulnerable communities (SESP – Risk 6) – Community health & safety (SESP – Risks 12 & 13) – Occupational Health & Safety	Moderate L = 3 I = 3 Moderate L = 2 I = 3 Moderate L = 2 I = 3	and Project Results Framework). The project will optimize the use of gender-responsive training materials to project stakeholders and project interventions Risk 3 - As mentioned in the Environmental and Social Management Framework (ESMF), targeted assessments will be conducted to consider potential social and environmental impacts associated with the replication plan for scaling up energy efficient buildings and public lighting in Tunisia (Output 3.4) and the Strategy and Action Plan that will be developed under Output 1.3 to enhance the value chain of locally-fabricated bricks. . The SEP has been developed to fully account for the needs and expectations of national and local stakeholders. In addition, the SEP provides for modalities for engagement of all stakeholders including vulnerable groups and 'local populations'. Risk 6 - As mentioned in the ESMF, a targeted assessment will be undertaken for the rehabilitation works at the municipal building (Output 1.6). This assessment will tackle this risk and propose mitigation measures in the resulting site-specific ESMP. The targeted assessment for the street light replacement activity (Output 2.3) will analyze this risk and propose mitigation measures to reduce it. These measures will be incorporated in the ESMP that will be developed Risk 12 – The targeted assessment that will be undertaken for the construction works at the new municipal building (Output 1.6) will assess the risks to workers such that mitigation measures will be incorporated into the resulting site-specific ESMP that will include an Occupational Health and Safety Plan. The targeted assessment for the replacement of public light bulbs (Output 2.3) will assess electrocution, falling and other occupational health and safety risks and proposed measures to mitigate them, which shall be incorporated into an Occupational Health and Safety Plan as part of the overall ESMP for this activity. In addition, Labour Management Procedures (LMP) will be developed for the project to ensure that labour standards and rights are upheld for project workers and for workers that are recruited by contractors working on various activities. Details on the LMP are provided in the ESMF. Risk 13 - As outlined in the ESMF, a targeted and strategic assessment to consider potential downstream social and environmental impacts will be conducted to inform the strategic plan for developing of value chain for the local production of traditional construction materials (Output 1.3) which will ensure that the issue of occupational health and safety is considered when assessing the options.

52. Stakeholder engagement: Table 15 lists key project partners required to achieve the desired results of the project at the output level. The description of their participation in the project is directly related to the assumptions in the ToC. Details of partnerships and stakeholder engagement can be found in Annex 10 (Stakeholder Engagement Plan). This annex also gives a mobilization and communication plan with stakeholders. In addition, a grievance redress mechanism will be set up in accordance with UNDP procedures. Stakeholder engagement was also carried out during project design and conceptualization and is reflected in Annex 12 and Annex 14. All the stakeholders will be involved in Outputs 3.1 and 3.3.

Table 15. Stakeholder involvement in the implementation of the project.

Stakeholder	Contributions	Relevant project outputs
Municipality of Kairouan	The Municipality of Kairouan will be the main beneficiary of the project and it will co-Chair the Project Steering Committee with the Ministry of Environment (Directorate of Sustainable Development). All project interventions will take place in the City of Kairouan. The Municipality is the local authority for issuing building licenses and permits for the construction of new buildings, and it has the mandate to ensure reliable public lighting. The municipality will use its Headquarters in the district of Keblia as baseline project for energy efficiency enhancements in relation to the implementation of the RTNB and the adoption of the ECOBat label. The project design and choice of locations for GEF investments were made in close collaboration with the Municipality. The Municipality as the main arm of local government is also the owner of the Sustainable Development Strategy of the City of Kairouan with which the GEF-financed project is aligned. The Municipality will host the Project Coordinator.	• Output 1.1: An operational manual for new building licensing and permits process to operationalise • Output 1.2: Enforcement capabilities of municipality strengthened for ensuring new building compliance with RTNB • Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials • Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated • Output 1.6: Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label • Output 2.1: Capacity of municipality staff to implement efficient lighting projects strengthened • Output 2.3: A smart usage system for municipal lighting demonstrated • Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.2: Data management system for benchmarking Kairouan City sustainability established and operational • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries • Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed.
Ministry of Environment (Directorate of Sustainable Development)	The Ministry of Environment is the National Climate Change Focal Point, and also serves as the GEF Operational Focal Point. The Directorate of Sustainable Development is the owner of the National Programme for Sustainable Cities in Tunisia, and it interacts with local authorities to support the formulation, implementation and monitoring and	• Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials • Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated • Output 2.2: National standards for public lighting technologies developed and adopted

Stakeholder	Contributions	Relevant project outputs
	evaluation of sustainable development strategies in cities, such as the Sustainable Development Strategy of the City of Kairouan . The Ministry, through the Directorate, will be the Implementing Partner of the project, and it will also be Chair of the Project Steering Committee. The Ministry is also provider of USD 500,000 million cash co-financing. The Directorate of Sustainable Development will host the Project Manager and the Administrative Assistant.	• Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.2: Data management system for benchmarking Kairouan City sustainability established and operational • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries • Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed
National Agency for Energy Conservation (Agence Nationale de Maitrise de l'Energie, ANME)	ANME is the technical body operating under the aegis of the Ministry of Energy, Mines and Energy Transitions to design and implement Tunisian national policy in the field of energy management. This is achieved through the promotion of sound energy usage, the development of renewable energies, and facilitating the energy transition. The ANME was instrumental in developing the RTNB and the ECOBat energy label for buildings. It also makes available tool such as CLIP. ANME is also supporting local authorities to adopt energy efficiency measures through the ACTE project (see Section IV in the Project Document). It also manages the Energy Transition Fund (that will provide grant co-financing to the project). The ANME is also responsible for the accreditation of energy auditors and it plays a key role in the verification process of planned energy use of new buildings or building extension. ANME will be squarely involved in project implementation through support it is already providing to the City of Kairouan through the ACTE project, implementation of RTNB and application of ECOBat label, and technical specifications for building thermal insulation and energy efficient lights, among others. The ANME will also be involved in the use of the CLIP model for the analysis of energy savings from	• Output 1.2: Enforcement capabilities of municipality strengthened for ensuring new building compliance with RTNB • Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials • Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated • Output 1.5: National standards for building materials are developed and adopted • Output 1.6: Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label • Output 2.2: National standards for public lighting technologies developed and adopted • Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries • Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed

Stakeholder	Contributions	Relevant project outputs
	proof-of-concept building during the PPG stage.	
Institut National de la Normalisation et de la Propriété Industrielle (INNOPRI)	The Institut National de la Normalisation et de la Propriété Industrielle (INNOPRI) is a non-administrative public institution operating under the supervision of the Ministry of Industry, Energy and Small and Medium-sized enterprises. It is in charge of standardisation, product and quality systems certification, quality promotion and protection of industrial property. INNOPRI is, therefore, the key stakeholder for establishing standards and carrying out conformity tests for building materials and public lighting equipment. It will be involved in the implementation of mainly Output 1.5 and Output 2.2.	• Output 1.5: National standards for building materials are developed and adopted • Output 2.2: National standards for public lighting technologies developed and adopted • Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries
Association of Architects	The Association of Architects plays an important role in the entire value chain for building construction starting from architectural design to construction, including the building materials supply chain. Hence, they are directly involved and impacted by the operationalization of the RTNB and the successful application of ECOBat energy label. It also has a significant voice in advocacy and policy debates regarding regulations that influence the conceptualisation and design of buildings, and the construction of these buildings (for example through the choice of building materials including those that are adopted for enhancing the thermal efficiency of building envelopes). Architects are an important link in the process of obtaining licenses and permits for building construction.	• Output 1.1: An operational manual for new building licensing and permits process to operationalise • Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials • Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated • Output 1.5: National standards for building materials are developed and adopted • Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries • Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed
Professional Organisation Tunisian Green	The TGBC is the local chapter of the Green Building Council that supports the development of green buildings, and sustainable communities and cities in Tunisia. It has a strong pedagogical	• Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and

Stakeholder	Contributions	Relevant project outputs
Building Council (TGBC)	approach of coordinating public and private stakeholders and providing technical support. Based on its mandate and interests, it will be involved in project implementation, especially regarding project activities to overcome perception and awareness barriers. It also has a significant voice in advocacy and policy debates regarding regulations that influence the conceptualisation and design of buildings, and the construction of these buildings, especially through the adoption of building standards and labels, which have building energy use as one component, among others.	disseminated • Output 1.5: National standards for building materials are developed and adopted • Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries • Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed
Development partners (EU, GIZ, USAID, Swiss Cooperation)	Bilateral development partners are supporting different initiatives in the baseline in Section IV of the Project Document. These partners will be involved in the project through coordination with those baseline initiatives.	• Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries • Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed
Local citizens	The citizens of the city are beneficiaries of services provided by the Municipality. For instance, they will use the enhanced municipal building in the District of Keblian, and they will also benefit from the energy efficient public lighting that the project will demonstrate across the city. As shown by the SESP screening, the local communities and citizens will also be impacted during the construction of the new building and replacement of baseline lighting technologies from various forms of pollution and waste that will be generated within the project boundary. The concerns and expectations of the citizens and local communities have been taken into account at PPG stage. Their needs, interests and perceptions about the technology value chain, as well as the energy efficiency technologies that will be used are crucial for project success. In	• Output 1.1: An operational manual for new building licensing and permits process to operationalise • Output 1.2: Enforcement capabilities of municipality strengthened for ensuring new building compliance with RTNB • Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials • Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated • Output 1.6: Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label • Output 2.3: A smart usage system for municipal lighting demonstrated • Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting

Stakeholder	Contributions	Relevant project outputs
	particular, the needs, interests and perceptions of women were captured and integrated in the project design through the Gender Analysis and Gender Action Plan (Annex 12). Most project outputs that deliver results or involve the local communities are central to the needs, interests and perceptions of local communities, women and vulnerable groups.	Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries
Civil Society Organisations / Local Associations	Associations play a central role in local community development initiatives. In particular, there are associations focused on a number of areas of relevance to the project including environmental, energy, cultural and women. They will want to play an active role in tracking and providing input into the implementation of the first two components of the project. As mentioned above, the Local Associations / Civil Society Organisations will place a key role in communicating with local citizens on relevant aspects of the project. The Associations/CSO that will participate in project implementation are: Association des Bassins des Aghlabides des Sports pour Tous (ABAST); Association des jeunes et sciences ; Association de Protection de la Nature et de l'Environnement à Kairouan (APNEK) ; Association de Sauvegarde de la Médina de Kairouan et Conservateur de musée de Rakkada.	• Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials • Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated • Output 1.6: Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label • Output 2.3: A smart usage system for municipal lighting demonstrated • Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries
Scientific and technical institutions	The project will work with scientific and technical institutions such as CTMCCV on technical training and outreach activities, among others. Their expertise will be useful in providing input into the first two components. They will also be interested in providing input and gathering information through the third component related to knowledge management and the collection of lessons learned.	• Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials • Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated • Output 1.6: Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label • Output 2.2: National standards for public lighting technologies developed and adopted

Stakeholder	Contributions	Relevant project outputs
		• Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries • Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed
Private sector	Private sector companies were engaged and will participate in the project in two ways: (1) members of the TGBC and the Association of Architects are private sector operators in the construction industry. In this sense, the companies comprising these two organisations will contribute to the project as detailed above. Also, the technical committees that are used by INNORPI to develop national standards also comprise of private sector operators. The persons who are involved in the fabrication of bricks in Kairouan are also private operators who were consulted and will benefit from the project interventions; and (2) a private sector real estate developer, namely 'the Proportion of Gold' that is developing a commercial building in the City of Kairouan will participate in the project as co-financer. Through technical assistance, the UNDP-GEF project will support the commercial building to get ECOBat labelling. However, any private sector involvement in the project pilots will be through a transparent tendering process. Besides, no GEF finance will be invested in the private development.	• Output 1.1: An operational manual for new building licensing and permits process to operationalise • Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials • Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated • Output 1.5: National standards for building materials are developed and adopted • Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label • Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Final Evaluation carried out • Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries • Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed

53. South-South cooperation: South-South cooperation will take place in a bidirectional way – i.e. lessons learned from the UNDP-GEF project shared and disseminated in with other countries through Output 3.3, and by drawing lessons learned from similar projects that have or are being implemented in similar cities/countries. In order to facilitate dissemination, the Project Manager will take necessary steps for the Municipality of Kairouan to join the PANORAMA partnership[29]. The project will furthermore provide opportunities for regional

[29] https://panorama.solutions/en

cooperation with countries that are implementing initiatives on sustainable cities and sustainable urbanization in geopolitical, social and environmental contexts relevant to the proposed project in Iraq and Jordan as detailed in the next paragraph.

54. The GEF is also financing a number of city-wide approaches to energy efficiency such as the Sustainable Cities Impact Programme in 23 cities in nine countries around the world. In closer alignment with the regional scope and also scope of project interventions, the proposed project will collaborate with the UNDP-GEF funded project in the City of Amman, Jordan entitled "A systemic approach to sustainable urbanization and resource efficiency in Greater Amman Municipality (GAM)", and the UNDP-GEF project in Iraq entitled "Promoting carbon reduction through energy efficiency (EE) techniques in Baghdad City". Collaboration will also be carried out with the Marrakech Sustainable Cities Child project entitled "Child Project Title: Strengthening Marrakech's sustainable development through innovative planning and financing". In order to advance the case of net-zero buildings, the project will also collaborate with the UNEP-GEF project entitled "Zero-Carbon Buildings for All: from energy efficiency to decarbonization".

55. Gender equality and Women's Empowerment: During the preparatory phase, the project team conducted a gender analysis including gender-disaggregated socioeconomic data collected during a field mission in Kairouan and Tunis. A gender action plan (GAP) was developed by integrating measures, indicators, targets and budget in the project activities. The gender analysis and GAP are found in Annex 12 and highlights the gender dimension in relation to the status in the family and in society, capacity and participation in decision making. The main findings of the gender analysis are:
    - Women's high rate of unemployment and limited participation in economic life, making them more vulnerable to poverty, especially in rural areas;
    - Persistent social norms and cultural beliefs exacerbate violence against women and infringe on their right to free and dignified lives;
    - Limited participation of women in politics, leadership and public life;
    - Inadequate implementation of laws and measures to protect women from all forms of violence;
    - Discriminatory laws and policies that infringe on women's rights, access to justice and economic prosperity; and
    - Limited engagement of men and boys as advocates for gender equality, resulting in increasing unpaid care work for women and girls.

56. Cognizant of these findings, and the main conclusions regarding gender issues listed in Table 16, a gender action plan (GAP) has been developed to integrate gender equality and equity in the project design (Table 17). The proposed measures aim to ensure that gender disparities are minimized and to enhance women's empowerment. If these proposed measures are not taken, then there is a risk of the project increasing gender disparities. The implementation and monitoring and evaluation of the GAP will be under the responsibility of the Project Manager. Also, the project will advocate the mainstreaming of gender equality among its staff so that they are conversant with gender-related issues in the project design and attentive to issues gender mainstreaming. Capacity building activities for gender mainstreaming will take into account women's multiple responsibilities and time constraints. The project also will work with UNDP gender experts in order to integrate their knowledge in the development and implementation of GEF-funded projects. Table 16 and Table 17 show that the project includes gender-responsive measures to address gender gaps or promote gender equality and women's empowerment, in order to improve women's participation and decision making; and for generating socio-economic benefits or services for women. The project's results framework includes sex-responsive indicators.

Table 16. Summary of the main conclusions regarding gender issues, as well as suggested interventions.

Activities/ Tasks	Present status	Suggested project interventions

Reducing Women's Vulnerability to Gender-Based Violence	The municipality has limited capacity to provide public lighting as well as to ensure that is available in areas that are vulnerable to women, particularly those who have to work after dark or who exercise in the evening.	In the short term, the project's focus on energy efficient public lighting will seek to select an area that involves high female traffic such as the location where women exercise or where female factory workers travel. It also has the potential to serve as a model for the expansion of projects or funding for efforts to improve the overall presence of public lighting in a bid to create a more safe and secure space for women and other vulnerable people such as the disabled and the elderly while contributing to the reduction of GBV.
Facilitating Equal Participation of Men and Women as well as Vulnerable Populations	In Kairouan, women have a much heavier work load than men, particularly at the household level. They are also often excluded from consultations and decision-making processes.	The project will seek to ensure women's equal participation as well as as to ensure the engagement with relevant vulnerable populations such as rural women. Participatory approaches will be considered such as mapping exercises to identify vulnerable populations and to ways to accommodate barriers such as the high rates of illiteracy with some members of the female population in Kairouan. When consultations are organized, efforts will be made to identify times and locations that take into account the heavy household burden of women as well as any transport barriers that could prevent their participation. If appropriate separate consultations with women and men will be considered to accommodate women's reluctance to speak at mixed gender meetings. Attendance sheets and surveys following meetings can provide quantitative and qualitative ways to assess the number and level of satisfaction with the participation of women and other vulnerable groups. Measures will be undertaken to give space for women and other vulnerable populations to actively participate in the implementation of the project such as capacity building and awareness raising which will include efforts to enable women to play a leadership role where possible such as on project committees.
Leveraging Women's Concern for Environmental Issues	While men also have concerns about environmental issues including climate change, because of women's role at the household level and the direct impacts they face as a result of climate change, they often have greater concerns and engage to a greater extent on environmental issues.	The project will work to leverage women's interest and engagement on environmental issues as a means of raising awareness about the importance and merits of adopting energy efficient technologies and building measures. Efforts will be made to link up with local NGOs working on environmental issues that have previous experience integrating gender considerations to seek their assistance to facilitate the consultation and engagement of

		women and other vulnerable populations about their needs and perspectives.
Facilitating Employment Opportunities for Women	In the project area, women have not always been able to participate when employment opportunities are created.	The project will encourage women to attend capacity building programs as well as to be selected for employment connected to the project at various levels including be setting specific employment targets. Where possible, efforts will be made to facilitate leadership opportunities as well as to facilitate equal pay for equal work. Other barriers to employment such as socio-cultural beliefs about women's role in the workplace will also be considered including identifying possible female role models such as the female engineers and architect who have been recruited to work at the municipality.

Table 17. Gender Action Plan.

Objective	Action	Indicator and Targets	Responsible Institution	Timeline	Allocated Budget (US$)
Component 1: Institutional strengthening for regulating the market for new efficient buildings					
Output 1.2: Enforcement capabilities of municipality strengthened for ensuring new building compliance with RTNB	Ensuring that women municipality members have the skills, knowledge, and self-confidence to participate in the energy sector's consultations and decision-making processes. The actions are: 1) Develop tailored gender and age-inclusive targeting criteria 2) Implementing and designing gender-sensitive training methodologies and content that consider the different learning styles and preferences of women and men, and the different needs and experiences of women and men in both technical and non-technical training 3) Facilitating a safe enabling environment during capacity-building activities ensures that all individuals, regardless of	Baseline: (a) 0; (b) 0% Indicator: (a) Number of sex-differentiated beneficiaries; (b) % of women report more confidence in engaging in the energy sector Target: (a) at least 40% of members of the municipality receiving capacity building are women; (b) at least 75% level of satisfaction among women who benefited from capacity enhancement activities	Municipality of Kairouan and ANME	Year 1, Q3 – Year 2, Q3	5,000

	gender and age, can fully participate and benefit from the learning.				
Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials	Ensuring that diverse women's, and youth's needs and voices are represented and accounted for in developed local strategic plans. The actions are: 1). Ensure the inclusion of women entrepreneurs and businesses into the value chain 2) Ensure active participation of diverse gender and age groups in consultation and planning meetings. 3) Strengthened representation of diverse gender needs in local plans and solutions	Baseline: 0 Indicator: (a) % of people involved in the development of the strategy are women (b) # of gender and women empowerment-related issues in the strategic plans Target: (a) At least 30% of the people involved in the development of the replication plan are women; (b) at least 5 issues covered (e.g. job creation; safety at work; increased knowledge; participation in the EE value chains; security)	Municipality of Kairouan and Ministry of Environment	Year 1, Q4 to Year 2, Q2	5,000
Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated.	Sensitisation to expand awareness of women's associations and other relevant stakeholders to participate in the consultations related to the development of the	Baseline: (a) 0; (b) no data Indicator: (a) Number of women consulted; (b) sex, age and physical ability data are	ANME and Municipality of Kairouan	Year 1, Q3 – Year 2, Q3	2,500

	online tool. Ensure that the comparative socio-economic and environmental analysis 0n-line tool is accessible and a friendly user tool for diversified gender groups. The Action: 1) ensure that the online tool has a flexible outreach process to engage different age and gender groups 2) partnership/collaboration with local women, and youth networks, CSOs, bank tanks to maximize different gender group participation. 3) integrate gender and age-disaggregated indicators in the socio-economic and environmental analysis	available <u>Target:</u> (a) At least 50% of people consulted for the development of the tool are women; (b) subsidiary to (a)			
Component 2: Energy efficient public lighting					
Output 2.1: Capacity of municipality staff to implement efficient lighting projects strengthened	Ensuring that women municipality members have the technical skills, knowledge, and self-confidence to implement efficient lighting projects strengthened. The actions are: 1) Implementing and	<u>Baseline</u>: 0 <u>Indicator:</u> Number of sex-differentiated beneficiaries. <u>Target:</u> at least 40% women receiving capacity building are	Municipality of Kairouan	Year 1, Q3 – Year 2, Q3	2,000

	designing gender-sensitive training methodologies and content that consider the different learning styles and preferences of women and men, and the different needs and experiences of women and men in both technical and non-technical training 2) Facilitating a safe enabling environment during capacity-building activities ensures that all individuals, regardless of gender and age, can fully participate and benefit from the learning experience—for instance provision of child-care facilities.	women.			
Component 3: Knowledge management, monitoring and evaluation, and scale-up strategy					
Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Terminal Evaluation carried out	M & E Reports prepared to discuss key findings and recommendations integrates gender and social inclusion considerations Development and integration of gender-sensitive indicators and collection of sex-disaggregated data for monitoring and evaluating project	Baseline: 0 Indicator: M & E reports include the integration of quantitative and qualitative information on sex-disaggregated participation and beneficiaries	Project Manager and Project Coordinator	Year 1-Year 3	10,000

	results. Monitor and evaluate gender equality and women's participation and their empowerment through project interventions	<u>Target:</u> At least 60% of indicators are sex disaggregated 100% of M & E reports integrate quantitative and qualitative information on sex-disaggregated participation and beneficiaries Sex disaggregated and gender responsive results and targets included in Results Framework and other monitoring and evaluation formats at various levels.			
Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries	Improve understanding of gender issues related to the energy-efficient technology, including access to public lighting, capacity building needs of women and policy issues at the municipal level related to energy efficient technologies, among	<u>Baseline</u>: 0 <u>Indicator:</u> % of project participants who are women. <u>Target:</u> At least 50% of the people	Municipality of Kairouan and Ministry of Environment	Year 2 &Year 3	5,000

	others. This will inform the output on lessons learned and disseminated.	involved in developing and disseminating experiences are women. At least 50% of recipients of information generated including lessons learned, are women			
Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed	Project results incorporated into the replication plan integrate gender considerations	Baseline: 0 Indicator: % of people involved in the development of the replication plan are women Target: At least 40% of the people involved in the development of the replication plan are women.	Municipality of Kairouan, ANME and Ministry of Environment	Year 3, Q3 & Q4	5,000

57. Innovativeness, sustainability and potential for scaling up: Sustainability of the project is based on the project strategy and timing, replication of the project results is also at the heart of the project strategy and design so that the project's replication strategy ensures that the project approach is distilled and actively disseminated to inform similar initiatives in Tunisia and elsewhere in the region. The mitigation options selected will enhance sustainable development through minimizing dependence on imported energy, minimizing energy costs to the economy, creating new employment opportunities and improving the local environment. As noted in the key indicators and results section, the sustainable development and socioeconomic development impacts of the project will be substantial and multi-faceted. The project will specifically help the Municipality of Kairouan mitigate the additional financial obligations and income losses resulting from energy use in municipal buildings and public lighting. It is also important to note the high levels of unemployment in Kairouan. Distribution of wealth is unequal, and its young demographic means that there is a high dependency ratio of non-active individuals to workers. The project will include a robust set of key indicators to monitor the socioeconomic impacts of project interventions.

58. The innovativeness of the proposed project stems from migrating from a conventional, project-based approach to a city-wide transformational approach that will also include the testing and implementation of novel approaches based on local institution capacity strengthening to scaling-up the diffusion of energy efficiency technologies (buildings and public lighting) in Tunisia. The UNDP-GEF project will ensure that a mechanism will be put in place for the regular update of the RTNB and accompanying regulations – i.e. racheting process - so that building energy codes will be updated to take into account best available technologies. The revision of the building energy codes will also extend the scope of application of regulations so that a larger cohort of building types (beyond the current restriction to commercial buildings and apartment blocks) that will energy a wider uptake of energy efficiency in buildings. Further, as per Output 3.4, a country-wide replication plan, accompanied by an investment plan and bankable Concept Note for mobilizing international climate finance, will be developed to support implementation of the National Programme for Sustainable Cities in Tunisia. The eventual outcome is expected to be larger global environmental benefits accruing from the application of more stringent building energy codes to a wider range of building types, as well as shifting public lighting using LEDs.

V. PROJECT RESULTS FRAMEWORK

Table 18. Project Results Framework.

This project will contribute to the following Sustainable Development Goal (s): ***SDG13 – Climate Action; SDG7 – Affordable and Clean Energy; SDG 11 – Sustainable Cities and Communities; SDG5 – Gender Equality; SDG17 – Partnerships to achieve the Goal***				
This project will contribute to the following country outcome (UNDAF): ***By 2025, all the actors involved ensure equitable, transparent and sustainable management of natural resources, ecosystems and territories, improving their resilience/adaptation as well as that of populations, especially the most vulnerable, in the face of crises and climate risks.***				
	Objective and Outcome Indicators (no more than a total of 20 indicators)	**Baseline[30]**	**Mid-term Target[31]**	**End of Project Target** *Expected level when terminal evaluation undertaken*
Project Objective: ***To promote green buildings and efficient lighting in alignment with the objectives of the enhanced Nationally Determined Contribution, the Sustainable Development Strategy of Kairouan and the National Programme for Sustainable Cities in Tunisia.***	**Mandatory** Indicator 1: **# direct project beneficiaries disaggregated by gender (individual people)**[32]	0	*n/a*	*187,000 (94,248 female; 92,752 male)*
	Mandatory GEF Core Indicators: *Indicator 2:* ***Greenhouse gas emissions mitigated (cumulative metric tons of carbon dioxide equivalent (tCO2e))***	*0*	*n/a*	*Direct: 20,219 tCO2e* *Indirect: 31,064 tCO2e*
Project component 1 (*no indicators required)*	***Institutional strengthening for regulating the market for new efficient buildings***			
Project Outcome[33] 1	*Indicator 3:* ***(a) Number of updated RTNB and***	*(a) 0*	*n/a*	*(a) 2*

[30] *Baseline, mid-term and end of project target levels must be expressed in the same neutral unit of analysis as the corresponding indicator. Baseline is the current/original status or condition and needs to be quantified. The baseline can be zero when appropriate given the project has not started. The baseline must be established before the project document is submitted to the GEF for final approval. The baseline values will be used to measure the success of the project through implementation monitoring and evaluation.*

[31] *Target is the change in the baseline value that will be achieved by the mid-term review and then again by the terminal evaluation.*

[32] *Provide total number of all direct project beneficiaries expected to benefit from all project activities until project closure. Separate the total number by female and male. This indicator captures the number of individual people who receive targeted support from a given GEF project and/or who use the specific resources that the project maintains or enhances. Support is defined as direct assistance from the project. Direct beneficiaries are all individuals receiving targeted support from a given project. Targeted support is the intentional and direct assistance of a project to individuals or groups of individuals who are aware that they are receiving that support and/or who use the specific resources.*

[33] *Outcomes are medium term results that the project makes a contribution towards, and that are designed to help achieve the longer-term objective. Achievement of outcomes will be influenced both by project outputs and additional factors that may be outside the direct control of the project.*

Institutional strengthening with enabling conditions, methodologies and tools for enforcing regulatory framework for new buildings	***associated regulations**; (b) **Number of municipal staff with competencies to apply operational guidelines for enforcement of updated RTNB (40% women / 60 % men)***	*(b) 0*		*(b) 20 (12 men + 8 women)*
	*Indicator 4: **# strategic plan for revitalizing the value chain for locally-fabricated bricks***	*0*	*n/a*	*1*
Outputs to achieve Outcome 1	→ *Output 1.1: An operational manual for new building licensing and permits process in the Municipality of Kairouan to operationalise RTNB (Réglementation Thermique des Nouveaux Bâtiments) is designed and adopted;* → *Output 1.2: Enforcement capabilities of municipality strengthened for ensuring new building compliance with RTNB;* → *Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials;* → *Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated;* → *Output 1.5: National standards for building materials are developed and adopted; and* → *Output 1.6: Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label*			
Project component 2 (*no indicators required)*	***Energy efficient public lighting***			
Outcome 2 ***Institutional strengthening for the implementation of an efficient municipal lighting system***	*Indicator 5: **Number of persons with enhanced capacity to implement efficient municipal lighting projects (50% women + 50% men)***	*0*	*n/a*	*10 (5 women + 5 men)*
	*Indicator 6: **Number of standards adopted for LED used in public lighting***	*0*	*n/a*	*1*
Outputs to achieve Outcome 2	→ *Output 2.1: Capacity of municipality staff to implement efficient lighting projects strengthened;* → *Output 2.2: National standards for public lighting technologies developed and adopted; and* → *Output 2.3: Energy efficient municipal lighting demonstrated.*			
Project component 3 *(no indicators required)*	***Knowledge management, monitoring and evaluation, and scale-up strategy***			
Outcome 3 ***Municipal learning organisation capable to implement and manage sustainable city strategy***	*Indicator 7: **Number of data management system in place for collecting city-wide data for tracking city sustainability performance according to ISO37120** (number of data management systems)*	*0*	*n/a*	*1*
	*Indicator 8: (a) **Number of replication plan, including detailed budget, for scaling up the deployment of energy efficient buildings and public lighting based on lessons learned** (number of plans); **(b) # of bankable Concept Notes developed to leverage international climate*	*0*	*n/a*	*1 (for both (a) and (b))*

	finance through the GCF			
Outputs to achieve Outcome 3	→ *Output 3.1 Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, (iii) Terminal Evaluation carried out;* → *Output 3.2: Data management system for benchmarking Kairouan City sustainability established and operational;* → *Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries; and* → *Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia developed.*			

VI. MONITORING AND EVALUATION (M&E) PLAN

60. Project-level monitoring and evaluation will be undertaken in compliance with UNDP requirements as outlined in the UNDP POPP (including guidance on GEF project revisions) and UNDP Evaluation Policy. **The UNDP Country Office is responsible for ensuring full compliance with all UNDP project M&E requirements including project monitoring, UNDP quality assurance requirements, quarterly risk management, and evaluation requirements.**

61. Additional mandatory GEF-specific M&E requirements will be undertaken in accordance with the GEF Monitoring Policy and the GEF Evaluation Policy and other relevant GEF policies[34]. The M&E plan and budget included below will guide the GEF-specific M&E activities to be undertaken by this project.

62. In addition to these mandatory UNDP and GEF M&E requirements, other M&E activities deemed necessary to support project-level adaptive management will be agreed – including during the Project Inception Workshop - and will be detailed in the Inception Report.

Minimum project monitoring and reporting requirements as required by the GEF:

63. Inception Workshop and Report: A project inception workshop will be held within 2 months from the First disbursement date, with the aim to:
    a. Familiarize key stakeholders with the detailed project strategy and discuss any changes that may have taken place in the overall context since the project idea was initially conceptualized that may influence its strategy and implementation.
    b. Discuss the roles and responsibilities of the project team, including reporting lines, stakeholder engagement strategies and conflict resolution mechanisms.
    c. Review the results framework and monitoring plan.
    d. Discuss reporting, monitoring and evaluation roles and responsibilities and finalize the M&E budget; identify national/regional institutes to be involved in project-level M&E; discuss the role of the GEF OFP and other stakeholders in project-level M&E.
    e. Update and review responsibilities for monitoring project strategies, including the risk log; SESP report, Social and Environmental Management Framework (where relevant) and other safeguard requirements; project grievance mechanisms; gender strategy; knowledge management strategy, and other relevant management strategies.
    f. Review financial reporting procedures and budget monitoring and other mandatory requirements and agree on the arrangements for the annual audit.
    g. Plan and schedule Project Board meetings and finalize the first-year annual work plan. Finalize the TOR of the Project Board.
    h. Formally launch the Project.

64. GEF Project Implementation Report (PIR): The annual GEF PIR covering the reporting period July (previous year) to June (current year) will be completed for each year of project implementation. UNDP will undertake quality assurance of the PIR before submission to the GEF. The PIR submitted to the GEF will be shared with the Project Board. UNDP will conduct a quality review of the PIR, and this quality review and feedback will be used to inform the preparation of the subsequent annual PIR.

65. GEF Core Indicators: The GEF Core indicators included as Annex 16 will be used to monitor global environmental benefits and will be updated for reporting to the GEF prior to MTR and TE. Note that the project team is responsible for updating the indicator status. The updated monitoring data should be shared with MTR/TE consultants prior to required evaluation missions, so these can be used for subsequent groundtruthing.

[34] See https://www.thegef.org/gef/policies_guidelines

The methodologies to be used in data collection have been defined by the GEF and are available on the GEF website.

66. Terminal Evaluation (TE): An independent terminal evaluation (TE) will take place upon completion of all major project outputs and activities. The terms of reference, the evaluation process and the final TE report will follow the standard templates and guidance for GEF-financed projects available on the UNDP Evaluation Resource Center. TE should be completed 3 months before the estimated operational closure date, set from the signature of the ProDoc and according to the duration of the project. Provisions should be taken to complete the TE in due time to avoid delay in project closure. Therefore, TE must start no later than 6 months to the expected date of completion of the TE (or 9 months prior to the estimated operational closure date).

67. The evaluation will be 'independent, impartial and rigorous'. The evaluators that UNDP will hire to undertake the assignment will be independent from organizations that were involved in designing, executing or advising on the project to be evaluated. Equally, the evaluators should not be in a position where there may be the possibility of future contracts regarding the project being evaluated.

68. The GEF Operational Focal Point and other stakeholders will be actively involved and consulted during the terminal evaluation process. Additional quality assurance support is available from the BPPS/NCE-VF Directorate.

69. The final TE report and TE TOR will be publicly available in English and posted on the UNDP ERC by 31 December 2026. A management response to the TE recommendations will be posted to the ERC within six weeks of the TE report's completion.

70. Final Report: The project's terminal GEF PIR along with the terminal evaluation (TE) report and corresponding management response will serve as the final project report package. The final project report package shall be discussed with the Project Board during an end-of-project review meeting to discuss lesson learned and opportunities for scaling up.

71. Agreement on intellectual property rights and use of logo on the project's deliverables and disclosure of information**:** To accord proper acknowledgement to the GEF for providing grant funding, the GEF logo will appear together with the UNDP logo on all promotional materials, other written materials like publications developed by the project, and project hardware. Any citation on publications regarding projects funded by the GEF will also accord proper acknowledgment to the GEF. Information will be disclosed in accordance with relevant policies notably the UNDP Disclosure Policy[35] and the GEF policy on public involvement[36].

72. **Monitoring Plan:** The project results, corresponding indicators end-of-project targets in the project results framework will be monitored by the Project Management Unit annually, and will be reported in the GEF PIR every year, and will be evaluated periodically during project implementation. If baseline data for some of the results indicators is not yet available, it will be collected during the first year of project implementation. Project risks, as outlined in the risk register, will be monitored quarterly.

[35] See http://www.undp.org/content/undp/en/home/operations/transparency/information_disclosurepolicy/

[36] See https://www.thegef.org/gef/policies_guidelines

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods[37]	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
Project objective: *To promote green buildings and efficient lighting in alignment with the objectives of the enhanced Nationally Determined Contribution, the Sustainable Development Strategy of Kairouan and the National Programme for Sustainable Cities in Tunisia.*	*Indicator 1:* *# direct project beneficiaries disaggregated by gender (individual people)*	*MT: n/a* *EoP: 187,000 (94,248 female; 92,752 male)*	*The direct beneficiaries will be the entire population of the city of Kairouan who will use the municipal building at Keblia and efficient street lights that will be installed across the city as per map shown in Annex 2.*	*Official demographic statistics of the City of Kairouan and of the National Institute of Statistics (INS)*	Annually Reported in DO tab of the GEF PIR	*National Institute of Statistics* *Project consultant carrying out annual lessons learned reports*	*National statistics report*	*<u>Risks</u>:* *Political and institutional instability prevents the minimum governance structures to be put in place for project implementation; low level coordination between project partners; changes in the deployment plan of LED lights in the city that results in geographical concentration compared to the geographical spread shown in Annex 2; delays in completion of municipal building in Keblia.* *<u>Assumptions</u>:* *The investments proposed under Output 1.6 and 2.3 are successful; continued*

[37] Data collection methods should outline specific tools used to collect data and additional information as necessary to support monitoring. The PIR cannot be used as a source of verification.

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
								political and institutional commitment from all project stakeholders; mechanism for enforcing the application of RTNB is adequate
	Indicator 2: *Greenhouse gas emissions mitigated (cumulative metric tons of carbon dioxide equivalent (tCO2e))*	*MT: n/a* *EoP:* *Direct: 20,219 tCO2e* *Indirect: 31,064 tCO2e*	*The expected direct and indirect emission reductions from the generation of renewable electricity as calculated in Annex 13 (Part A) and the GEF GHG measurement methodology for energy efficiency measures*	*Efficient lighting: The Municipality will launch a bid for procurement of LED lights with general specifications given in Table 5 in the Project Document. The Municipality will record the type and location of lamps replaced, and it will use its calibrated metering system that is disaggregated by lighting sectors for measuring changes in electricity consumption before and after lamp replacements.* *Municipal building: The performance*	Annually Reported in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	*Calculations carried out by international consultant carrying out Terminal Evaluation*	*Risks: Same as for Indicator 1* *Assumptions: Same as for Indicator 1*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
				of the enhanced building will first be based on the CLIP software model, and compared with a building without energy efficiency measures. Building electricity consumption will be monitored during the project lifetime, and a comparison made with the consumption of a old typical municipal building				
Project Outcome 1: *Institutional strengthening with enabling conditions, methodologies and tools for enforcing regulatory framework for new buildings*	*Indicator 3: Number of municipal staff with competencies to apply operational guidelines for enforcement of RTNB (40% women / 60 % men)*	*MT: n/a* *EoP: 20 (12 men + 8 women)*	*A mechanism will be established to support the enforcement of RTNB with a combination of capacity building (Output 1.2) and development of an operational manual for the licensing and permits*	*Data will be collected through regular surveys that will form part of the annual lessons learned studies.*	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	Project reports and reports to/minutes of the PSC	*<u>Risks</u>: Competing demands on the time of municipality staff because of several parallel initiatives leads to low level participation in capacity building activities* *<u>Assumptions</u>: Low turnover of*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
			process (Output 1.1)					*municipality staff ensures institutional and human capacity strengthening*
	Indicator 4: *# strategic plan for revitalizing the value chain for locally-fabricated bricks*	*MT: n/a* *EoP: 1*	*Output 1.3 will develop a strategic and action plan for reinvigorating the value chain of locally-fabricated bricks, including a focus on livelihoods and gender while taking into account risks (Annex 6)*	*Publication and dissemination of the strategic and action plan, and captured in the process of producing lessons learned report.*	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	Project reports and reports to/minutes of the PSC	<u>Risks</u>: *Same as for Indicator 1.* <u>Assumptions</u>: *Same as from Indicator 1.*
Project Outcome 2: *Institutional strengthening for the implementation of an efficient municipal lighting system*	*Indicator 5:* *Number of persons with enhanced capacity to implement efficient municipal lighting projects (50% women + 50% men)*	*MT: n/a* *EoP: 10 (5 women + 5 men)*	*Municipal staff will be capacitated to design, cost and implement efficient lighting projects.*	Project reports on attendance and participation of municipal staff in training that will include a competency test at the end of the training.	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	Project reports and reports to/minutes of the PSC	<u>Risks</u>: As for Indicator 3. <u>Assumptions</u>: As for Indicator 3.
	Indicator 6: *Number of standards adopted*	*MT: n/a* *EoP: 1*	*A national standard will be developed for LEDs used for public*	*Adoption and publication of the national standards, and captured in the process of producing*	Annually Reported in project	*Project Manager with support from project consulta*	Project reports and reports to/minutes of the	<u>Risks</u>: Standard is developed through a parallel initiative by the time the

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
	for LED used in public lighting		*lighting (Output 2.2)*	*lessons learned report.*	reports and also in DO tab of the GEF PIR	*nt carrying out annual lessons learned reports*	PSC	UNDP-GEF project is implemented in Q4 (2023) or Q1 (2024). Assumptions: Technical group for developing or adopting international standard is established;
Project Outcome 3: *Municipal learning organisation capable to implement and manage sustainable city strategy*	*Indicator 7:* *Number of data management system in place for collecting city-wide data for tracking city sustainability performance according to ISO37120 (number of data management systems)*	*MT: 0* *EoP: 1*	None of the cities in Tunisia have data management system to allow then to track and report on multi-dimensional sustainable city performance. The City of Kairouan will be supported to put in place a data management system for the adoption of ISO37120.	Membership of the Municipality of Kairouan to the WCCD, and reporting on indicators done through operational data management system established under Output 3.2 as captured by the process of producing lessons learned reports.	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	Project reports and reports to/minutes of the PSC	Risks: Lack of internal coordination and participation of persons from different departments and divisions within the municipality; changes in leadership (e.g. Mayor) that dampens leadership for the Municipality of Kairouan to become member of the WCCD Assumptions: Low turnover of municipal staff; fruitful South-South cooperation, especially with the City of Amman that is already a member of

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
								the WCCD
	Indicator 8: *(a) Number of replication plan, including detailed budget, for scaling up the deployment of energy efficient buildings and public lighting based on lessons learned (number of plans)* *(b) # of bankable concept notes developed to leverage international climate finance through the GCF*	*MT: n/a* *EoP: 1 (for both indicators)*	*Based on lessons learned (Output 3.3) and knowledge shared through South-South Cooperation, a replication plan will be developed for scaling up the energy efficiency actions across other cities in Tunisia; a bankable Concept Note for the GCF will be developed*	*Validation, publication and dissemination of the Replication Plan; submission of the Concept Note to the GCF as captured in the process of lessons learned reporting*	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	Project reports and reports to/minutes of the PSC	*<u>Risks</u>: Same as for Indicator 1.* *<u>Assumptions</u>: Same as from Indicator 1.*
Add indicators included in gender action plan, stakeholder engagement plan or other monitoring plans as needed.								
Output 1.3: A strategic plan is developed for developing a local	*% of people involved in the development of the strategy are*	*MT: n/a* *EoP: At least 30% of the people involved in the*	*Ensuring that a gender-differentiated approach is adopted in developing*	*Data will be collected through regular surveys that will form part of the annual lessons*	Annually Reported in project reports and also	*Project Manager with support from project consultant*	*Project reports and reports to/minutes of the PSC*	*<u>Risks</u>: The project replicates gender biases and gender-based discrimination as per Risk*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
value chain for sustainable and low-carbon building materials	*women*	*development of the replication plan are women*	*the strategic plan*	*learned studies.*	in DO tab of the GEF PIR	*carrying out annual lessons learned reports*		*10 in UNDP Risk Register (SESP – Risk 2)* *<u>Assumptions</u>: The GAP (Annex 10) is implemented as an integral part of the project*
Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Terminal Evaluation carried out	*M &E reports include the integration of quantitative and qualitative information on gender*	*MT: n/a* *EoP: At least 60% of indicators are sex disaggregated* *100% of M & E reports integrate quantitative and qualitative information on gender*	*M & E Reports prepared to discuss key findings and recommendations integrates gender and social inclusion considerations* *Development and integration of gender-sensitive indicators and collection of sex-disaggregated data for monitoring and evaluating project results.*	*Data will be collected through regular surveys that will form part of the annual lessons learned studies.*	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	*Project reports and reports to/minutes of the PSC*	*<u>Risks</u>: The project replicates gender biases and gender-based discrimination as per Risk 10 in UNDP Risk Register (SESP – Risk 2)* *<u>Assumptions</u>: The GAP (Annex 10) is implemented as an integral part of the project*
Output 3.3: Lessons learnt, experiences and best practices related to	*% of project participants who are women*	*MT: n/a* *EoP: At least 50% of the people involved in the*	*Improve understanding of gender issues related to the energy efficient technology,*	*Data will be collected through regular surveys that will form part of the annual lessons*	Annually Reported in project reports and also	*Project Manager with support from project consultant*	*Project reports and reports to/minutes of the PSC*	*<u>Risks</u>: The project replicates gender biases and gender-based discrimination as per Risk*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/ Assumptions
the project are compiled and disseminated in other cities of Tunisia and MENA countries		*development and dissemination of experiences are women.* *At least 50% of recipients of information generated including lessons learned are women*	*including access to public lighting, capacity building needs of women and policy issues at the municipal level related to energy efficient technologies, among others. This will inform the output on lessons learned and disseminated.*	*learned studies.*	in DO tab of the GEF PIR	*carrying out annual lessons learned reports*		*10 in UNDP Risk Register (SESP – Risk 2)* <u>Assumptions</u>: *The GAP (Annex 10) is implemented as an integral part of the project*
Environmental and Social risks related studies, as relevant[38].	N/A at this stage	N/A at this stage	N/A at this stage	As per ESMF	Annually	Project Manager with the support of Project Coordinator	Annual lessons learned reports and project terminal evaluation report	<u>Risks</u>: Grievance mechanism not publicized / introduced among project partners; <u>Assumptions</u>: ESIA and SESA proposed in the ESMF and integrated in Outputs 1.3, 1.6, 2.3 and 3.4 applied adequately; Stakeholder Engagement Plan implemented adequately

[38] See Section VI and Annex 9: Environmental and Social Management Framework in the ProDoc.

Table 19. Monitoring and evaluation plan and budget.

Monitoring and Evaluation Budget for project execution:		
GEF M&E requirements to be undertaken by Project Management Unit (PMU)	**Indicative costs (US$)**	**Time frame**
Inception Workshop and Report	*2,500*	Inception Workshop within 2 months of the First Disbursement
M&E required to report on progress made in reaching GEF core indicators and project results included in the project results framework	*4,500*	Annually and at closure.
Preparation of the annual GEF Project Implementation Report (PIR)	*None* *[Covered by the salary of the Project Manager]*	Annually typically between June-August
Monitoring of *SES, gender action plan, stakeholder engagement plan, project risks*	*3,845*	On-going.
Supervision missions	*None*	Annually
Learning missions	*None*	As needed
Independent Mid-term Review (MTR):	*n/a*	*n/a*
Independent Terminal Evaluation (TE):	*31,500*	31 December 2026
TOTAL indicative COST	***42,345***	

VII. GOVERNANCE AND MANAGEMENT ARRANGEMENTS

Section 1: General roles and responsibilities in the projects' governance mechanism

73. Implementing Partner: The Implementing Partner for this project is the Ministry of Environment (through the Directorate of Sustainable Development). The Implementing Partner is the entity to which the UNDP Administrator has entrusted the implementation of UNDP assistance specified in this signed project document along with the assumption of full responsibility and accountability for the effective use of UNDP resources and the delivery of outputs, as set forth in this document.

74. The Implementing Partner is responsible for executing this project. Specific tasks include:
    - Project planning, coordination, management, monitoring, evaluation and reporting. This includes providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.
    - Overseeing the management of project risks as included in this project document and new risks that may emerge during project implementation.
    - Procurement of goods and services, including human resources.
    - Financial management, including overseeing financial expenditures against project budgets.
    - Approving and signing the multiyear workplan.
    - Approving and signing the combined delivery report at the end of the year; and,
    - Signing the financial report or the funding authorization and certificate of expenditures.

75. Project stakeholders and target groups: The project stakeholders and target groups are identified in Table 14 (Section IV) above with more details given in Annex 10 – Stakeholder Engagement Plan. The involvement of the key stakeholders in project management is shown below in project organization structure.

76. UNDP: UNDP is accountable to the GEF for the implementation of this project. This includes overseeing project execution undertaken by the Implementing Partner to ensure that the project is being carried out in accordance with UNDP and GEF policies and procedures and the standards and provisions outlined in the Delegation of Authority (DOA) letter for this project. **The UNDP GEF Executive Coordinator, in consultation with UNDP Bureaus and the Implementing Partner, retains the right to revoke the project DOA, suspend or cancel this GEF project.** UNDP is responsible for the Project Assurance function in the project governance structure and presents to the Project Board and attends Project Board meetings as a non-voting member.

77. A firewall will be maintained between the delivery of project oversight and quality assurance performed by UNDP and charged to the GEF Fee and any support to project execution performed by UNDP (as requested by and agreed to by both the Implementing Partner and GEF) and may be charged to the GEF project management costs (only if approved by GEF). The segregation of functions and firewall provisions for UNDP in this case is described in the next section.

Section 2: Project governance structure

78. The project will be implemented using the Support to NIM modality where UNDP is providing country support services to the Implementing Partner via the Tunis Country Office as agreed in the LOA included in Annex 4, and following approval by the GEF at the time of CEO Endorsement. The governance structure is shown in Figure 10.

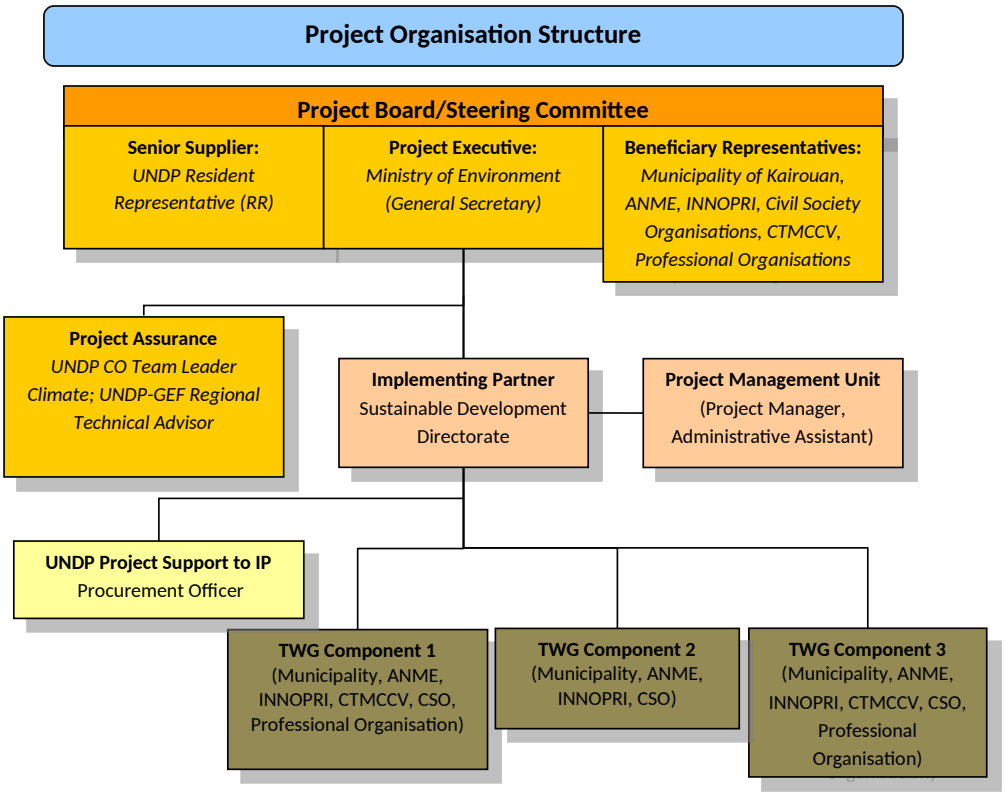


Figure 10. Project Organisational Structure.

79. The UNDP Resident Representative assumes full responsibility and accountability for oversight and quality assurance of this Project and ensures its timely implementation in compliance with the GEF-specific requirements and UNDP's Programme and Operations Policies and Procedures (POPP), its Financial Regulations and Rules and Internal Control Framework. A representative of the UNDP Country Office will assume the assurance role and will present assurance findings to the Project Board, and therefore attends Project Board meetings as a non-voting member.

80. **UNDP project support:** The Implementing Partner and GEF OFP have requested UNDP to provide support services in the amount of **USD 18,086** for the full duration of the project, and the GEF has agreed for UNDP to provide such execution support services and for the cost of these services to be charged to the project budget. The execution support services – whether financed from the project budget or other sources - have been set out in detail and agreed between UNDP Country Office and the Implementing Partner in a Letter of Agreement (LOA). This LOA is attached to this Project Document (Annex 4).

81. To ensure the strict independence required by the GEF and in accordance with the UNDP Internal Control Framework, these execution services will be delivered independent from the GEF-specific oversight and quality assurance services.

Section 3: Segregation of duties and firewalls vis-à-vis UNDP representation on the project board:

82. As noted in the Minimum Fiduciary Standards for GEF Partner Agencies, in cases where a GEF Partner Agency (i.e. UNDP) carries out both implementation oversight and execution of a project, the GEF Partner Agency (i.e. UNDP) must separate its project implementation oversight and execution duties, and describe in the relevant project document a: 1) Satisfactory institutional arrangement for the separation of implementation oversight

and executing functions in different departments of the GEF Partner Agency; and 2) Clear lines of responsibility, reporting and accountability within the GEF Partner Agency between the project implementation oversight and execution functions.

83. In this case, UNDP's implementation oversight role in the project – as represented in the project board and via the project assurance function – is performed by the Team Leader Climate Change, Environment, Energy and Disaster Risk Reduction and the UNDP-GEF RTA (Arab States Regional Hub). UNDP's execution role in the project (as requested by the implementing partner and approved by the GEF) is performed by the Procurement Officer**,** who will report to the Head of Operations**.**

Section 4: Roles and Responsiblities of the Project Organization Strucutre:

84. **Project Board:** All UNDP projects must be governed by a multi-stakeholder board or committee established to review performance based on monitoring and evaluation, and implementation issues to ensure quality delivery of results. The Project Board (also called the Project Steering Committee) is the most senior, dedicated oversight body for a project.

85. The two main (mandatory) roles of the project board are as follows:

1) **High-level oversight of the execution of the project by the Implementing Partner** (as explained in the "Provide Oversight" section of the POPP). This is the primary function of the project board and includes annual (and as-needed) assessments of any major risks to the project, and decisions/agreements on any management actions or remedial measures to address them effectively. The Project Board reviews evidence of project performance based on monitoring, evaluation and reporting, including progress reports, evaluations, risk logs and the combined delivery report. The Project Board is responsible for taking corrective action as needed to ensure the project achieves the desired results.
2) **Approval of strategic project execution decisions of the Implementing Partner** with a view to assess and manage risks, monitor and ensure the overall achievement of projected results and impacts and ensure long term sustainability of project execution decisions of the Implementing Partner (as explained in the "Manage Change" section of the POPP).

86. **Requirements to serve on the Project Board**:
- ✓ Agree to the Terms of Reference of the Board and the rules on protocols, quorum and minuting.
- ✓ Meet annually; at least once.
- ✓ Disclose any conflict of interest in performing the functions of a Project Board member and take all measures to avoid any real or perceived conflicts of interest. This disclosure must be documented and kept on record by UNDP.
- ✓ Discharge the functions of the Project Board in accordance with UNDP policies and procedures.
- ✓ Ensure highest levels of transparency and ensure Project Board meeting minutes are recorded and shared with project stakeholders.

87. **Responsibilities of the Project Board**:
- ✓ Consensus decision making:
  - The project board provides overall overall guidance and direction to the project, ensuring it remains within any specified constraints, and providing overall oversight of the project implementation.
  - Review project performance based on monitoring, evaluation and reporting, including progress reports, risk logs and the combined delivery report;

- The project board is responsible for making management decisions by consensus.
- In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value money, fairness, integrity, transparency and effective international competition.
- In case consensus cannot be reached within the Board, the UNDP representative on the board will mediate to find consensus and, if this cannot be found, will take the final decision to ensure project implementation is not unduly delayed.

✓ Oversee project execution:
- Agree on project manager's tolerances as required, within the parameters outlined in the project document, and provide direction and advice for exceptional situations when the project manager's tolerances are exceeded.
- Appraise annual work plans prepared by the Implementing Partner for the Project; review combined delivery reports prior to certification by the implementing partner.
- Address any high-level project issues as raised by the project manager and project assurance;
- Advise on major and minor amendments to the project within the parameters set by UNDP and the donor and refer such proposed major and minor amendments to the UNDP BPPS Nature, Climate and Energy Executive Coordinator (and the GEF, as required by GEF policies);
- Provide high-level direction and recommendations to the project management unit to ensure that the agreed deliverables are produced satisfactorily and according to plans.
- Track and monitor co-financed activities and realisation of co-financing amounts of this project.
- Approve the Inception Report, GEF annual project implementation reports, mid-term review and terminal evaluation reports.
- Ensure commitment of human resources to support project implementation, arbitrating any issues within the project.

✓ Risk Management:
- Provide guidance on evolving or materialized project risks and agree on possible mitigation and management actions to address specific risks.
- Review and update the project risk register and associated management plans based on the information prepared by the Implementing Partner. This includes risks related that can be directly managed by this project, as well as contextual risks that may affect project delivery or continued UNDP compliance and reputation but are outside of the control of the project. For example, social and environmental risks associated with co-financed activities or activities taking place in the project's area of influence that have implications for the project.
- Address project-level grievances.

✓ Coordination:
- Ensure coordination between various donor and government-funded projects and programmes.
- Ensure coordination with various government agencies and their participation in project activities.

88. **Composition of the Project Board**: The composition of the Project Board must include individuals assigned to the following three roles:

1. **Project Executive:** This is an individual who represents ownership of the project and chairs (or co-chairs) the Project Board. The Executive usually is the senior national counterpart for nationally implemented projects (typically from the same entity as the Implementing Partner), and it must be UNDP for projects that are direct implementation (DIM). In exceptional cases, two individuals from different entities can co-share this role and/or co-chair the Project Board. If the project executive co-chairs the project board with

representatives of another category, it typically does so with a development partner representative. The Project Executive is the Secretary General of the Ministry of Environment.

2. **Beneficiary Representative(s):** Individuals or groups representing the interests of those groups of stakeholders who will ultimately benefit from the project. Their primary function within the board is to ensure the realization of project results from the perspective of project beneficiaries. Often representatives from civil society, industry associations, or other government entities benefiting from the project can fulfil this role. There can be multiple beneficiary representatives in a Project Board. The Beneficiary representative (s) is/are: Municipality of Kairouan, ANME, INNOPRI, Civil Society Organisations, CTMCCV, Professional Organisations (Table 19, Annex 10).

3. **Development Partner(s):** Individuals or groups representing the interests of the parties concerned that provide funding, strategic guidance and/or technical expertise to the project. The Development Partner(s) is the UNDP Resident Representative or Deputy Resident Representative who will ensure the policies of UNDP and the GEF are complied with.

89. **Project Assurance:** Project assurance is the responsibility of each project board member; however, UNDP has a distinct assurance role for all UNDP projects in carrying out objective and independent project oversight and monitoring functions. UNDP performs quality assurance and supports the Project Board (and Project Management Unit) by carrying out objective and independent project oversight and monitoring functions, including compliance with the risk management and social and environmental standards of UNDP. The Project Board cannot delegate any of its quality assurance responsibilities to the Project Manager. Project assurance is totally independent of project execution.

90. A designated representative of UNDP playing the project assurance role is expected to attend all board meetings and support board processes as a non-voting representative. It should be noted that while in certain cases UNDP's project assurance role across the project may encompass activities happening at several levels (e.g. global, regional), at least one UNDP representative playing that function must, as part of their duties, specifically attend board meeting and provide board members with the required documentation required to perform their duties. The UNDP representative playing the main project assurance function are: the Team Leader Climate Change, Environment, Energy and Disaster Risk Reduction and the UNDP-GEF RTA (Arab States Regional Hub).

91. **Project Management – Execution of the Project:** The Project Manager (PM) (also called project coordinator) is the senior most representative of the Project Management Unit (PMU) and is responsible for the overall day-to-day management of the project on behalf of the Implementing Partner, including the mobilization of all project inputs, supervision over project staff, responsible parties, consultants and sub-contractors. The project manager typically presents key deliverables and documents to the board for their review and approval, including progress reports, annual work plans, adjustments to tolerance levels and risk registers.

92. The roles and responsibilities of the PMU members are detailed Annex 9. A designated representative of the PMU is expected to attend all board meetings and support board processes as a non-voting representative.

93. The primary PMU representative attending board meetings is the Project Manager.

VIII. FINANCIAL PLANNING AND MANAGEMENT

94. The total cost of the project is ***USD 4,069,906***. This is financed through a GEF grant of USD 846,906 administered by UNDP, USD 500,000 in cash co-financing to be administered by UNDP and additional support of *USD* ***2,723,000.*** UNDP, as the GEF Implementing Agency, is responsible for the oversight of the GEF resources and the cash co-financing transferred to UNDP bank account only.

95. **Co-financing**: The actual realization of project co-financing amounts will be monitored by the UNDP Country Office and the PMU on an annual basis in the GEF PIF and will be reported to the GEF during the terminal evaluation process as follows:

Table 20. Table of co-financing.

Co-financing source	Co-financing type	Co-financing amount
Government (Ministry of Environment)	*Cash*	*500,000*
Government (Ministry of Environment)	*In-kind*	*150,000*
Government (Municipality of Kairouan)	*Cash*	*825,000*
Government (Municipality of Kairouan)	*In-kind*	*350,000*
Private Sector (the Proportion of Gold)	*Cash (grant)*	*133,000*
GEF Agency (UNDP)	*Cash*	*415,000*
Private Sector (the Proportion of Gold)	*Cash (equity/debt)*	*850,000*

96. **Budget Revision and Tolerance**: As per UNDP POPP, the project board may agree with the project manager on a tolerance level for each detailed plan under the overall multi-year workplan. The agreed tolerance should be written in the project document or approved project board meeting minutes. It should normally not exceed 10 percent of the agreed annual budget at the activity level, but within the overall approved multi-year workplan at the activity level. Within the agreed tolerances, the project manager can operate without intervention from the project board. Restrictions apply as follows:

97. Should the following deviations occur, the Project Manager/IP through UNDP Country Office will seek the approval of the BPPS/NCE-VF team to ensure accurate reporting to the GEF. It is **strongly encouraged** to maintain the expenditures within the approved budget at the budgetary account and at the component level:

a) Budget reallocations must prove that the suggested changes in the budget will not lead to material changes in the results to be achieved by the project. A strong justification is required and will be approved on an exceptional basis. Budget re-allocations among the components (including PMC) of the approved Total Budget and Work Plans (TBWP) that represent a value greater than 10% of the total GEF grant.
b) Introduction of new outputs/activities (i.e. budget items) that were not part of the agreed project document and TBWP that represent a value greater than 5% of the total GEF grant. The new budget items must be eligible as per the GEF and UNDP policies.
c) Project management cost (PMC): budget under PMC component is capped and cannot be increased.

98. Any over expenditure incurred beyond the available GEF grant amount must be absorbed by non-GEF resources (e.g. UNDP TRAC or cash co-financing).

99. **Project extensions:** The UNDP Resident Representative and the UNDP-GEF Executive Coordinator must approve all project extension requests. Note that all extensions incur costs and the GEF project budget cannot be increased. A single extension may be granted on an exceptional basis and subject to the conditions and maximum durations set out in the UNDP POPP; the project management costs during the extension period must remain within the originally approved amount, and any increase in PMC costs will be covered by non-GEF

resources; the additional UNDP oversight costs during the extension period must be covered by non-GEF resources, in accordance with UNDP's guidance set out in UNDP POPP.

100. Audit: The project will be audited as per UNDP Financial Regulations and Rules and applicable audit policies. Audit cycle and process must be discussed during the Inception workshop. If the Implementing Partner is an UN Agency, the project will be audited according to that Agencies applicable audit policies.

101. **Project Closure**: Project closure will be conducted as per UNDP requirements outlined in the UNDP POPP. All costs incurred to close the project must be included in the project closure budget and reported as final project commitments presented to the Project Board during the final project review. The only costs a project may incur following the final project review are those included in the project closure budget.

102. **Operational completion**: The project will be operationally completed when the last UNDP-financed inputs have been provided and the related activities have been completed. This includes the final clearance of the Terminal Evaluation Report (that will be available in English) and the corresponding management response, and the end-of-project review Project Board meeting. **Operational closure must happen at the end date calculated by the approved duration after the Project Document signature or at the revised operational closure date as approved in the project extension. Any expected activity after the operational date requires project extension approval.** The Implementing Partner through a Project Board decision will notify the UNDP Country Office when operational closure has been completed. At this time, the project should have completed the transfer or disposal of any equipment that is still the property of UNDP.

103. **Transfer or disposal of assets**: In consultation with the Implementing Partner and other parties of the project, UNDP is responsible for deciding on the transfer or other disposal of assets. Transfer or disposal of assets is recommended to be reviewed and endorsed by the project board following UNDP rules and regulations. Assets may be transferred to the government for project activities managed by a national institution at any time during the life of a project (it is strongly encouraged to be done before the operational closure date). In all cases of transfer, a transfer document must be prepared and kept on file[39]. The transfer should be done before Project Management Unit complete their assignments.

104. **Financial completion (closure):** The project will be financially closed when the following conditions have been met: a) the project is operationally completed or has been cancelled; b) the Implementing Partner has reported all financial transactions to UNDP; c) UNDP has closed the accounts for the project; d) UNDP and the Implementing Partner have certified a final Combined Delivery Report (which serves as final budget revision).

105. The project will be financially completed **within 6 months of operational closure or after the date of cancellation**. If Operational Closure is delayed for any justified and approved reason, the Country Office should do all efforts to Financially Close the project within 9 months after TE is completed. Between operational and financial closure, the implementing partner will identify and settle all financial obligations and prepare a final expenditure report. The UNDP Country Office will send the final signed closure documents including confirmation of final cumulative expenditure and unspent balance to the BPPS/NCE-VF Unit for confirmation before the project will be financially closed in Atlas by the UNDP Country Office.

106. Refund to GEF: Should a refund of unspent funds to the GEF be necessary, this will be managed directly by the BPPS/NCE-VF Directorate in New York. No action is required by the UNDP Country Office on the actual refund from UNDP project to the GEF Trustee.

[39] See https://popp.undp.org/_layouts/15/WopiFrame.aspx?sourcedoc=/UNDP_POPP_DOCUMENT_LIBRARY/Public/PPM_Project%20Management_Closing.docx&action=default.

IX. TOTAL BUDGET AND WORK PLAN

Total Budget and Work Plan			
Atlas Award ID:	01000110	Atlas Output Project ID:	000xxxxx TBC
Atlas Proposal or Award Title:	Towards Sustainable Cities in Tunisia		
Atlas Business Unit	TUN10		
Atlas Primary Output Project Title	Towards Sustainable Cities in Tunisia		
UNDP-GEF PIMS No.	6686		
Implementing Partner	Ministry of Environment		

Atlas Activity (GEF Component)	**Atlas Implementing Agent (Responsible Party, IP, or UNDP)**	**Atlas Fund ID**	**Donor Name**	**Atlas Budgetary Account Code**	**ATLAS Budget Account Description**	**Amount Year 1 (USD)**	**Amount Year 2 (USD)**	**Amount Year 3 (USD)**	**Total (USD)**	**See Budget Note**
OUTCOME 1 Institutional strengthening for regulating the market for new efficient buildings	**UNDP** (to provide support service to **MoE**)	**62000**	**GEF**	72100	Contractual Services – Companies	39,000	359,458	4,000	402,458	1
				71400	Contractual Services - Individ	6,960	6,960	6,960	20,880	2
				74200	Audio Visual & Print Prod Costs	6,012	4,008	-	10,020	3
				75700	Workshops and Confer	18,700	6,700	5,700	31,100	4
	UNDP (to provide support service to **MoE**)	**30071**	**MOE**		**Outcome 1 - GEF**	**70,672**	**377,126**	**16,660**	**464,458**	
				72100	Contractual Services – Companies	39,000	59,000	4,000	102,000	5
				71400	Contractual Services - Individ	3,600	3,600	3,600	10,800	6
				75700	Workshops and Confer	3,800	3,800	2,000	9,600	7

					Outcome 1 - MOE	**46,400**	**66,400**	**9,600**	**122,400**	
					Outcome 1 (GEF + MOE)	**117,072**	**443,526**	**26,260**	**586,858**	
	UNDP (to provide support service to **MoE**)	**62000**	**GEF**	72100	Contractual Services – Companies	34,500	34,500	-	69,000	8
				71400	Contractual Services - Individ	2,400	2,400	2,400	7,200	9
				72200	Equipment & Furniture	105,457	0	0	105,457	10
				74200	Audio Visual & Print Prod Costs	3,400	3,400	-	6,800	11
				75700	Training, Workshops and Confer	8,500	8,500	-	17,000	12
					Outcome 2 - GEF	**154,257**	**48,800**	**2,400**	**205,457**	
OUTCOME 2 Energy efficient public lighting	**UNDP** (to provide support service to **MoE**)	**30071**	**MOE**	72100	Contractual Services – Companies	30,000	-	-	30,000	13
				71400	Contractual Services - Individ	6,000	6,000	6,000	18,000	14
				72800	IT Equipment	3,000	-	-	3,000	15
				72200	Equipment and Furniture	5,000	-	-	5,000	16
				74200	Audio Visual & Print Prod Costs	5,000	4,000	958	9,958	17
				75700	Training, Workshops and Confer	5,000	-	-	5,000	18
					Outcome 2 – MOE	**54,000**	**10,000**	**6,958**	**70,958**	
					Outcome 2 (GEF + MOE)	**208,257**	**58,800**	**9,358**	**276,415**	
OUTCOME 3: Knowledge management,	**UNDP** (to provide support service to **MoE**)	**62000**	**GEF**	71200	International Consultants	-	-	13,500	13,500	19
				71400	Contractual Services - Individ	1,750	1,750	1,755	5,255	20

monitoring and evaluation, and scale-up strategy				71600	Travel	4,000	6,000	6,000	16,000	21
				74200	Audio Visual & Print Prod Costs	4,000	6,000	6,000	16,000	22
				75700	Training, Workshops and Confer		-	6,900	6,900	23
					Total KM	**9,750**	**13,750**	**34,155**	**57,655**	
				71200	International Consultants			31,500	31,500	24
				71300	Local Consultant	3,000	3,000	2,345	8,345	25
				75700	Training, Workshops and Confer	2,500			2,500	26
					Total M&E	**5,500**	**3,000**	**33,845**	**42,345**	
					Outcome 3 - GEF	**15,250**	**16,750**	**68,000**	**100,000**	
	UNDP (to provide support service to **MoE**)	**30071**	**MOE**	71200	International Consultants	18,000	33,000	50,700	101,700	27
				71300	Local Consultants	16,000	16,000	11,500	43,500	28
				71400	Contractual Services - Individ	1,200	1,200	1,200	3,600	29
				75700	Training, Workshops and Confer	1,200	1,200	600	3,000	30
				72100	Contractual Services – Companies	-	-	9,000	9,000	31
					Outcome 3 - MOE	**36,400**	**51,400**	**73,000**	**160,800**	
					Outcome 3 (GEF + MOE)	**51,650**	**68,150**	**141,000**	**260,800**	
Project Management	**UNDP** (to provide support service to **MoE**)	**62000**	**GEF**	71400	Contractual Services - Individ	15,360	15,360	15,360	46,080	32
				72800	Information Technology Equipment	5,000	-	-	5,000	33
				74100	Professional Services	2,608	2,608	2,609	7,825	34

				74596	Direct Project Costs	8,016	5,035	5,035	18,086	35
					PMU GEF	**30,984**	**23,003**	**23,004**	**76,991**	
	UNDP (to provide support service to **MoE**)	**30071**	**MOE**	71400	Contractual Services - Individ	19,680	19,680	19,680	59,040	36
				72500	Supplies	5,000	0	0	5,000	37
				74596	Direct Project Costs	17,833	17,833	17,834	53,500	38
				75100	GMS	9,434	9,434	9,434	28,302	39
					PMU MOE	**51,947**	**46,947**	**46,948**	**145,842**	
					PMU (GEF + MOE)	**82,931**	**69,950**	**69,952**	**222,833**	
				PROJECT TOTAL (GEF)		**271,163**	**465,679**	**110,064**	**846,906**	
				PROJECT TOTAL (MOE)		**188,747**	**174,747**	**136,506**	**500,000**	
				PROJECT TOTAL (GEF + MOE)		**459,910**	**640,426**	**246,570**	**1,346,906**	

Budget note number	Comments
1	A Company (thereafter Company A) will be recruited to provide technical assistance services for Outputs 1.1, 1.2, 1.3 and 1.6. The services will require 88 days of work @USD 1,000 per day (all-inclusive cost) – i.e. USD 88,000 total. The allocation is as follows: Output 1.1: 20 days (USD 20,000) for the review and update of existing licensing and permits process, and formulation of operational manual for the municipality; Output 1.2: 16 days 9USD 16,000) operationalization of the cross-institutional mechanism for the enforcement of the RTNB and review of the RTNB; Output 1.3: 37 days (USD 37,000) for formulating the strategic plan for developing a value chain for locally-fabricated bricks; Output 1.6: 15 days (USD 15,000) for carrying out a targeted environmental and social risk assessment for municipal building at Keblia and for developing a ESMP. As per the scheduling of activities in the work plan, the yearly allocation is estimated as follows: **USD 88,000** -Yr1 – USD 39,000 (Output 1.1 – 20 days; Output 1.2 – 9 days; Output 1.6 – 10 days); Yr2 – USD 45,000 (Output 1.2 – 3 days; Output 1.3 – 37 days; Output 1.6 – 5 days); Yr3 – 4,000 (Output 1.2 – 4 days). The technical assistance financed using GEF funding will be complemented with the co-financing from MOE as described in note 6. For Outputs 1.1, 1.2, 1.3 and 1.6, the Company A will provide the following expertise as detailed in Annex 9. • Strategic Value Chain Expert (37 days) – Output 1.3 • Environmental & Social Risk Expert (15 days) – Output 1.6 • Institutional et Regulatory Expert (36 days) – Outputs 1.1 and 1.2

	Please see note 6 for complementary expertise provided by Company A. The Environmental & Social Risk Expert will also carry out targeted assessments according to SESP and ESMF for Output 2.3 (note 15) and strategic assessments for Output 3.4 (note 31) GEF investments will be used to enhanced the energy efficiency of the Municipal Building at Keblia that will yield global environmental benefits (Annex 14 and Section IV of the Project Document), as well as the operationalization of the procedure for enforcing the application of the building energy code (RTNB). Through the GEF investments totaling **USD 314,458**, the newly developed ECOBat label in Tunisia will be sought for the energy efficiency-enhanced municipal building.
2	Partial salary of the Project Manager (20% of USD 72,000 = USD 14,400) and of the Project Associate/Administrative Assistance (15% of USD 43,200 = USD 6,480) are covered under Outcome 1 for technical work. The total of **USD 20,880** is spread equally over 3 years (i.e. USD 6,960 per year). The Project Manager will carry out technical duties as specified in the ToR given in Annex 9 that is imputed to GEF-financing under the project components for a total of USD 25,200 – i.e. 35% of total PM salary. The largest share – i.e. USD 14,400 – is allocated to component 1 because of the highest level of time inputs for technical work.
3	A total budget of **USD 10,020** is allocated for printing expenses for the operational manual (Output 1.1) and the manual for the online tool (Output 1.4). The quality standards for bricks (Output 1.5) will also be published. Since the bulk of these activities take place in Yr1 but straddle Yr2, the allocation is USD 6.012 (Yr1) and USD 4,008 (Yr2).
4	A total budget of **USD 31,100** is allocated for coordinating project stakeholders for the following: Output 1.1 – USD 11,000 for training of municipal staff on the use of operational manual; Output 1.2 – USD 3,600 for training of municipal staff for carrying out enforcement of construction sites; Output 1.3 – USD 7,500 for stakeholder consultations for developing strategic plan; Output 1.6 – USD 9,000 for consultations for targeted social and environmental risks assessments as per SESP and ESMF. The yearly allocation is: Yr1 – USD 18,700; Yr2 – USD 6,700; Yr 3 – 5,700.
5	The co-financing of MOE will be used to complement the supply of technical services by Company A for implementing Output 1.4, Output 1.5 and Output 1.6 for a total of 90 days @USD 1,000 per day – i.e. USD 90,000. The allocation by output is: Output 1.4: 40 days (USD 40,000) for developing online tool following needs gaps analysis, as well as training on its use; Output 1.5: 25 days (USD 25,000) for developing quality standards for locally-fabricated bricks; Output 1.6: 25 days (USD 25,000) for carrying out a targeted environmental and social risk assessment for municipal building at Keblia and for developing a ESMP (complements GEF financing). As per the scheduling of activities in the work plan, the yearly allocation is estimated as follows: **USD 90,000** -Yr1 – USD 35,000 (Output 1.4 – 20 days; Output 1.6 – 15 days); Yr2 – USD 55,000 (Output 1.4 – 20 days; Output 1.5 – 25 days; Output 1.6 – 10 days). The technical assistance financed using MOE funding will be complemented GEF-funded activities described in note 1. For Outputs 1.4 and 1.5, the Company A will provide the following complementary expertise as detailed in Annex 9. • Building Energy Efficiency Expert (40 days) – Output 1.4 • Quality Standards Expert (25 days) – Output 1.5 The Quality Standards Expert will also provide technical assistance related to Output 2.2.

	Under Output 1.3, the public laboratory CTMCCV will be funded to carry out characterization of the physical and thermal properties of locally-fabricated bricks. These tests are needed to ensure quality standards of the bricks. It is expected that 20 samples will be tested each year for a unit cost of USD 200 – i.e. USD 4,000 per year over three years. The total budget is **USD 12,000**.
6	Part of the salary of the Project Coordinator (20% of USD 54,000 = **USD 10,800**) is allocated to Outcome 2 using MOE co-financing. This budget is allocated equally (USD 3,600) over three years. It is pointed out that 34% of total salary budget (i.e. USD 91,440) for project management staff is paid using MOE cash co-financing.
7	This budget complements the budget at note 4. A total budget of **USD 9,600** is allocated for coordinating project stakeholders for the following: Output 1.4 – USD 3,600 for stakeholders' needs analysis and training on the use of online tool; Output 1.5 – USD 6,000 for technical committee that will be established at INNORPI to develop quality standards for bricks. The annual budget allocation is: Yr1 =Yr2 = USD 3,800; Yr3 = 2,000.
8	A Services Company (Company A) will be recruited to carry out training for the municipality staff for the designing, implementation and monitoring and evaluation of intelligent public lighting under Output 2.1 (Public Lighting Efficiency Expert). The will also develop all necessary training materials based on a needs gaps analysis in close collaboration with the municipality. This is budgeted at USD 47,000 is for 47 days of services delivered at USD 1,000 per day (all-inclusive cost). The same company will be requested to provide technical support on the development of national standards (Quality Standards Expert) for public lamps under Output 2.2 for a total cost of USD 22,000 (20 days). The total cost of **USD 69,000** will be spread equally between Yr1 and Yr2 for USD 34,500 each.
9	Part salary of the Project Manager (10% of USD 72,000 = **USD 7,200**) is budgeted for technical work as per the technical duties given in the ToR in Annex 9.
10	GEF investments of **USD 105,457** will be used to replace 574 conventional street lights with energy efficient LEDS with remote sensing function. The average unit cost of each LED lamp (and casing) is USD 184. The replacement of lamps will yield global environmental benefits as calculated in Annex 14 and given in Section IV of the Project Document.
11	Training materials will be developed and printed for Output 2.1 and Output 2.2 for a total cost of **USD 6,800**. Each output will receive half the budget and spread equally between Yr1 and Yr2 (USD 3,400 for each year).
12	A total budget of **USD 17,000** is for training workshops for Output 2.1 will be spread equally over Yr1 and Yr2 at USD 4,500 for each year. For Output 2.2, four workshops (@USD 2,000 each) are planned for developing standards for efficient public lamps. The cost of USD 8,000 will be split equally over Yr1 and Yr2 – i.e. USD 4,000 per year. This gives total budget of Yr1=Yr2=USD 4,500.
13	A services company (Company A) will be recruited for providing Environmental and Social Risk Expert for carrying out targeted assessments for lighting substitution as per SESP and ESMF (Output 2.3). It is for 30 days @ USD 1,000 per day (all-inclusive cost) – i.e. **USD 30,000**.
14	As per budget note 6, partial salary of the Project Manager (10% of USD 72,000 = USD 7,200) and of the Project Coordinator (20% of USD 54,000 = USD 10,800) are covered under Outcome 2 using MOE co-financing. The total of **USD 18,000** is spread equally over 3 years.
15	At the beginning of the project three computers (USD 2,000 each) and one printer (USD 2,000) will be purchased for project management. One computer will be used by each of the Project Manager, Administrative Assistant and Project Coordinator. The total budget is USD 8,000 and the **USD 3,000** complements the USD 5,000 allocated at note 33 below.
16	Office furniture valued at **USD 5,000** will be bought for accommodating the PMU in Yr1.
17	A total of **USD 9,958** is allocated for developing audiovisual and printing materials for the training to be dispensed under Output 2.1 that will span mainly Yr1 (USD 5,000) and Yr 2 (USD 4,000). A small budget USD 958 in Yr3.
18	Five mini-workshops (@USD 1,000 / unit) will be carried out in Yr1 for developed the targeted assessments related to environmental and social risks in order to develop ESMP for lighting replacements. The total budget is **USD 5,000**.
19	An international consultant who is an expert in formulating concept notes for climate finance (International Climate Finance Expert) will be recruited in Yr3 for a total of **USD 13,500** (15 days @USD 900 per day (all costs inclusive)).

20	GEF funding will cover 5% of the salary of the Project Manager (USD 72,000) – i.e. **5,255** spread equally over three years; USD 1,750 per year. This is related to technical duties given in the ToR of Annex 9 for component 3.
21	Since the project is implemented in Kairouan and while the PMU will be located in Tunis, a total internal travel budget of **USD 16,000** has been allocated for the entire duration of the project. The travels will also serve the M&E purpose. The allocation is: Yr1 = USD 4,000; Yr2=Yr3= USD 6,000.
22	A lumpsum printing cost of **USD 16,000** is allocated for Outcome related mainly to lessons learned and knowledge products (Output 3.3); terminal evaluation report (Output 3.1); and replication plan and concept notes to leverage international climate finance (Output 3.4). The allocation is: Yr1 = USD 4,000; Yr2=Yr3= USD 6,000.
23	A country level workshop will be used to validate the replication plan over 2 days at a cost of **USD 6,900**. The total cost for these two workshops is USD 9,400.
24	An international consultant (International Evaluation Expert) will be recruited in Yr3 to carry out Terminal Evaluation of the project for a total of **USD 31,500** (35 days @USD 900 per day (all costs inclusive)).
25	A total of **USD 8,345** is for Local Consultant to prepare a report on progress made in reaching GEF core indicators and project results included in the project results framework and to monitor SES, gender action plan, stakeholder engagement plan, project risks
26	An inception workshop will take place in Yr1 (**USD 2,500**)
27	International consultants will be required for (i) developing a data management system (Output 3.2 – 40 days; International Data Management Expert); (ii) developing knowledge products (Output 3.3 – 43 days; International Knowledge Management Expert); (iii) developing replication plan (Output 3.4 – 30 days; International Knowledge Management Expert). At a lumpsum price of USD 900 per day (inclusive of all travel and accommodation expenses), the total sum is **USD 101,700** that is spread as follows: Yr1 – USD 18,000 (Output 3.2); Yr 2 – USD – 33,000 (Output 3.2 and Output 3.3); Yr3 – 59,700 (Output 3.3 & Output 3.4).
28	A local consultant (Knowledge and Data Management Expert) will be used to support the international consultants at note 27 and to process of developing bankable concept note under Output 3.4 for leveraging international climate finance for a total of 87 person days @USD 500 per day. The allocation by Output is: Output 3.2 – 45 days; Output 3.3 – 42 days. The work of the local consultants will be spread over the 3 years as: **USD 43,500**-Yr1=Yr2=USD 16,000 and Yr3 = 11,500
29	As per budget note 6, part of the cash co-financing of MOE is used to cover 5% of the salary of the Project Manager (USD 72,000) – i.e. **USD 3,600** spread equally over three years.
30	Five workshops are planned related to (i) lessons learned under Output 3.3, and (ii) the development of a data inventory system in the municipality of Kairouan under Output 3.2. The average cost of workshop is USD 600 (i.e. **total USD 3,000**): Yr1 and Yr 2 (2 workshops); Yr3 (1 workshop).
31	Company's A (see note 1) Environmental and Social Risk Expert will provide 9 days of technical support (i.e. **USD 9,000**) in Yr3 to carry out strategic assessment of potential environmental and social risks associated with the Replication Plan that will be developed under Output 3.4.
32	The total of **USD 46,080** is for the partial salary of the Project Manager (25% of USD 72,000 = USD 18,000) and the Project Associate / Administrative Assistant (65% of USD 43,200 = USD 28,080). The total is allocated equally over 3 years – i.e. USD 15,360 per year.
33	At the beginning of the project three computers (USD 2,000 each) and one printer (USD 2,000) will be purchased for project management. One computer will be used by each of the Project Manager, Administrative Assistant and Project Coordinator. The total budget is **USD 5,000** and complements the USD 3,000 allocated at note 15 above.
34	The project will undergo an independent financial audit every year for a total of **USD 7,825.**
35	The project is implemented using Support to NIM and the UNDP will charge for a total of **USD 18,086** for support services as per details given in Annex 4.
36	The MOE co-financing will be used to cover the partial salaries of the Project Manager (25% of USD 72,000 = USD 18,000); Project Coordinator (60% of USD 54,000 = USD 32,400); and the Project Associate / Administrative Assistant (20% of USD 43,200 = USD 8,640). The total cost of **USD 59,040** is spread equally

	over the project duration.
37	A lumpsum cost of **USD 5,000** is allocated in Yr1 for the operation of the PMU.
38	The UNDP will host the PMU and the total rental and utility costs of **USD 30,000** are spread equally over 3 years. Direct Project Costs have been estimated by UNDP at **USD 23,500.** This is spread equally over the project lifetime at around USD 7,833 per year (with a rounding off to USD 7,834 in Yr3.
39	The UNDP charges 6% GMS on the MOE co-financing that totals **USD 28,302**. This is allocated equally to each project year – USD 9,434 per year.

X. LEGAL CONTEXT

107. This project document shall be the instrument referred to as such in Article 1 of the Standard Basic Assistance Agreement between the Government of Tunisia and UNDP, signed on 25 April 1987. All references in the SBAA to "Executing Agency" shall be deemed to refer to "Implementing Partner."

108. This project will be implemented by The Ministry of Environment (through the Directorate of Sustainable Development) in accordance with its financial regulations, rules, practices and procedures only to the extent that they do not contravene the principles of the Financial Regulations and Rules of UNDP. Where the financial governance of an Implementing Partner does not provide the required guidance to ensure best value for money, fairness, integrity, transparency, and effective international competition, the financial governance of UNDP shall apply.

109. The designations employed and the presentation of material on this map do not imply the expression of any opinion whatsoever on the part of the Secretariat of the United Nations or UNDP concerning the legal status of any country, territory, city or area or its authorities, or concerning the delimitation of its frontiers or boundaries.

XI. RISK MANAGEMENT

110. **Option a. Implementing Partner is a Government Entity (NIM)**

1. Consistent with the Article III of the SBAA *[or the Supplemental Provisions to the Project Document]*, the responsibility for the safety and security of the Implementing Partner and its personnel and property, and of UNDP's property in the Implementing Partner's custody, rests with the Implementing Partner. To this end, the Implementing Partner shall:
   a) put in place an appropriate security plan and maintain the security plan, taking into account the security situation in the country where the project is being carried;
   b) assume all risks and liabilities related to the Implementing Partner's security, and the full implementation of the security plan.

2. UNDP reserves the right to verify whether such a plan is in place, and to suggest modifications to the plan when necessary. Failure to maintain and implement an appropriate security plan as required hereunder shall be deemed a breach of the Implementing Partner's obligations under this Project Document.

3. The Implementing Partner agrees to undertake all reasonable efforts to ensure that no UNDP funds received pursuant to the Project Document are used to provide support to individuals or entities associated with terrorism and that the recipients of any amounts provided by UNDP hereunder do not appear on the list maintained by the Security Council Committee established pursuant to resolution 1267 (1999). The list can be accessed via http://www.un.org/sc/committees/1267/aq_sanctions_list.shtml.

4. The Implementing Partner acknowledges and agrees that UNDP will not tolerate sexual harassment and sexual exploitation and abuse of anyone by the Implementing Partner, and each of its responsible parties, their respective sub-recipients and other entities involved in Project implementation, either as contractors or subcontractors and their personnel, and any individuals performing services for them under the Project Document.

(a) In the implementation of the activities under this Project Document, the Implementing Partner, and each of its sub-parties referred to above, shall comply with the standards of conduct set forth in the Secretary General's Bulletin ST/SGB/2003/13 of 9 October 2003, concerning "Special measures for protection from sexual exploitation and sexual abuse" ("SEA").

(b) Moreover, and without limitation to the application of other regulations, rules, policies and procedures bearing upon the performance of the activities under this Project Document, in the implementation of activities, the Implementing Partner, and each of its sub-parties referred to above, shall not engage in any form of sexual harassment ("SH"). SH is defined as any unwelcome conduct of a sexual nature that might reasonably be expected or be perceived to cause offense or humiliation, when such conduct interferes with work, is made a condition of employment or creates an intimidating, hostile or offensive work environment.

5. a) In the performance of the activities under this Project Document, the Implementing Partner shall (with respect to its own activities), and shall require from its sub-parties referred to in paragraph 4 (with respect to their activities) that they, have minimum standards and procedures in place, or a plan to develop and/or improve such standards and procedures in order to be able to take effective preventive and investigative action. These should include: policies on sexual harassment and sexual exploitation and abuse; policies on whistleblowing/protection against retaliation; and complaints, disciplinary and investigative mechanisms. In line with this, the Implementing Partner will and will require that such sub-parties will take all appropriate measures to:

   i. Prevent its employees, agents or any other persons engaged to perform any services under this Project Document, from engaging in SH or SEA;

   ii. Offer employees and associated personnel training on prevention and response to SH and SEA, where the Implementing Partner and its sub-parties referred to in paragraph 4 have not put in place its own training regarding the prevention of SH and SEA, the Implementing Partner and its sub-parties may use the training material available at UNDP;

   iii. Report and monitor allegations of SH and SEA of which the Implementing Partner and its sub-parties referred to in paragraph 4 have been informed or have otherwise become aware, and status thereof;

   iv. Refer victims/survivors of SH and SEA to safe and confidential victim assistance; and

   v. Promptly and confidentially record and investigate any allegations credible enough to warrant an investigation of SH or SEA. The Implementing Partner shall advise UNDP of any such allegations received and investigations being conducted by itself or any of its sub-parties referred to in paragraph 4 with respect to their activities under the Project Document, and shall keep UNDP informed during the investigation by it or any of such sub-parties, to the extent that such notification (i) does not jeopardize the conduct of the investigation, including but not limited to the safety or security of persons, and/or (ii) is not in contravention of any laws applicable to it. Following the investigation, the Implementing Partner shall advise UNDP of any actions taken by it or any of the other entities further to the investigation.

   b) The Implementing Partner shall establish that it has complied with the foregoing, to the satisfaction of UNDP, when requested by UNDP or any party acting on its behalf to provide such confirmation. Failure of the Implementing Partner, and each of its sub-parties referred to in paragraph 4, to comply of the foregoing, as determined by UNDP, shall be considered grounds for suspension or termination of the Project.

6. Social and environmental sustainability will be enhanced through application of the UNDP Social and Environmental Standards (http://www.undp.org/ses) and related Accountability Mechanism (http://www.undp.org/secu-srm).

7. The Implementing Partner shall: (a) conduct project and programme-related activities in a manner consistent with the UNDP Social and Environmental Standards, (b) implement any management or mitigation plan prepared for the project or programme to comply with such standards, and (c) engage in a constructive and timely manner to address any concerns and complaints raised through the Accountability Mechanism. UNDP will seek to ensure that communities and other project stakeholders are informed of and have access to the Accountability Mechanism.

8. All signatories to the Project Document shall cooperate in good faith with any exercise to evaluate any programme or project-related commitments or compliance with the UNDP Social and Environmental Standards. This includes providing access to project sites, relevant personnel, information, and documentation.

9. The Implementing Partner will take appropriate steps to prevent misuse of funds, fraud or corruption, by its officials, consultants, responsible parties, subcontractors and sub-recipients in implementing the project or using UNDP funds. The Implementing Partner will ensure that its financial management, anti-corruption and anti-fraud policies are in place and enforced for all funding received from or through UNDP.

10. The requirements of the following documents, then in force at the time of signature of the Project Document, apply to the Implementing Partner: (a) UNDP Policy on Fraud and other Corrupt Practices and (b) UNDP Office of Audit and Investigations Investigation Guidelines. The Implementing Partner agrees to the requirements of the above documents, which are an integral part of this Project Document and are available online at www.undp.org.

11. In the event that an investigation is required, UNDP has the obligation to conduct investigations relating to any aspect of UNDP projects and programmes in accordance with UNDP's regulations, rules, policies and procedures. The Implementing Partner shall provide its full cooperation, including making available personnel, relevant documentation, and granting access to the Implementing Partner's (and its consultants', responsible parties', subcontractors' and sub-recipients') premises, for such purposes at reasonable times and on reasonable conditions as may be required for the purpose of an investigation. Should there be a limitation in meeting this obligation, UNDP shall consult with the Implementing Partner to find a solution.

12. The signatories to this Project Document will promptly inform one another in case of any incidence of inappropriate use of funds, or credible allegation of fraud or corruption with due confidentiality.

    Where the Implementing Partner becomes aware that a UNDP project or activity, in whole or in part, is the focus of investigation for alleged fraud/corruption, the Implementing Partner will inform the UNDP Resident Representative/Head of Office, who will promptly inform UNDP's Office of Audit and Investigations (OAI). The Implementing Partner shall provide regular updates to the head of UNDP in the country and OAI of the status of, and actions relating to, such investigation.

13. UNDP shall be entitled to a refund from the Implementing Partner of any funds provided that have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document. Such amount may be deducted by UNDP from any payment due to the Implementing Partner under this or any other agreement. Recovery of such amount by UNDP shall not diminish or curtail the Implementing Partner's obligations under this Project Document.

    Where such funds have not been refunded to UNDP, the Implementing Partner agrees that donors to UNDP (including the Government) whose funding is the source, in whole or in part, of the funds for the activities under this Project Document, may seek recourse to the Implementing Partner for the recovery of any funds

determined by UNDP to have been used inappropriately, including through fraud or corruption, or otherwise paid other than in accordance with the terms and conditions of the Project Document.

Note: The term "Project Document" as used in this clause shall be deemed to include any relevant subsidiary agreement further to the Project Document, including those with responsible parties, subcontractors and sub-recipients.

14. Each contract issued by the Implementing Partner in connection with this Project Document shall include a provision representing that no fees, gratuities, rebates, gifts, commissions or other payments, other than those shown in the proposal, have been given, received, or promised in connection with the selection process or in contract execution, and that the recipient of funds from the Implementing Partner shall cooperate with any and all investigations and post-payment audits.

15. Should UNDP refer to the relevant national authorities for appropriate legal action any alleged wrongdoing relating to the project, the Government will ensure that the relevant national authorities shall actively investigate the same and take appropriate legal action against all individuals found to have participated in the wrongdoing, recover and return any recovered funds to UNDP.

16. The Implementing Partner shall ensure that all of its obligations set forth under this section entitled "Risk Management" are passed on to each responsible party, subcontractor and sub-recipient and that all the clauses under this section entitled "Risk Management Standard Clauses" are included, *mutatis mutandis*, in all sub-contracts or sub-agreements entered into further to this Project Document.

111. **Special Clauses**.

1. The schedule of payments and UNDP bank account details.
2. The value of the payment, if made in a currency other than United States dollars, shall be determined by applying the United Nations operational rate of exchange in effect on the date of payment. Should there be a change in the United Nations operational rate of exchange prior to the full utilization by the UNDP of the payment, the value of the balance of funds still held at that time will be adjusted accordingly. If, in such a case, a loss in the value of the balance of funds is recorded, UNDP shall inform the Government with a view to determining whether any further financing could be provided by the Government. Should such further financing not be available, the assistance to be provided to the project may be reduced, suspended or terminated by UNDP.
3. The above schedule of payments takes into account the requirement that the payments shall be made in advance of the implementation of planned activities. It may be amended to be consistent with the progress of project delivery.
4. UNDP shall receive and administer the payment in accordance with the regulations, rules and directives of UNDP.
5. All financial accounts and statements shall be expressed in United States dollars.
6. If unforeseen increases in expenditures or commitments are expected or realized (whether owing to inflationary factors, fluctuation in exchange rates or unforeseen contingencies), UNDP shall submit to the government on a timely basis a supplementary estimate showing the further financing that will be necessary. The Government shall use its best endeavors to obtain the additional funds required.
7. If the payments referred above are not received in accordance with the payment schedule, or if the additional financing required in accordance with paragraph [] above is not forthcoming from the

Government or other sources, the assistance to be provided to the project under this Agreement may be reduced, suspended or terminated by UNDP.

8. Any interest income attributable to the contribution shall be credited to UNDP Account and shall be utilized in accordance with established UNDP procedures.

In accordance with the decisions and directives of UNDP's Executive Board:

The contribution shall be charged:

(a) [6%] cost recovery for the provision of general management support (GMS) by UNDP headquarters and country offices

(b) Direct cost for implementation support services (ISS) provided by UNDP and/or an executing entity/implementing partner.

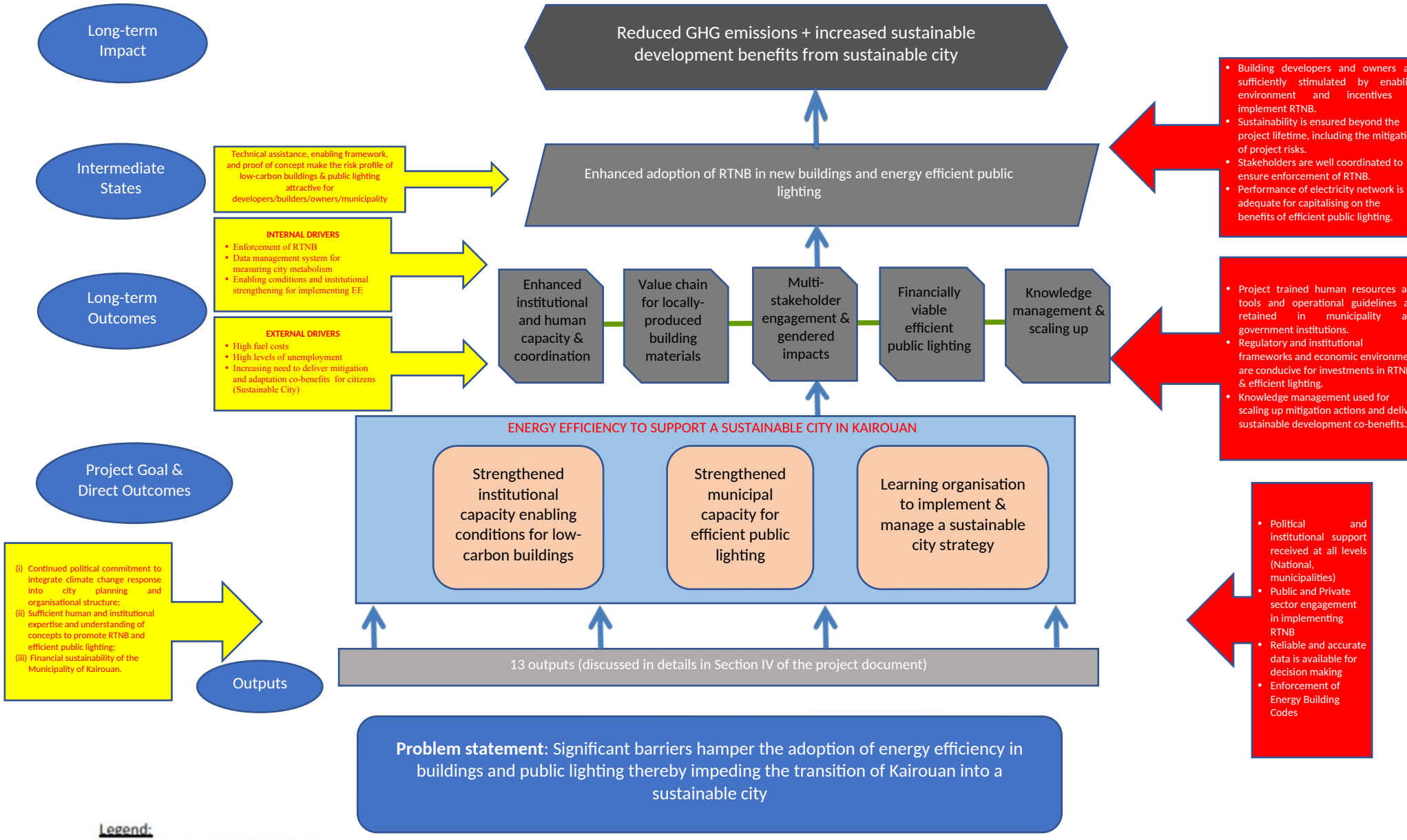
9. Ownership of equipment, supplies and other properties financed from the contribution shall vest in UNDP. Matters relating to the transfer of ownership by UNDP shall be determined in accordance with the relevant policies and procedures of UNDP.

10. The contribution shall be subject exclusively to the internal and external auditing procedures provided for in the financial regulations, rules and directives of UNDP."

XII. MANDATORY ANNEXES

1. Theory of Change Diagram
2. Project Map and geospatial coordinates of the project area
3. GEF Budget template
4. GEF Execution Letter
5. Multiyear Workplan
6. Monitoring Plan
7. Social and Environmental Screening Procedure (SESP)
8. UNDP Atlas Risk Register
9. Overview of technical consultancies/subcontracts
10. Stakeholder Engagement Plan
11. Environmental Social Management Framework (ESMF)
12. Gender Analysis and Gender Action Plan
13. Procurement Plan – for first year of implementation especially
14. GEF focal area specific annexes (e.g. METT, GHG calculations, target landscape profile, feasibility study, other technical reports)
15. Additional agreements: such as cost sharing agreements, project cooperation agreements signed with NGOs (where the NGO is designated as the "executing entity"), letters of financial commitments etc..
16. GEF and/or LDCF/SCCF Core indicators (see template below)
17. GEF Taxonomy (see template below)
18. Partners Capacity Assessment Tool and HACT assessment
19. UNDP Project Quality Assurance Report (to be completed in UNDP online corporate planning system)
20. Signed LOA between UNDP and IP requesting UNDP Support Services (if required on exceptional basis and authorized by the GEF)

Annex 1: Project Theory of Change Diagram



Annex 2: Project map and Geospatial Coordinates of project sites

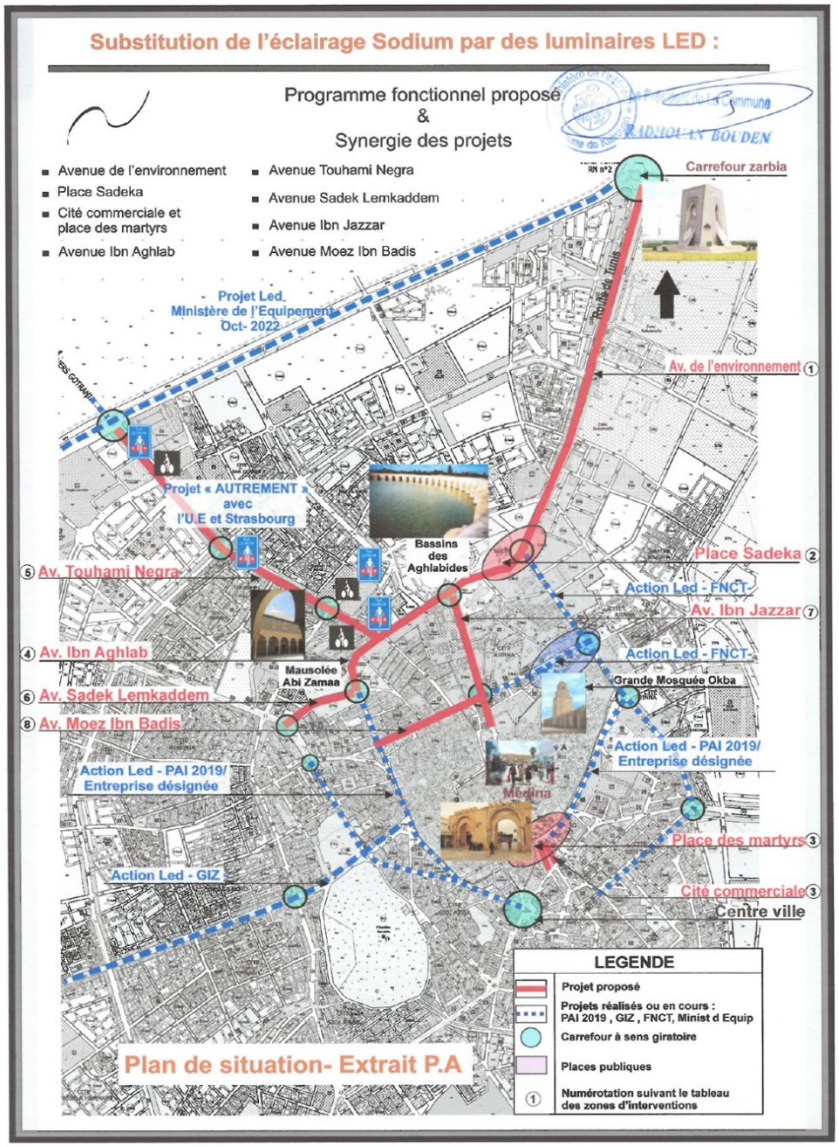
The aerial picture shows the geographical location of the baseline municipal building that will be enhanced (energy efficient through application of the RTNB and ECOBat label) using GEF investments.

Address: Intersection of Avenue 3 septembre 1934 and Avenue 02 Mars 1934 ;

Coordinates: Latitude - 35°40'11.49"N; Longitude - 10° 6'8.16"E.



Map of the City of Kairouan showing the locations for efficient public lighting interventions.


Substitution de l'éclairage Sodium par des luminaires LED :
Programme fonctionnel proposé
&
Synergie des projets
Avenue de l'environnement
Place Sadeka
Cité commerciale et place des martyrs
Avenue Ibn Aghlab
Avenue Touhami Negra
Avenue Sadek Lemkaddem
Avenue Ibn Jazzar
Avenue Moez Ibn Badis
Carrefour zarbia
Projet Led Ministère de l'Equipement Oct- 2022
Av. de l'environnement 1
Projet « AUTREMENT » avec l'U.E et Strasbourg
Bassins des Aghlabides
Place Sadeka 2
Av. Touhami Negra 5
Action Led - FNCT
Av. Ibn Jazzar 7
Av. Ibn Aghlab 4
Action Led - FNCT
Mausolée Abi Zamaa
Grande Mosquée Okba
Av. Sadek Lemkaddem 6
Av. Moez Ibn Badis 8
Action Led - PAI 2019/ Entreprise désignée
Action Led - PAI 2019/ Entreprise désignée
Médina
Place des martyrs 3
Action Led - GIZ
Cité commerciale 3
Centre ville
LEGENDE
Projet proposé
Projets réalisés ou en cours : PAI 2019 , GIZ , FNCT, Minist d Equip
Carrefour à sens giratoire
Places publiques
Numérotation suivant le tableau des zones d'interventions
Plan de situation- Extrait P.A

Annex 3: GEF Budget Template

Annexed separately.

Annex 4: GEF Execution Support Letter

REPUBLIC OF TUNISIA
***
MINISTRY OF ENVIRONEMENT
***

07/02/2023

To: **Pradeep Kurukulasuriya,**
Executive Coordinator & Director
Global Environmental Finance
Bureau for Policy and Programme Support (BPPS)/
Global Policy Network
United Nations Development programme
304 East 45th Street, NY10017, USA

Subject: Letter of Support to request GEF Agency Execution for: Towards a Sustainable City through Energy Efficiency in Kairouan - GEF ID#11040

1. In my capacity as GEF Operational Focal Point for Tunisia, I hereby request UNDP, the GEF implementing agency for the aforementioned project, to also carry out execution services for the above project/program, on an exceptional basis.

2. The execution services provided by UNDP are expected to include:

   - Finance: Payments process; creation of vendor forms; issuing cheques; adjustment of financial transaction; monitoring and reviewing financial transaction; managing vendor profiles; reimbursement of travel expenses.
   - Procurement of goods and services: Review of the terms of reference; preparation of bidding documents; advertising, bidding evaluation, contract issuing and management; issuing the Purchase orders.
   - Travel: Travel Management: travel authorization; ticketing, settlement/reimbursement of travel claims.
   - Human resources: Staff selection and recruitment process (Advertising, Long/Short-listing, Interviews), Staff benefits administration and management, recurrent personnel management services: staff Payroll, banking administration and management

3. The execution services to be provided by the Ministry of Environment are expected to include:

   - Project planning, coordination, management, monitoring, evaluation and reporting. This includes providing all required information and data necessary for timely, comprehensive and evidence-based project reporting, including results and financial data, as necessary. The Implementing Partner will strive to ensure project-level M&E is undertaken by national institutes and is aligned with national systems so that the data used and generated by the project supports national systems.

- Overseeing the management of project risks as included in this project document and new risks that may emerge during project implementation.
- Financial management, including overseeing financial expenditures against project budgets.
- Approving and signing the multiyear work plan.
- Approving and signing the combined delivery report at the end of the year; and,
- Signing the financial report or the funding authorization and certificate of expenditures.

4. Execution activities, including those provided by UNDP will be described in detail in the GEF CEO Endorsement/Approval request and accompanying project/program documents, including the project/program budget.

Sincerely,

Sabria Bnouni

GEF Operational Focal point

Annex 5: Multi Year Work Plan

Outcomes	Outputs	Activities	Year 1				Year 2				Year 3			
			Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Outcome 1 Institutional strengthening with enabling conditions, methodologies and tools for enforcing regulatory framework for new buildings	Output 1.1: An operational manual for new building licensing and permits process in the Municipality of Kairouan to operationalise RTNB is designed and adopted	Activity 1.1.1: Review of existing licensing and permits process												
		Activity 1.1.2: Operational manual formulated												
		Activity 1.1.3: Training of staff												
	Output 1.2: Enforcement capabilities of municipality strengthened for ensuring new building compliance with RTNB	Activity 1.2.1. Cross-institutional enforcement mechanism established												
		Activity 1.2.2. Cross-institutional enforcement mechanism operationalised												
	Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials	Activity 1.3.1. Strategy and action plan												
		Activity 1.3.2. Characterisation of the properties of bricks												
	Output 1.4: An online tool for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology is developed and disseminated	Activity 1.4.1. Assess the functional needs of end users												
		Activity 1.4.2. Design and validate the tool												
		Activity 1.4.3. Dissemination of tool												
	Output 1.5: National standards for building materials are developed and adopted	Activity 1.5.1. Standard for bricks used for building envelope												
		Activity 1.5.2.												
	Output 1.6: Towards a net-zero municipal building through energy efficiency and adoption of the ECOBat label	Activity 1.6.1. ESIA for the building extension												
		Activity 1.6.2. Enhancing energy efficiency of municipal building												
		Activity 1.6.3. Process to get ECOBat label												

Outcome 2 Institutional strengthening for the implementation of an efficient municipal lighting system	Output 2.1: Capacity of municipality staff to implement efficient lighting projects strengthened	Activity 2.1.1. Training provided to municipality technicians for the adoption of efficient public lighting														
		Activity 2.1.2. Mechanism for the safe disposal of lamps														
	Output 2.2: National standards for public lighting technologies developed and adopted	Activity 2.2.1. Standards for LED lamps														
		Activity 2.2.2.														
	Output 2.3: Energy efficient municipal lighting demonstrated	Activity 2.3.1. Environmental and Social Management Plan														
		Activity 2.3.2. Procurement of LED lamps														
		Activity 2.3.3. Replacement of SHP lamps with LEDs														
Outcome 3 Municipal learning organisation capable to implement and manage sustainable city strategy	Output 3.1 Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, (iii) Terminal Evaluation carried out	Activity 3.1.1. Inception workshop														
		Activity 3.1.2. Monitoring and Evaluation														
		Activity 3.1.3. Terminal Evaluation														
	Output 3.2: Data management system for benchmarking Kairouan City sustainability established and operational	Activity 3.2.1. Establish a coordinating structure for data collection														
		Activity 3.2.2. Establish data management system														
		Activity 3.2.3. Supporting process of WCCD membership														
	Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries	Activity 3.3.1. Knowledge product on lessons learned														
		Activity 3.3.2. Adhesion to PANORAMA														
	Output 3.4: Replication plan for scaling up energy efficient buildings and public lighting in Tunisia	Activity 3.4.1. Replication plan														
		Activity 3.4.2. Bankable concept note for climate finance														

	developed														

Annex 6: Monitoring Plan

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods[40]	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
Project objective: *To promote green buildings and efficient lighting in alignment with the objectives of the enhanced Nationally Determined Contribution, the Sustainable Development Strategy of Kairouan and the National Programme for Sustainable Cities in Tunisia.*	*Indicator 1:* *# direct project beneficiaries disaggregated by gender (individual people)*	*MT: n/a* *EoP:* *187,000 (94,248 female; 92,752 male)*	*The direct beneficiaries will be the entire population of the city of Kairouan who will use the municipal building at Keblia and efficient street lights that will be installed across the city as per map shown in Annex 2.*	*Official demographic statistics of the City of Kairouan and of the National Institute of Statistics (INS)*	Annually Reported in DO tab of the GEF PIR	*National Institute of Statistics* *Project consultant carrying out annual lessons learned reports*	*National statistics report*	*Risks:* *Political and institutional instability prevents the minimum governance structures to be put in place for project implementation; low level coordination between project partners; changes in the deployment plan of LED lights in the city that results in geographical concentration compared to the geographical spread shown in Annex 2; delays in completion of municipal building in Keblia.* *Assumptions:* *The investments proposed under Output 1.6 and 2.3 are successful; continued political and institutional commitment from all project stakeholders; mechanism for enforcing the application of RTNB is adequate*
	Indicator 2: *Greenhouse gas emissions mitigated*	*MT: n/a* *EoP: Direct:*	*The expected direct and indirect emission reductions from*	*Efficient lighting: The Municipality will launch a bid for procurement of LED*	Annually Reported in	*Project Manager with support from project consultant*	*Calculations carried out by international consultant*	*Risks: Same as for Indicator 1* *Assumptions: Same as for Indicator*

[40] Data collection methods should outline specific tools used to collect data and additional information as necessary to support monitoring. The PIR cannot be used as a source of verification.

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
	(cumulative metric tons of carbon dioxide equivalent (tCO2e))	*20,219 tCO2e* *Indirect: 31,064 tCO2e*	*the generation of renewable electricity as calculated in Annex 13 (Part A) and the GEF GHG measurement methodology for energy efficiency measures*	*lights with general specifications given in Table 5 in the Project Document. The Municipality will record the type and location of lamps replaced, and it will use its calibrated metering system that is disaggregated by lighting sectors for measuring changes in electricity consumption before and after lamp replacements.* *Municipal building: The performance of the enhanced building will first be based on the CLIP software model, and compared with a building without energy efficiency measures. Building electricity consumption will be monitored during the project lifetime, and a comparison made with the consumption of a old typical municipal building*	DO tab of the GEF PIR	*carrying out annual lessons learned reports*	*carrying out Terminal Evaluation*	*1*
Project Outcome 1:	*Indicator 3:* *Number of*	*MT: n/a*	*A mechanism will be established to support the*	*Data will be collected through regular surveys that will form*	Annually	*Project Manager with support from project*	Project reports and reports to/minutes of	*Risks:* *Competing demands on the time of*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
Institutional strengthening with enabling conditions, methodologies and tools for enforcing regulatory framework for new buildings	*municipal staff with competencies to apply operational guidelines for enforcement of RTNB (40% women / 60 % men)*	*EoP: 20 (12 men + 8 women)*	*enforcement of RTNB with a combination of capacity building (Output 1.2) and development of an operational manual for the licensing and permits process (Output 1.1)*	*part of the annual lessons learned studies.*	Reported in project reports and also in DO tab of the GEF PIR	*consultant carrying out annual lessons learned reports*	the PSC	*municipality staff because of several parallel initiatives leads to low level participation in capacity building activities* *<u>Assumptions</u>:* *Low turnover of municipality staff ensures institutional and human capacity strengthening*
	Indicator 4: *# strategic plan for revitalizing the value chain for locally-fabricated bricks*	*MT: n/a* *EoP: 1*	*Output 1.3 will develop a strategic and action plan for reinvigorating the value chain of locally-fabricated bricks, including a focus on livelihoods and gender while taking into account risks (Annex 6)*	*Publication and dissemination of the strategic and action plan, and captured in the process of producing lessons learned report.*	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	Project reports and reports to/minutes of the PSC	*<u>Risks</u>: Same as for Indicator 1.* *<u>Assumptions</u>: Same as from Indicator 1.*
Project Outcome 2: *Institutional strengthening for the implementation of an efficient municipal*	*Indicator 5:* *Number of persons with enhanced capacity to implement efficient municipal lighting projects (50%*	*MT: n/a* *EoP: 10 (5 women + 5 men)*	*Municipal staff will be capacitated to design, cost and implement efficient lighting projects.*	Project reports on attendance and participation of municipal staff in training that will include a competency test at the end of the training.	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	Project reports and reports to/minutes of the PSC	<u>Risks</u>: As for Indicator 3. <u>Assumptions</u>: As for Indicator 3.

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
lighting system	*women + 50% men)*							
	Indicator 6: *Number of standards adopted for LED used in public lighting*	*MT: n/a* *EoP: 1*	*A national standard will be developed for LEDs used for public lighting (Output 2.2)*	*Adoption and publication of the national standards, and captured in the process of producing lessons learned report.*	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	Project reports and reports to/minutes of the PSC	Risks: Standard is developed through a parallel initiative by the time the UNDP-GEF project is implemented in Q4 (2023) or Q1 (2024). Assumptions: Technical group for developing or adopting international standard is established;
Project Outcome 3: *Municipal learning organisation capable to implement and manage sustainable city strategy*	*Indicator 7:* *Number of data management system in place for collecting city-wide data for tracking city sustainability performance according to ISO37120 (number of data management systems)*	*MT: 0* *EoP: 1*	None of the cities in Tunisia have data management system to allow then to track and report on multi-dimensional sustainable city performance. The City of Kairouan will be supported to put in place a data management system for the adoption of ISO37120.	Membership of the Municipality of Kairouan to the WCCD, and reporting on indicators done through operational data management system established under Output 3.2 as captured by the process of producing lessons learned reports.	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	Project reports and reports to/minutes of the PSC	Risks: Lack of internal coordination and participation of persons from different departments and divisions within the municipality; changes in leadership (e.g. Mayor) that dampens leadership for the Municipality of Kairouan to become member of the WCCD Assumptions: Low turnover of municipal staff; fruitful South-South cooperation, especially with the City of Amman that is already a member of the WCCD
	Indicator 8: *(a) Number of replication plan, including detailed*	*MT: n/a* *EoP: 1 (for both indicators)*	*Based on lessons learned (Output 3.3) and knowledge shared through South-South Cooperation, a*	*Validation, publication and dissemination of the Replication Plan; submission of the Concept Note to the GCF as captured in the process of lessons*	Annually Reported in project reports and also in DO	*Project Manager with support from project consultant carrying out annual lessons*	Project reports and reports to/minutes of the PSC	*Risks: Same as for Indicator 1.* *Assumptions: Same as from Indicator 1.*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
	budget, for scaling up the deployment of energy efficient buildings and public lighting based on lessons learned (number of plans) *(b) # of bankable concept notes developed to leverage international climate finance through the GCF*		*replication plan will be developed for scaling up the energy efficiency actions across other cities in Tunisia; a bankable Concept Note for the GCF will be developed*	*learned reporting*	tab of the GEF PIR	*learned reports*		
Add indicators included in gender action plan, stakeholder engagement plan or other monitoring plans as needed.								
Output 1.3: A strategic plan is developed for developing a local value chain for sustainable and low-carbon building materials	*% of people involved in the development of the strategy are women*	*MT: n/a* *EoP: At least 30% of the people involved in the development of the replication plan are women*	*Ensuring that a gender-differentiated approach is adopted in developing the strategic plan*	*Data will be collected through regular surveys that will form part of the annual lessons learned studies.*	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	*Project reports and reports to/minutes of the PSC*	*<u>Risks</u>: The project replicates gender biases and gender-based discrimination as per Risk 10 in UNDP Risk Register (SESP – Risk 2)* *<u>Assumptions</u>: The GAP (Annex 10) is implemented as an integral part of the project*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
Output 3.1: Monitoring and Evaluation (M&E) and Reporting, including (i) Conducting Inception workshop and preparing report, (ii) Ongoing M&E, and (iii) Terminal Evaluation carried out	*M &E reports include the integration of quantitative and qualitative information on gender*	*MT: n/a* *EoP: At least 60% of indicators are sex disaggregated* *100% of M & E reports integrate quantitative and qualitative information on gender*	*M & E Reports prepared to discuss key findings and recommendations integrates gender and social inclusion considerations* *Development and integration of gender-sensitive indicators and collection of sex-disaggregated data for monitoring and evaluating project results.*	*Data will be collected through regular surveys that will form part of the annual lessons learned studies.*	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	*Project reports and reports to/minutes of the PSC*	*<u>Risks</u>: The project replicates gender biases and gender-based discrimination as per Risk 10 in UNDP Risk Register (SESP – Risk 2)* *<u>Assumptions</u>: The GAP (Annex 10) is implemented as an integral part of the project*
Output 3.3: Lessons learnt, experiences and best practices related to the project are compiled and disseminated in other cities of Tunisia and MENA countries	*% of project participants who are women*	*MT: n/a* *EoP: At least 50% of the people involved in the development and dissemination of experiences are women.* *At least 50% of recipients of*	*Improve understanding of gender issues related to the energy efficient technology, including access to public lighting, capacity building needs of women and policy issues at the municipal level related to energy efficient technologies, among others. This will inform*	*Data will be collected through regular surveys that will form part of the annual lessons learned studies.*	Annually Reported in project reports and also in DO tab of the GEF PIR	*Project Manager with support from project consultant carrying out annual lessons learned reports*	*Project reports and reports to/minutes of the PSC*	*<u>Risks</u>: The project replicates gender biases and gender-based discrimination as per Risk 10 in UNDP Risk Register (SESP – Risk 2)* *<u>Assumptions</u>: The GAP (Annex 10) is implemented as an integral part of the project*

Monitoring	Indicators	Targets	Description of indicators and targets	Data source/Collection Methods	Frequency	Responsible for data collection	Means of verification	Risks/Assumptions
		information generated including lessons learned are women	*the output on lessons learned and disseminated.*					
Environmental and Social risks related studies, as relevant[41].	N/A at this stage	N/A at this stage	N/A at this stage	As per ESMF	Annually	Project Manager with the support of Project Coordinator	Annual lessons learned reports and project terminal evaluation report	Risks: Grievance mechanism not publicized / introduced among project partners; Assumptions: ESIA and SESA proposed in the ESMF and integrated in Outputs 1.3, 1.6, 2.3 and 3.4 applied adequately; Stakeholder Engagement Plan implemented adequately

[41] See Section VI and Annex 9: Environmental and Social Management Framework in the ProDoc.

Annex 7: UNDP Social and Environmental Screening Procedure (SESP)

Social and Environmental Screening Template (2021 SESP Template, Version 1)

The completed template, which constitutes the Social and Environmental Screening Report, must be included as an annex to the Project Document at the design stage. Note: this template will be converted into an online tool. The online version will guide users through the process and will embed relevant guidance.

Project Information

Project Information	
1. Project Title	Towards a Sustainable City through Energy Efficiency in Kairouan
2. Project Number (i.e. Atlas project ID, PIMS+)	6686
3. Location (Global/Region/Country)	Tunisia
4. Project stage (Design or Implementation)	Design
5. Date	16 February 2023

Part A. Integrating Programming Principles to Strengthen Social and Environmental Sustainability

QUESTION 1: How Does the Project Integrate the Programming Principles in Order to Strengthen Social and Environmental Sustainability?
Briefly describe in the space below how the project mainstreams the human rights-based approach
The project is based on, and specifically responds to, a request from the Municipality of Kairouan and the Directorate of Sustainable Development of the Ministry of Environment to develop energy efficiency measures in new buildings and public lighting. The City of Kairouan has started implementation of a city-wide sustainable development strategy, which provides an overall vision for the growth of the city until 2030 with a clear overarching focus on climate-resilient development, the creation of green jobs, and a strive for resource efficiency in all aspects of municipal planning and investments. The interventions proposed in the GEF funded project are also fully supportive of the updated Nationally Determined Contribution that seeks to have large investments in economy-wide energy efficiency. Since 77% of interventions are conditional upon financial support, the GEF funding augurs well. Against a backdrop of deteriorating socioeconomic conditions in Tunisia, the socioeconomic situation in the City of Kairouan is below the country average. For instance, the unemployment rate is 18.7% and is higher among the female population at 28.25% compared to 14.04% for the male population. The illiteracy rate of 17.64% is also higher than the national average of 13%. Kairouan also have a population growth rate of 1.6% per annum that exceeds the national average of 1%. The project is implemented in a city characterized by an unfavorable socio-economic situation. The project will seek to revitalize the local value chain for ecological building materials that has potential for new job creation. By catalyzing investments in energy efficiency measures, the local business value chain for energy efficient materials and supplies will be enhanced with further potential for job creation. By investing in energy efficiency in municipal buildings and public lighting, the savings on energy bill by the municipality can be invested in the provision of other social services

to improve the socio-economic condition in the City of Kairouan. The overall project will contribute to improve the quality of life of the people.

Briefly describe in the space below how the project is likely to improve gender equality and women's empowerment

As mentioned above, the adverse socio-economic situation in the City of Kairouan has strong gender biases against the female population. First, all institutional capacity building that will be provided by the project will ensure that a minimum 40% of women are beneficiaries. Job creation will take place mostly by re-invigorating the construction of locally available ecological building materials. The project will put in place a strategy, including appropriate training, to ensure that at least 40% of jobs created are for women. The replication plan that will be developed under Component 3 of the project will be gender-differentiated. Also, dissemination of lessons learned and knowledge products will be gendered. A detailed Gender Analysis and Gender Action Plan has been prepared to ensure that gender issues have been mainstreamed throughout project activities..

Briefly describe in the space below how the project mainstreams sustainability and resilience

The project is a GEF-funded climate change mitigation project and therefore environmental sustainability is the core of the project objective: "To promote green buildings and efficient lighting in alignment with the objectives of the enhanced Nationally Determined Contribution, the Sustainable Development Strategy of Kairouan and the National Programme for Sustainable Cities in Tunisia".

The building sector is the main final energy consumer sector (37% in 2019, including biomass-energy). It is also involved through the building materials industries which are among the most GHG-emitting sectors. Taken as a whole, the building sector would represent – from upstream to downstream – around 50% of final energy demand in Tunisia, and at least 55% of GHG emissions attributable to energy in 2019. Out of the 13.8 MtCO2 emission reductions to 2030 (relative to the business-as-usual scenario) expected from the energy sector, ~2MtCO2 would be obtained from energy efficiency in buildings. The selection of Kairouan as pilot city also stems from the fact that it already implementing a sustainable energy plan (PAED), implying strong buy-in and ownership of the proposed project by the local government. Based on projections the total emissions were expected to reach 402,762 tCO2 by 2020.

The proposed project will assist the Municipality of Kairouan to achieve a low-carbon pathway via targeted energy efficiency interventions in the municipal buildings and street lighting sectors. The building sector has been identified as having the largest potential for energy efficiency (EE) interventions in the country, and as such is the main focus of this project. Further the project will provide more general support for the application of low-carbon planning and performance tools building off the existing Sustainable Development Strategy for the City of Kairouan under the ambit of the National Programme for Sustainable Cities in Tunisia, and informed by lessons learned from the GEF-financed Sustainable Cities IAP and other projects (e.g. Jordan and Iraq).

The project will also enhance sustainable development through minimising dependence on imported energy, minimising energy costs to the municipality and the larger economy, creating new employment opportunities and improving the local environment. The sustainable development and socioeconomic development impacts of the project will be substantial and multi-faceted. Potential indicators to be considered in the final results framework include:

- Performance of city (measured ex ante and ex post) based on ISO 37120
- Direct and indirect reductions in greenhouse gases
- Energy intensity/improved efficiency of buildings
- Total number of direct and indirect beneficiaries; Number of beneficiaries relative to total population
- Economic co-benefits – aggregate annual monetary savings (US$ equivalent) to the municipal budget from EE public street lighting and more efficient energy consumption in public buildings and public lighting.

Briefly describe in the space below how the project strengthens accountability to stakeholders

A multi-stakeholder process was used to design the project. Hence, the submission of the proposed PIF was approved by the key project stakeholders including the Municipality of Kairouan, the Directorate of Sustainable Development (Ministry of Environment) and the National Agency for Energy Conservation (ANME). A detailed stakeholder analysis and Stakeholder Engagement Plan (SEP) has been developed to fully account for the needs and expectations of national and local stakeholders. The SEP provides for the modalities for the project to engage with the stakeholders. Importantly, the project Steering Committee (PSC) will oversee project implementation. The composition of the PSC will be informed by the stakeholder analysis, and it will be the main institutional arrangement for project accountability to stakeholders. Due care will be exercised to ensure participation of all stakeholders in project activities. The SEP also provides a grievance mechanism that any stakeholder that feels aggrieved by the project may use to seek redress.

Part B. Identifying and Managing Social and Environmental Risks

QUESTION 2: What are the Potential Social and Environmental Risks? *Note: Complete SESP Attachment 1 before responding to Question 2.*	**QUESTION 3: What is the level of significance of the potential social and environmental risks?** *Note: Respond to Questions 4 and 5below before proceeding to Question 5*			**QUESTION 6: Describe the assessment and management measures for each risk rated Moderate, Substantial or High**
Risk Description (broken down by event, cause, impact)	***Impact and Likelihood (1-5)***	***Significance (Low, Moderate Substantial, High)***	***Comments (optional)***	***Description of assessment and management measures for risks rated as Moderate, Substantial or High***
Risk 1: Poor communities are economically affected by new building licensing and permits process Related to: • Human Rights; P.2, P.3, P.5 • Accountability; P.13, P.14 • Standard 6: Indigenous Peoples; 6.1	I = 4 L = 1	Low	The Governorate of Kairouan is considered the 3rd poorest governorate in Tunisia whose poverty rate recorded in 2015 is 29.3%, which is significantly higher than the national average rate (15.3%)[42]. The introduction of an operational manual for new building licensing and permits	The RTNB applies only to commercial buildings and apartment blocks and therefore the operational manual would be limited to these. Nevertheless, the Stakeholder Engagement Plan (SEP) that has been prepared for the project will be used to give priority to community engagement to ensure that No-one is Left Behind. This will imply a proactive attitude to reach out to vulnerable people and groups and consider their concerns.

[42] INS & BM. (2020). Map of Poverty in Tunisia, September 2020. Retrieved from http://www.ins.tn/publication/carte-de-la-pauvrete-en-tunisie-septembre-2020 (Accessed 4 January 2023)

			(Output 1.1) may affect poor communities, including 'local populations', who might be economically disadvantaged by these regulations through the need to use expensive building materials, fixtures or equipment. However, it is pointed out that the RTNB is not applicable to all buildings, and it only covers commercial buildings and collective residential buildings (apartment blocks). In 2019, only 11% of residential buildings were covered by the existing regulation (RTNB). Note: 'Local populations' meet the criteria of SES Standard 6, but are not referred to as 'Indigenous populations' in the Tunisian context. Both the Stakeholder Engagement Plan (Annex 8) and the Gender Action Plan (Annex 10) provide description of how local populations are engaged.	
Risk 2: Limiting women's opportunities to benefit from the project Related to:	I = 3 L = 3	Moderate	There is low level participation of women across the entire energy efficiency value chain for buildings and public lighting.	The project Gender Action Plan contains actions to contribute towards overcoming gender-related disparities through four considerations, namely: (i) facilitating equitable participation, (ii) ensuring male engagement to prevent backlash against women, (iii) leveraging existing initiatives, and (iv) differing levels of

• Gender Equality and Women's Empowerment; P.10 • Standard 7: Labour and Working Conditions; 7.5			Therefore, their opportunities to benefit from the project may be limited by restricting or prohibiting them from engaging in the policy and decision making or in the design and management of an operational manual for new building licensing and permits and national standards for building materials. The project will also create new green jobs and will conduct training and awareness campaigns. Women may not be well represented in these activities and may not receive the training and awareness raising offered.	literacy and education. The project will make use of gender-disaggregated indicators (Gender Action Plan and Project Results Framework). The project will optimize the use of gender-responsive training materials to project stakeholders and project interventions
Risk 3: Discrimination against and marginalization of vulnerable communities due to implementation of the replication plan for scaling up energy efficient buildings and public lighting Related to: • Human Rights; P.5 • Accountability; P.13, P.14	I = 3 L = 3	Moderate	Stakeholders, including vulnerable groups and Local Populations, who may be affected by or benefit from new energy-efficient buildings and public lighting may be marginalized and unable to voice their concerns, thereby not benefiting from the project and the replication plan or even being negatively affected by it.	As mentioned in the Environmental and Social Management Framework (ESMF), targeted assessments will be conducted to consider potential social and environmental impacts associated with the replication plan for scaling up energy efficient buildings and public lighting in Tunisia (Output 3.4) and the Strategy and Action Plan that will be developed under Output 1.3 to enhance the value chain of locally-fabricated bricks. The SEP has been developed to fully account for the needs and expectations of national and local stakeholders. In addition, the SEP provides for modalities for engagement of all stakeholders including vulnerable

			While 'local populations' are not located in the City of Kairouan and are therefore not directly impacted by project activities, there is the potential that the 'local populations' who are found mainly in the southern parts of Tunisia will be impacted during replication of the project.	groups and 'local populations'. The SEP includes a grievance mechanism that any stakeholder that feels aggrieved by the project may use to seek redress.
Risk 4: Improper storage, transport and disposal of generated waste and hazardous material from retrofitting of municipal building and public street lighting can have an adverse effect on community health and the environment Related to: • Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management; 1.1, 1.7 • Standard 3: Community Health, Safety and Security; 3.1, 3.2, 3.4, 3.5, 3.6, 3.7 • Standard 8: Pollution Prevention and Resource Efficiency; 8.1, 8.2, 8.6	I = 3 L = 3	Moderate	The municipal building is located in a mixed zone area where there are commercial activities and residences. Community health and safety might be affected due to the improper transport, storage, and/or disposal of hazardous material and waste resulting from the replacement of non-EE items in the municipal building. Chemicals and hazardous materials can include paints for refurbishing of walls and facades, low-pressure mercury-vapor gas-discharge lamps (fluorescents), and use of new insulation materials. There is one controlled landfill within the governorate of Kairouan. This	A targeted assessment will be undertaken for the rehabilitation works of the municipal building (Output 1.6) to address this risk. The assessment will result in an Environmental and Social Management Plan (ESMP) that will include a Waste Management Plan that adopts international best practices. In Tunisia, the collection, transport, treatment and dumping of hazardous and special waste is regulated. The list of companies authorized by the Minister of the Environment for the management of hazardous waste is available on the ANGed website[44] and will be used to contract the company if hazardous waste needs to be disposed.

			landfill is located 11 km west of the city of Kairouan. The landfill treats an average of 165 tons of waste per day[43].	
Risk 5: Project outputs are impacted by climate change and natural disasters Related to: • Human Rights: P.7 • Standard 2: Climate Change and Disaster Risks; 2.1, 2.2, 2.3	I = 2 L = 2	Low	The ambient temperature in Tunisia is expected to increase in the future due to climate change. As discussed, energy building codes related specifically to building envelope insulation will be an effective form of adaptation to this climate impact. The accentuated impacts of future climate change are expected to increase political interest in addressing the drivers of such change through large-scale implementation of building codes that will lead to scaling-up of mitigation actions.	Measures that have been incorporated into this project include; 1) introduction of climate-responsive building techniques and elements to reduce the effect of heat and reduce demand on energy for cooling (under Output 1.3 that will support the uptake of locally-fabricated bricks, as well as characterize the thermos-physical properties of these bricks that are typically used in the maintenance and renovation of buildings of cultural heritage value, such as the Medina); 2) raising awareness on the long-term benefits of energy efficiency (Output 1.4 will develop an interactive tool to quantify the benefits of investments in energy efficiency in buildings); 3) application of building energy codes and support to choose building insulation to reduce energy demand (Output 1.1 and Output 1.2 with demonstration under Output 1.6). A site-specific ESMP (informed by a targeted assessment) will be conducted to inform the retrofitting of the municipality building (Output 1.6) and will assess this risk and recommend measures that ensure that the construction works do not result in maladaptive practices.
Risk 6: Community health and safety risks from noise and air emissions during retrofitting of municipal building and replacement of street	I = 3 L = 2	Moderate	Increased noise level (noise from the mechanical machinery and equipment, vehicles, removal of the	As mentioned in the ESMF, a targeted assessment will be undertaken for the rehabilitation works at the municipal building (Output 1.6). This assessment will tackle this risk and propose mitigation measures in the resulting site-

[43] Environmental Assessment and Management (EAM). (2022). Environmental and Social Impact Assessment (ESIA) 225 kV Overhead Transmission Line (OHTL) 100 MW Kairouan Solar Power Project – Tunisia. Retrieved from https://www.afdb.org/sites/default/files/environmental-social-impact-assessment-ohtl-en.pdf (Accessed 4 January 2023)

[44] ANGed (2022). November 2022: List of authorized hazardous waste companies. Retrieved from http://www.anged.nat.tn/Novembre_2022Liste_societes_autorisees_dechets_dangereux.html (Accessed 4 January 2023)

lighting Related to: • Standard 3: Community Health, Safety and Security; 3.1, 3.2,			old buildings equipment, etc.) may occur during the construction works at the municipal building Air pollution may be caused by emissions from vehicles, dismantling of the old equipment and constructions, transportation of demolition / construction materials and also during the final interior and exterior works. The types of infrastructure that are found in the vicinity of public lighting in the project boundary are: educational institutions (primary and secondary schools; public and private); health institutions (hospitals and clinics; public and private); residential houses; markets; commercial buildings (offices, small businesses); municipal buildings.	specific ESMP. The targeted assessment for the street light replacement activity (Output 2.3) will analyze this risk and propose mitigation measures to reduce it. These measures will be incorporated in the ESMP that will be developed.
Risk 7: Emissions from local production of traditional construction materials resulting from implementation of the strategic plan for the value chain Related to:	I = 3 L = 4	Moderate	Used tires and shoes are currently burned in furnaces to fire the bricks. There are residential areas that are estimated to be within 1-2 km of the brick production site, and depending on wind	As outlined in the ESMF, a targeted and strategic assessment to consider potential social and environmental impacts will be undertaken for the strategic plan for development of a value chain for the local production of traditional construction materials (Output 1.3) that will address this issue to ensure proposed solutions do not lead to air pollution.

• Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management; 1.1, 1.7 • Standard 3: Community Health, Safety and Security; 3.2, 3.4, 3.5, 3.6 • Standard 8: Pollution Prevention and Resource Efficiency; 8.1, 8.2, 8.3			direction, there is possibility that these households are affected by the smokes and fumes from furnaces if alternative methods are not proposed.	
Risk 8: Improper disposal of obsolete lamps after replacement with LEDs can release pollutants into the environment Related to: • Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management; 1.1, 1.7 • Standard 3: Community Health, Safety and Security; 3.2, 3.4, 3.5, 3.6 • Standard 8: Pollution Prevention and Resource Efficiency; 8.1, 8.2, 8.3	I = 3 L = 3	Moderate	An estimated 574 units of sodium lamps will be replaced with LEDs (Output 2.3). High pressure sodium lamps contain mercury[45] and therefore if improperly disposed, would like to mercury contamination in nearby water bodies, air and soil.	A targeted assessment will be undertaken prior to commencement of the replacement activities (Output 2.3) to analyze this risk and develop a Waste Management Plan (as part of the ESMP) that will ensure safe storage or disposal of these lamps. This may include storing them at public storage facilities to be reused elsewhere.
Risk 9: Improper disposal of hazardous waste (obsolete lamps and construction waste) resulting from implementation of the replication plan for scaling up energy efficient buildings and public lighting leading to an adverse impact on the environment and public health Related to:	I = 3 L = 3	Moderate	If not properly disposed, hazardous waste can have adverse impacts on the environment and public health due to the lack of adequate solid waste infrastructure in Tunisia[46].	As outlined in the ESMF, a targeted assessment will be developed for the replication plan for scaling up energy efficient buildings and public lighting (Output 3.4) that will consider environmentally feasible alternatives to the disposal of obsolete lamps and construction waste, including potentially exporting them in line with the Basel Convention.

[45] EPA (2016). Basic Information about Recycling Mercury-Containing Light Bulbs (Lamps). Retrieved from https://archive.epa.gov/epawaste/hazard/web/html/basic.html (Accessed 4 January 2023)

[46] Middle East Institute (2022). Solving Tunisia's growing waste management problem. Retrieved from https://www.mei.edu/publications/solving-tunisias-growing-waste-management-problem (Accessed 4 January 2023)

• Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management; 1.1, 1.7 • Standard 3: Community Health, Safety and Security; 3.2, 3.4, 3.5, 3.6 • Standard 8: Pollution Prevention and Resource Efficiency; 8.1, 8.2, 8.3				
Risk 10: Damage to biodiversity, natural resources and cultural heritage sites due to local brick production Related to: • Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management; 1.1, 1.7 • Standard 3: Community Health, Safety and Security; 3.2, 3.4, 3.5, 3.6 • Standard 4: Cultural Heritage; 4.1 • Standard 8: Pollution Prevention and Resource Efficiency; 8.1, 8.2, 8.3	I = 3 L = 2	Moderate	The location of brick fabrication activities is not close to environmentally sensitive areas, including biodiversity protected areas, wetlands, cultural heritage sites and water sources. However, the types of brick that may be produced after implementing the strategic plan for developing a value chain for the local production of traditional construction materials (Output 1.3) may have an impact on the cultural heritage value of the city of Kairouan if not in line with aesthetics of the existing architecture.	Tunisia has important mining resources that constitute the raw materials for the manufacture of the Tozeur brick and plaster. The UNDP-GEF project will draw from lessons learned from those two experiences to propose a strategy and action plan for the enhancement of the local value chain for bricks produced in Kairouan. The strategic plan for development of the value chain will consider these risks and integrate measures to avoid and minimize such risks. As outlined in the ESMF, a targeted and strategic assessment will be conducted to ensure potential downstream social and environmental impacts are considered in the development of the strategic plan for developing a value chain for the local production of traditional construction materials (Output 1.3).
Risk 11: Project activities could inadvertently impact 'local population' Related to: • Standard 6: Indigenous Peoples; 6.1	I = 1 L = 1	Low	The SESP has identified that there are 'local populations' present in Tunisia. However, Standard 6 is not triggered given that impacts on the human rights, lands, natural resources, territories, and traditional livelihoods of 'local populations' are not	

			anticipated given the nature of project activities (including scaling up energy efficient buildings and public lighting). Public street lighting activities will be undertaken throughout the city of Kairouan (Output 2.3) during project implementation and throughout Tunisia during implementation of the replication plan for scaling up energy efficient buildings and public lighting (Output 3.4) that will be prepared by the project. While these activities may take place in areas claimed or used by "local population", as they only include replacement of light bulbs, they are not expected to have any significant impact on these populations. The activity has a small footprint and can be done in a very short period.	
Risk 12: Working conditions for project activities do not meet national labour laws and international labour standards Related to: • Standard 7: Labour and Working Conditions; 7.1, 7.2, 7.3, 7.5	I = 3 L = 2	Moderate	Workers involved in the retrofitting of the building and replacement of public light bulbs may be exposed to working conditions not aligned with labour laws and health and safety standards. Forced labor and child labor are also possible.	The targeted assessment that will be undertaken for the construction works at the new municipal building (Output 1.6) will assess the risks to workers such that mitigation measures will be incorporated into the resulting site-specific ESMP that will include an Occupational Health and Safety Plan. The targeted assessment for the replacement of public light bulbs (Output 2.3) will assess electrocution, falling

			The national law provides for the right of workers to organize, form, and join unions, and to bargain collectively. The law also prohibits and criminalizes all the worst forms of child labor and provides for a minimum age of employment[47].	and other occupational health and safety risks and proposed measures to mitigate them, which shall be incorporated into an Occupational Health and Safety Plan as part of the overall ESMP for this activity. In addition, Labour Management Procedures (LMP) will be developed for the project to ensure that labour standards and rights are upheld for project workers and for workers that are recruited by contractors working on various activities. Details on the LMP are provided in the ESMF.
Risk 13: Occupational health and safety risks associated with local brick production Related to: • Standard 7: Labour and Working Conditions; 7.1, 7.6	I = 3 L= 2	Moderate	The operations are carried out manually using unskilled labour. There is no investment to improve working conditions such as the use of protective equipment (PPE).	As outlined in the ESMF, a targeted and strategic assessment to consider potential downstream social and environmental impacts will be conducted to inform the strategic plan for developing of value chain for the local production of traditional construction materials (Output 1.3) which will ensure that the issue of occupational health and safety is considered when assessing the options.

	QUESTION 4: What is the overall project risk categorization?		
	Low Risk	☐	
	Moderate Risk	**X**	
	Substantial Risk	☐	
	High Risk	☐	
	QUESTION 5: Based on the identified risks and risk categorization, what requirements of the SES are triggered? (check all that apply)		

[47] US Department of State. (2022). Tunisia 2021 Human Rights Report. Retrieved from https://www.state.gov/wp-content/uploads/2022/03/313615_TUNISIA-2021-HUMAN-RIGHTS-REPORT.pdf (Accessed 4 January 2023)

	Question only required for Moderate, Substantial and High Risk projects				
	Is assessment required? (check if "yes")	**X**			***Status? (completed, planned)***
	if yes, indicate overall type and status		**X**	Targeted assessment(s)	Completed (Gender and Stakeholder Assessment), Planned to inform outputs 1.3, 1.6, 2.3, 3.4
				ESIA (Environmental and Social Impact Assessment)	
				SESA (Strategic Environmental and Social Assessment)	
	Are management plans required? (check if "yes)	**X**			
	If yes, indicate overall type		**X**	Targeted management plans (e.g. Gender Action Plan, Waste Management Plan, others)	Completed (Gender Action Plan, Stakeholder Engagement Plan)
			X	ESMP (Environmental and Social Management Plan which may include range of targeted plans)	Planned
			X	ESMF (Environmental and Social Management Framework)	Completed
	Based on identified risks, which Principles/Project-level Standards triggered?		**Comments (not required)**		

	Overarching Principle: Leave No One Behind		
	Human Rights	**X**	
	Gender Equality and Women's Empowerment	**X**	
	Accountability	**X**	
	1. Biodiversity Conservation and Sustainable Natural Resource Management	**X**	
	2. Climate Change and Disaster Risks		
	3. Community Health, Safety and Security	**X**	
	4. Cultural Heritage	X	
	5. Displacement and Resettlement		
	6. Indigenous Peoples		
	7. Labour and Working Conditions	**X**	
	8. Pollution Prevention and Resource Efficiency	**X**	

Final Screening at the design-stage is not complete until the following signatures are included

Final Sign Off

Signature	***Date***	***Description***
QA Assessor		UNDP staff member responsible for the project, typically a UNDP Programme Officer. Final signature confirms they have "checked" to ensure that the SESP is adequately conducted.
QA Approver		UNDP senior manager, typically the UNDP Deputy Country Director (DCD), Country Director (CD), Deputy Resident Representative (DRR), or Resident Representative (RR). The QA Approver cannot also be the QA Assessor. Final signature confirms they have "cleared" the SESP prior to submittal to the PAC.
PAC Chair		UNDP chair of the PAC. In some cases PAC Chair may also be the QA Approver. Final signature confirms that the SESP was considered as part of the project appraisal and considered in recommendations of the PAC.

SESP Attachment 1. Social and Environmental Risk Screening Checklist

Checklist Potential Social and Environmental Risks	
INSTRUCTIONS: The risk screening checklist will assist in answering Questions 2-6 of the Screening Template. Answers to the checklist questions help to (1) identify potential risks, (2) determine the overall risk categorization of the project, and (3) determine required level of assessment and management measures. Refer to the SES toolkit for further guidance on addressing screening questions.	
Overarching Principle: Leave No One Behind **Human Rights**	**Answer (Yes/No)**
P.1 Have local communities or individuals raised human rights concerns regarding the project (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
P.2 Is there a risk that duty-bearers (e.g. government agencies) do not have the capacity to meet their obligations in the project?	Yes
P.3 Is there a risk that rights-holders (e.g. project-affected persons) do not have the capacity to claim their rights?	Yes
Would the project potentially involve or lead to:	
P.4 adverse impacts on enjoyment of the human rights (civil, political, economic, social or cultural) of the affected population and particularly of marginalized groups?	No
P.5 inequitable or discriminatory impacts on affected populations, particularly people living in poverty or marginalized or excluded individuals or groups, including persons with disabilities? [48]	Yes
P.6 restrictions in availability, quality of and/or access to resources or basic services, in particular to marginalized individuals or groups, including persons with disabilities?	No
P.7 exacerbation of conflicts among and/or the risk of violence to project-affected communities and individuals?	No
Gender Equality and Women's Empowerment	

[48] Prohibited grounds of discrimination include race, ethnicity, sex, age, language, disability, sexual orientation, gender identity, religion, political or other opinion, national or social or geographical origin, property, birth or other status including as an indigenous person or as a member of a minority. References to "women and men" or similar is understood to include women and men, boys and girls, and other groups discriminated against based on their gender identities, such as transgender and transsexual people.

P.8	Have women's groups/leaders raised gender equality concerns regarding the project, (e.g. during the stakeholder engagement process, grievance processes, public statements)?	No
Would the project potentially involve or lead to:		
P.9	adverse impacts on gender equality and/or the situation of women and girls?	No
P.10	reproducing discriminations against women based on gender, especially regarding participation in design and implementation or access to opportunities and benefits?	Yes
P.11	limitations on women's ability to use, develop and protect natural resources, taking into account different roles and positions of women and men in accessing environmental goods and services? *For example, activities that could lead to natural resources degradation or depletion in communities who depend on these resources for their livelihoods and well being*	No
P.12	exacerbation of risks of gender-based violence? *For example, through the influx of workers to a community, changes in community and household power dynamics, increased exposure to unsafe public places and/or transport, etc.*	No
Sustainability and Resilience: Screening questions regarding risks associated with sustainability and resilience are encompassed by the Standard-specific questions below		
Accountability		
Would the project potentially involve or lead to:		
P.13	exclusion of any potentially affected stakeholders, in particular marginalized groups and excluded individuals (including persons with disabilities), from fully participating in decisions that may affect them?	Yes
P.14	grievances or objections from potentially affected stakeholders?	Yes
P.15	risks of retaliation or reprisals against stakeholders who express concerns or grievances, or who seek to participate in or to obtain information on the project?	No
Project-Level Standards		
Standard 1: Biodiversity Conservation and Sustainable Natural Resource Management		
Would the project potentially involve or lead to:		

1.1	adverse impacts to habitats (e.g. modified, natural, and critical habitats) and/or ecosystems and ecosystem services? *For example, through habitat loss, conversion or degradation, fragmentation, hydrological changes*	Yes
1.2	activities within or adjacent to critical habitats and/or environmentally sensitive areas, including (but not limited to) legally protected areas (e.g. nature reserve, national park), areas proposed for protection, or recognized as such by authoritative sources and/or indigenous peoples or local communities?	No
1.3	changes to the use of lands and resources that may have adverse impacts on habitats, ecosystems, and/or livelihoods? (Note: if restrictions and/or limitations of access to lands would apply, refer to Standard 5)	No
1.4	risks to endangered species (e.g. reduction, encroachment on habitat)?	No
1.5	exacerbation of illegal wildlife trade?	No
1.6	introduction of invasive alien species?	No
1.7	adverse impacts on soils?	Yes
1.8	harvesting of natural forests, plantation development, or reforestation?	No
1.9	significant agricultural production?	No
1.10	animal husbandry or harvesting of fish populations or other aquatic species?	No
1.11	significant extraction, diversion or containment of surface or ground water? *For example, construction of dams, reservoirs, river basin developments, groundwater extraction*	No
1.12	handling or utilization of genetically modified organisms/living modified organisms?[49]	No
1.13	utilization of genetic resources? (e.g. collection and/or harvesting, commercial development)[50]	No
1.14	adverse transboundary or global environmental concerns?	Yes
Standard 2: Climate Change and Disaster Risks		

[49] See the Convention on Biological Diversity and its Cartagena Protocol on Biosafety.

[50] See the Convention on Biological Diversity and its Nagoya Protocol on access and benefit sharing from use of genetic resources.

Would the project potentially involve or lead to:		
2.1	areas subject to hazards such as earthquakes, floods, landslides, severe winds, storm surges, tsunami or volcanic eruptions?	Yes
2.2	outputs and outcomes sensitive or vulnerable to potential impacts of climate change or disasters? *For example, through increased precipitation, drought, temperature, salinity, extreme events, earthquakes*	Yes
2.3	increases in vulnerability to climate change impacts or disaster risks now or in the future (also known as maladaptive or negative coping practices)? *For example, changes to land use planning may encourage further development of floodplains, potentially increasing the population's vulnerability to climate change, specifically flooding*	Yes
2.4	increases of greenhouse gas emissions, black carbon emissions or other drivers of climate change?	No
Standard 3: Community Health, Safety and Security		
Would the project potentially involve or lead to:		
3.1	construction and/or infrastructure development (e.g. roads, buildings, dams)? (Note: the GEF does not finance projects that would involve the construction or rehabilitation of large or complex dams)	Yes
3.2	air pollution, noise, vibration, traffic, injuries, physical hazards, poor surface water quality due to runoff, erosion, sanitation?	Yes
3.3	harm or losses due to failure of structural elements of the project (e.g. collapse of buildings or infrastructure)?	No
3.4	risks of water-borne or other vector-borne diseases (e.g. temporary breeding habitats), communicable and noncommunicable diseases, nutritional disorders, mental health?	Yes
3.5	transport, storage, and use and/or disposal of hazardous or dangerous materials (e.g. explosives, fuel and other chemicals during construction and operation)?	Yes
3.6	adverse impacts on ecosystems and ecosystem services relevant to communities' health (e.g. food, surface water purification, natural buffers from flooding)?	Yes
3.7	influx of project workers to project areas?	Yes

3.8	engagement of security personnel to protect facilities and property or to support project activities?	No
Standard 4: Cultural Heritage		
Would the project potentially involve or lead to:		
4.1	activities adjacent to or within a Cultural Heritage site?	Yes
4.2	significant excavations, demolitions, movement of earth, flooding or other environmental changes?	No
4.3	adverse impacts to sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture (e.g. knowledge, innovations, practices)? (Note: projects intended to protect and conserve Cultural Heritage may also have inadvertent adverse impacts)	No
4.4	alterations to landscapes and natural features with cultural significance?	No
4.5	utilization of tangible and/or intangible forms (e.g. practices, traditional knowledge) of Cultural Heritage for commercial or other purposes?	No
Standard 5: Displacement and Resettlement		
Would the project potentially involve or lead to:		
5.1	temporary or permanent and full or partial physical displacement (including people without legally recognizable claims to land)?	No
5.2	economic displacement (e.g. loss of assets or access to resources due to land acquisition or access restrictions – even in the absence of physical relocation)?	No
5.3	risk of forced evictions?[51]	No
5.4	impacts on or changes to land tenure arrangements and/or community based property rights/customary rights to land, territories and/or resources?	No

[51] Forced eviction is defined here as the permanent or temporary removal against their will of individuals, families or communities from the homes and/or land which they occupy, without the provision of, and access to, appropriate forms of legal or other protection. Forced evictions constitute gross violations of a range of internationally recognized human rights.

Standard 6: Indigenous Peoples		
Would the project potentially involve or lead to:		
6.1	areas where indigenous peoples are present (including project area of influence)?	Yes
6.2	activities located on lands and territories claimed by indigenous peoples?	No
6.3	impacts (positive or negative) to the human rights, lands, natural resources, territories, and traditional livelihoods of indigenous peoples (regardless of whether indigenous peoples possess the legal titles to such areas, whether the project is located within or outside of the lands and territories inhabited by the affected peoples, or whether the indigenous peoples are recognized as indigenous peoples by the country in question)? *If the answer to screening question 6.3 is "yes", then the potential risk impacts are considered significant and the project would be categorized as either Substantial Risk or High Risk*	No
6.4	the absence of culturally appropriate consultations carried out with the objective of achieving FPIC on matters that may affect the rights and interests, lands, resources, territories and traditional livelihoods of the indigenous peoples concerned?	No
6.5	the utilization and/or commercial development of natural resources on lands and territories claimed by indigenous peoples?	No
6.6	forced eviction or the whole or partial physical or economic displacement of indigenous peoples, including through access restrictions to lands, territories, and resources? *Consider, and where appropriate ensure, consistency with the answers under Standard 5 above*	No
6.7	adverse impacts on the development priorities of indigenous peoples as defined by them?	No
6.8	risks to the physical and cultural survival of indigenous peoples?	No
6.9	impacts on the Cultural Heritage of indigenous peoples, including through the commercialization or use of their traditional knowledge and practices? *Consider, and where appropriate ensure, consistency with the answers under Standard 4 above.*	No
Standard 7: Labour and Working Conditions		
Would the project potentially involve or lead to: (note: applies to project and contractor workers)		
7.1	working conditions that do not meet national labour laws and international commitments?	Yes

7.2	working conditions that may deny freedom of association and collective bargaining?	Yes
7.3	use of child labour?	Yes
7.4	use of forced labour?	Yes
7.5	discriminatory working conditions and/or lack of equal opportunity?	Yes
7.6	occupational health and safety risks due to physical, chemical, biological and psychosocial hazards (including violence and harassment) throughout the project life-cycle?	Yes
Standard 8: Pollution Prevention and Resource Efficiency		
Would the project potentially involve or lead to:		
8.1	the release of pollutants to the environment due to routine or non-routine circumstances with the potential for adverse local, regional, and/or transboundary impacts?	Yes
8.2	the generation of waste (both hazardous and non-hazardous)?	Yes
8.3	the manufacture, trade, release, and/or use of hazardous materials and/or chemicals?	Yes
8.4	the use of chemicals or materials subject to international bans or phase-outs? *For example, DDT, PCBs and other chemicals listed in international conventions such as the Montreal Protocol, Minamata Convention, Basel Convention, Rotterdam Convention, Stockholm Convention*	No
8.5	the application of pesticides that may have a negative effect on the environment or human health?	No
8.6	significant consumption of raw materials, energy, and/or water?	No

Annex 8: UNDP Risk Register

#	Description	Risk Category	Impact & Probability	Risk Treatment / Management Measures	Risk Owner
	Enter a brief description of the risk. Risk description should include future event and cause. Risks identified through HACT, PCAT, SES, Private Sector Due Diligence, and other assessments should be included.	Social and Environmental Financial Operational Organizational Political Regulatory Strategic Other Subcategories for each risk type should be consulted to understand each risk type (see UNDP Enterprise Risk Management Policy)	Describe the potential **effect** on the project if the future event were to occur. Enter **likelihood** based on 1-5 scale (1 = Not likely; 5 = Expected) Enter **impact** based on 1-5 scale (1 = Negligible 5 = Extreme) *Based on Likelihood and Impact, use the Risk Matrix to identify the Risk Level (high, Substantial, Moderate or Low)*	What actions have been taken/will be taken to manage this risk.	The person or entity with the responsibility to manage the risk.
1	COVID-19 pandemic is a national issue arising from a sanitary and health crisis that has negative socio-economic impacts, thereby resulting in project implementation constraints.	National risk giving rise to an Operational risk	As discussed in paragraph 2, there has been a significant drop in new COVID-19 cases except for isolated spikes that are well contained. Tunisia has experienced a very strong uptake in COVID-19 vaccination and sanitary measures (especially for foreign visitors) are functioning well to prevent new cases. However, it is pointed out that Kairouan was one of the hardest hit cities at the peak of the pandemic in July 2021 because of its weaker health infrastructure.[52] Hence, there is an onus on the project to put in place measures to minimise	The following project design provide mitigation actions that reduce the need for physical travel and meetings: • The biggest threat relates to the movement of international consultants. The project has been designed to make space for the use of local expertise as far as practicable by using the bundled approach for the recruitment of consultants that would favour local firms. • Data and information sharing will be shared via the website of the municipality and using its social media platforms (Annex 8). The use of virtual meetings to carry out stakeholder meetings and consultations will be	Project Manager / Project Coordinator

[52] https://www.rfi.fr/en/tunisia-hospitals-struggle-with-covid-tsunami - accessed 26 January 2023.

			potential infections arising from movement of project personnel (staff and consultants) and gatherings (workshops, trainings). Moderate L = 2 I = 3	used whenever practicable and desirable. For instance, the Inception Workshop for the PPG stage was carried out in mixed mode participation (virtual and physical), and this may be replicated during implementation. Regarding in-person workshops and technical working group meetings, the project will adopt precautionary measures such as social distancing and mandatory sanitary protocols, including wearing of face masks and hand-sanitizing among others.	
2	Since the Arab Spring, the country's socioeconomic situation has deteriorated. The discontent is especially high amongst the Tunisian youths as they resent the lack of economic opportunities. The situation has been aggravated by the Covid-19 pandemic. The level of autonomy for decentralized governance has also eroded in the past few years.	Political	The difficult socioeconomic situation translates into a volatile political environment. The positive side is that the socioeconomic situation has been stabilized through intervention by the International Monetary Fund (IMF). In the past few years, there has also been a trend for more centralization of decision-making, implying the decreasing autonomy of regional and local governments. The situation is dynamic and will require close monitoring. Changes in the governance arrangements can be subsumed in the overall sovereign risk. Moderate L = 4 I = 3	The sovereign risks are not within the control of the project. Nevertheless, the proposed GEF project will have socio-economic benefits at the local level in the form of job creation and catalyzing investments in energy efficiency value chains at the local level. Also, it is expected that the capacity building that is provided will enhance institutional capacities at the municipal level for better project management. Also, the Project Coordinator will be based in the Municipality of Kairouan to ensure good coordination with the PMU and national-level governance (Ministry of Environment acting as Implementation Partner).	Project Manager / Project Coordinator

3	There is a risk that the Implementing Partner – i.e. Ministry of Environment – does not have all the institutional capacity to implement the project under full NIM.	Organizational	The Ministry of Environment (Directorate of Sustainable Development) is the project Implementing Partner. The capacity assessment of the IP has revealed the IP to be 'moderate risk' overall with a combination of low, moderate and significant risks. The moderate and significant risks are:[53] -The level of risk is moderate for the following components: Program Management / Financial Reporting and Monitoring / Procurement -The level of risk is significant for the following areas: Organizational Structure and Staffing	In order to mitigate this risk, the Support to NIM implementation modality will be used with project administrative support provided by the UNDP CO. Further, the Ministry of Environment has requested the UNDP CO to host the Project Management Unit. As the project was initiated by the Directorate of Sustainable Development (Ministry of Environment) and the Municipality of Kairouan it is expected that they will be willing to build the capacity for project implementation. The institutional (municipality) appropriation for implementation of the proposed project is high.	Project Manager
4	Energy use has an incidence on emissions of greenhouse gases. Also, the construction of buildings through land use changes can cause maladaptation thereby exacerbating vulnerabilities to the impacts of climate change	Climate Change (Environmental) SESP – Risk 5	The ambient temperature in Tunisia is expected to increase in the future due to climate change. As discussed, energy building codes related specifically to building envelope insulation will be an effective form of adaptation to this climate impact. The accentuated impacts of future climate change are expected to increase political interest in addressing the drivers of such change through large-scale implementation of building codes that will lead to scaling-up	Measures that have been incorporated into this project include; 1) introduction of climate-responsive building techniques and elements to reduce the effect of heat and reduce demand on energy for cooling (under Output 1.3 that will support the uptake of locally-fabricated bricks, as well as characterize the thermos-physical properties of these bricks that are typically used in the maintenance and renovation of buildings of cultural heritage value, such as the Medina); 2) raising awareness on the long-term benefits of energy efficiency (Output 1.4 will develop an interactive	Project Manager / Project Coordinator

[53] PwC (2022) Rapport de Micro-évaluation – Ministère de l'Environnement.

			of mitigation actions. Low L = 2 I = 2	tool to quantify the benefits of investments in energy efficiency in buildings); 3) application of building energy codes and support to choose building insulation to reduce energy demand (Output 1.1 and Output 1.2 with demonstration under Output 1.6). A site-specific ESMP (informed by a targeted assessment) will be conducted to inform the retrofitting of the municipality building (Output 1.6) and will assess this risk and recommend measures that ensure that the construction works do not result in maladaptive practices.	
5	Improper storage, transport and disposal of generated waste and hazardous material from retrofitting of municipal building and public street lighting can have an adverse effect on community health and the environment	Environmental SESP – Risk 4	The municipal building is located in a mixed zone area where there are commercial activities and residences. Community health and safety might be affected due to the improper transport, storage, and/or disposal of hazardous material and waste resulting from the replacement of non-EE items in the municipal building. Chemicals and hazardous materials can include paints for refurbishing of walls and facades and use of new insulation materials. Moderate L = 3 I = 3	A targeted assessment will be undertaken for the rehabilitation works of the municipal building (Output 1.6) to address this risk. The assessment will result in an Environmental and Social Management Plan (ESMP) that will include a Waste Management Plan that adopts international best practices. In Tunisia, the collection, transport, treatment and dumping of hazardous and special waste is regulated. The list of companies authorized by the Minister of the Environment for the management of hazardous waste is available on the ANGed website[54] and will be used to contract the company if hazardous waste needs to be disposed.	Project Manager / Project Coordinator
6	Emissions from local	Environmental	Used tires and shoes are	As outlined in the ESMF, a targeted and	Project Manager / Project

[54] ANGed (2022). November 2022: List of authorized hazardous waste companies. Retrieved from http://www.anged.nat.tn/Novembre_2022Liste_societes_autorisees_dechets_dangereux.html (Accessed 4 January 2023)

	production of traditional construction materials resulting from implementation of the strategic plan for the value chain.	SESP – Risk 7	currently burned in furnaces to fire the bricks. There are residential areas that are estimated to be within 1-2 km of the brick production site, and depending on wind direction, there is possibility that these households are affected by the smokes and fumes from furnaces if alternative methods are not proposed. Moderate L = 4 I = 3	strategic assessment to consider potential social and environmental impacts will be undertaken for the strategic plan for development of a value chain for the local production of traditional construction materials (Output 1.3) that will address this issue to ensure proposed solutions do not lead to air pollution.	Coordinator
7	Improper disposal of obsolete lamps after replacement with LEDs can release pollutants into the environment	Environmental SESP – Risk 8	An estimated 574 units of sodium lamps will be replaced with LEDs (Output 2.3). High pressure sodium lamps contain mercury[55] and therefore if improperly disposed, would like to mercury contamination in nearby water bodies, air and soil. Moderate L = 3 I = 3	A targeted assessment will be undertaken prior to commencement of the replacement activities (Output 2.3) to analyze this risk and develop a Waste Management Plan (as part of the ESMP) that will ensure safe storage or disposal of these lamps. This may include storing them at public storage facilities to be reused elsewhere.	Project Manager / Project Coordinator
8	Improper disposal of hazardous waste (obsolete lamps and construction waste) resulting from implementation of the replication plan for scaling up energy efficient buildings and public lighting leading to an	Environmental SESP – Risk 9	If not properly disposed, hazardous waste can have adverse impacts on the environment and public health due to the lack of adequate solid waste infrastructure in Tunisia. Moderate L = 3	As outlined in the ESMF, a targeted assessment will be developed for the replication plan for scaling up energy efficient buildings and public lighting (Output 3.4) that will consider environmentally feasible alternatives to the disposal of obsolete lamps and construction waste, including potentially exporting them in line with the Basel Convention.	Project Manager / Project Coordinator

[55] EPA (2016). Basic Information about Recycling Mercury-Containing Light Bulbs (Lamps). Retrieved from https://archive.epa.gov/epawaste/hazard/web/html/basic.html (Accessed 4 January 2023)

	adverse impact on the environment and public health		I = 3		
9	Damage to biodiversity, natural resources and cultural heritage sites due to local brick production	Environmental SESP – Risk 10	The location of brick fabrication activities is not close to environmentally sensitive areas, including biodiversity, wetlands and water sources. However, the types of brick that may be produced after implementing the strategic plan for developing a value chain for the local production of traditional construction materials (Output 1.3) may have an impact on the cultural heritage value of the city of Kairouan if not in line with aesthetics of the existing architecture. Moderate L = 2 I = 3	Tunisia has important mining resources that constitute the raw materials for the manufacture of the Tozeur brick and plaster. The UNDP-GEF project will draw from lessons learned from those two experiences to propose a strategy and action plan for the enhancement of the local value chain for bricks produced in Kairouan. The strategic plan for development of the value chain will consider these risks and integrate measures to avoid and minimize such risks. As outlined in the ESMF, a targeted and strategic assessment will be conducted to ensure potential downstream social and environmental impacts are considered in the development of the strategic plan for developing a value chain for the local production of traditional construction materials (Output 1.3).	Project Manager / Project Coordinator
10	Limiting women's opportunities to benefit from the project	Social SESP – Risk 2	There is low level participation of women across the entire energy efficiency value chain for buildings and public lighting. The Gender Analysis has revealed several biases against women and described the relatively weaker socioeconomic participation of women than men (Annex 10). Therefore, their opportunities to benefit from the project may be limited by restricting or prohibiting them from engaging	The project Gender Action Plan contains actions to contribute towards overcoming gender-related disparities through four considerations, namely: (i) facilitating equitable participation, (ii) ensuring male engagement to prevent backlash against women, (iii) leveraging existing initiatives, and (iv) differing levels of literacy and education. The project will make use of gender-disaggregated indicators (Gender Action Plan and Project Results Framework). The project will optimize the use of	Project Manager / Project Coordinator

			in the policy and decision making or in the design and management of an operational manual for new building licensing and permits and national standards for building materials. The project will also create new green jobs and will conduct training and awareness campaigns. Women may not be well represented in these activities and may not receive the training and awareness raising offered. Moderate L = 3 I = 3	gender-responsive training materials to project stakeholders and project interventions	
11	Discrimination against and marginalization of vulnerable communities due to implementation of the replication plan for scaling up energy efficient buildings and public lighting.	Social SESP – Risk 3	Stakeholders, including vulnerable groups and Local Populations, who may be affected by or benefit from new energy-efficient buildings and public lighting may be marginalized and unable to voice their concerns, thereby not benefiting from the project and the replication plan or even being negatively affected by it. While 'local populations' are not located in the City of Kairouan and are therefore not directly impacted by project activities, there is the potential that the 'local populations' who are found mainly in the southern	As mentioned in the Environmental and Social Management Framework (ESMF), targeted assessments will be conducted to consider potential social and environmental impacts associated with the replication plan for scaling up energy efficient buildings and public lighting in Tunisia (Output 3.4) and the Strategy and Action Plan that will be developed under Output 1.3 to enhance the value chain of locally-fabricated bricks. The SEP has been developed to fully account for the needs and expectations of national and local stakeholders. In addition, the SEP provides for modalities for engagement of all stakeholders including vulnerable groups and 'local populations'.	Project Manager / Project Coordinator

			parts of Tunisia will be impacted during replication of the project. Moderate L = 3 I = 3	The SEP includes a grievance mechanism that any stakeholder that feels aggrieved by the project may use to seek redress.	
12	Community health and safety risks from noise and air emissions during retrofitting of municipal building and replacement of street lighting.	Social SESP – Risk 6	Increased noise level (noise from the mechanical machinery and equipment, vehicles, removal of the old buildings equipment, etc.) may occur during the construction works at the municipal building Air pollution may be caused by emissions from vehicles, dismantling of the old equipment and constructions, transportation of demolition / construction materials and also during the final interior and exterior works. The types of infrastructure that are found in the vicinity of public lighting in the project boundary are: educational institutions (primary and secondary schools; public and private); health institutions (hospitals and clinics; public and private); residential houses; markets; commercial buildings (offices, small businesses); municipal buildings. Moderate L = 2 I = 3	As mentioned in the ESMF, a targeted assessment will be undertaken for the rehabilitation works at the municipal building (Output 1.6). This assessment will tackle this risk and propose mitigation measures in the resulting site-specific ESMP. The targeted assessment for the street light replacement activity (Output 2.3) will analyze this risk and propose mitigation measures to reduce it. These measures will be incorporated in the ESMP that will be developed.	Project Manager / Project Coordinator
13	Working conditions for project activities do not	Social	Workers involved in the	The targeted assessment that will be undertaken for the construction works at	Project Manager / Project Coordinator

	meet national labour laws and international labour standards.	SESP – Risk 12	retrofitting of the building and replacement of public light bulbs may be exposed to working conditions not aligned with labour laws and health and safety standards. Forced labor and child labor are also possible. The national law provides for the right of workers to organize, form, and join unions, and to bargain collectively. The law also prohibits and criminalizes all the worst forms of child labor and provides for a minimum age of employment. Moderate L = 2 I = 3	the new municipal building (Output 1.6) will assess the risks to workers such that mitigation measures will be incorporated into the resulting site-specific ESMP that will include an Occupational Health and Safety Plan. The targeted assessment for the replacement of public light bulbs (Output 2.3) will assess electrocution, falling and other occupational health and safety risks and proposed measures to mitigate them, which shall be incorporated into an Occupational Health and Safety Plan as part of the overall ESMP for this activity. In addition, Labour Management Procedures (LMP) will be developed for the project to ensure that labour standards and rights are upheld for project workers and for workers that are recruited by contractors working on various activities. Details on the LMP are provided in the ESMF.	
14	Project activities could inadvertently impact 'local population'	Social SESP – Risk 11	The SESP has identified that there are 'local populations' present in Tunisia. However, Standard 6 is not triggered given that impacts on the human rights, lands, natural resources, territories, and traditional livelihoods of 'local populations' are not anticipated given the nature of project activities (including scaling up energy efficient buildings and public lighting).	No action planned but risk will be monitored.	Project Manager / Project Coordinator

			Low L = 1 I = 1		
15	Occupational health and safety risks associated with local brick production.	Social SESP – Risk 13	The operations are carried out manually using unskilled labour. There is no investment to improve working conditions such as the use of protective equipment (PPE). Moderate L = 2 I = 3	As outlined in the ESMF, a targeted and strategic assessment to consider potential downstream social and environmental impacts will be conducted to inform the strategic plan for developing of value chain for the local production of traditional construction materials (Output 1.3) which will ensure that the issue of occupational health and safety is considered when assessing the options.	Project Manager / Project Coordinator

16	Poor communities are economically affected by new building licensing and permits process.	Social SESP – Risk 1	The risk to vulnerable sections of the local community is already mitigated through the selective application of the RTNB. Nevertheless, the Stakeholder Engagement Plan (SEP) that has been prepared for the project will be used to give priority to community engagement to ensure that No-one is Left Behind. This will imply a proactive attitude to reach out to vulnerable people and groups and consider their concerns. Low L = 1 I = 4	The Governorate of Kairouan is considered the 3rd poorest governorate in Tunisia whose poverty rate recorded in 2015 is 29.3%, which is significantly higher than the national average rate (15.3%)[56]. The introduction of an operational manual for new building licensing and permits (Output 1.1) may affect poor communities, including 'local populations', who might be economically disadvantaged by these regulations through the need to use expensive building materials, fixtures or equipment. However, it is pointed out that the RTNB is not applicable to all buildings, and it only covers commercial buildings and collective residential buildings (apartment blocks). In 2019, only 11% of residential buildings were covered by the existing regulation (RTNB).	Project Manager / Project Coordinator
17	The higher costs associated with energy efficiency measures should not hinder socioeconomic development, especially regarding housing for lower income groups	Financial	The financial issue is related to SESP – Risk 3 described above (Risk #11). Discussions with leading private actors have shown that the financial risks associated with investing in the building energy code are low. However, it was also noted that care should be exercised to make sure that the enforcement of the code does not become a financial burden for buildings at the lower end of the market – i.e. for the poorer segment of the population.	Same as for SESP – Risk 3 (i.e. Risk #11 above) with the added information that the Energy Transition Fund operated by the ANME provides financial support in the form of grants for the adoption of energy efficiency in buildings, as well as support provided to municipalities through the ACTE project. The RTNB applies only to commercial buildings and apartment blocks (collective residential buildings) that comprise a very small portion of total buildings built.	Project Manager / Project Coordinator

[56] INS & BM. (2020). Map of Poverty in Tunisia, September 2020. Retrieved from http://www.ins.tn/publication/carte-de-la-pauvrete-en-tunisie-septembre-2020 (Accessed 4 January 2023)

			L = 1 I = 2		

Annex 9: Overview of Technical Consultancies

Consultant	Time Input	Tasks, Inputs and Outputs
For Project Management		
Local / National contracting		
Project Manager *Rate: $2,000/month*	*36 weeks / over 3 years*	*The Project Manager (PM) will be responsible for the overall management of the project, including the mobilization of all project inputs, supervision over project staff, consultants and sub-contractors.* *<u>Managerial duties and responsibilities</u>* • *Manage the overall conduct of the project.* • *Plan the activities of the project and monitor progress against the approved workplan.* • *Execute activities by managing personnel, goods and services, training and low-value grants, including drafting terms of reference and work specifications, and overseeing all contractors' work.* • *Monitor events as determined in the project monitoring plan, and update the plan as required.* • *Provide support for completion of assessments required by UNDP, spot checks and audits.* • *Manage requests for the provision of UNDP financial resources through funding advances, direct payments or reimbursement using the FACE form.* • *Monitor financial resources and accounting to ensure the accuracy and reliability of financial reports.* • *Monitor progress, watch for plan deviations and make course corrections when needed within project board-agreed tolerances to achieve results.* • *Ensure that changes are controlled and problems addressed.* • *Perform regular progress reporting to the project board as agreed with the board, including measures to address challenges and opportunities.* • *Prepare and submit financial reports to UNDP on a quarterly basis.* • *Manage and monitor the project risks – including social and environmental risks - initially identified and submit new risks to the Project Board for consideration and decision on possible actions if required; update the status of these risks by maintaining the project risks log;* • *Capture lessons learned during project implementation.* • *Prepare revisions to the multi-year workplan, as needed, as well as annual and quarterly plans if required.* • *Prepare the inception report no later than one month after the inception workshop.* • *Ensure that the indicators included in the project results framework are monitored annually in advance of the GEF PIR submission deadline so that progress can be reported in the GEF PIR.* • *Prepare the GEF PIR;* • *Assess major and minor amendments to the project within the parameters set by UNDP-GEF;* • *Monitor implementation plans including the gender action plan, stakeholder engagement plan, and any environmental and social management plans;* • *Monitor and track progress against the GEF Core indicators.*

Consultant	Time Input	Tasks, Inputs and Outputs
		• *Support the Terminal Evaluation process.* <u>Technical duties and responsibilities</u> • *Develop ToRs for the recruitment of consultants;* • *Oversee monitoring of progress in development/implementation of the project ESMP/ESMF ensuring that UNDPs SES policy is fully met and the reporting requirements are fulfilled;* • *Oversee/develop/coordinate implementation of all safeguard related plans;* • *Ensure social and environmental grievances are managed effectively and transparently;* • *Review the SESP annually, and update and revise corresponding risk log; mitigation/management plans as necessary;* • *Ensure full disclosure with concerned stakeholders;* • *Ensure environmental and social risks are identified, avoided, mitigated and managed throughout project implementation;* • *Work with the Project coordinator to ensure reporting, monitoring and evaluation fully address the safeguard issues of the project;* • *Monitor progress in implementation of the project Gender Action Plan ensuring that targets are fully met and the reporting requirements are fulfilled;* • *Oversee/develop/coordinate implementation of all gender-related work;* • *Review the Gender Action Plan annually, and update and revise corresponding management plans as necessary;* • *Work with the M&E officer and Safeguards Officer to ensure reporting, monitoring and evaluation fully address the gender issues of the project;*
Project Coordinator *$ 1,500 / month*	*36 weeks / over 3 years*	*The Project Coordinator will support the PM in the overall management of the project, and, will in paricular be responsble for activities implemented in Kairouan. The Project Coordinator will directly support the implementation and M&E of the Gender Action Plan, Stakeholder Engagement Plan, and all project risks (including environmental and social safeguards).* *Under the guidance and supervision of the Project Manager, the Project Coordinator will have the following specific responsibilities:* • *Coordinate all activities in project sites in Kairouan* • *Provide direct assistance to the PM in relation to all project M&E activities, including among others the following:* ✓ *Monitor project progress and participate in the production of progress reports ensuring that they meet the necessary reporting requirements and standards;* ✓ *Ensure project's M&E meets the requirements of the Government, the UNDP Country Office, and UNDP-GEF; develop project-specific M&E tools as necessary;*

Consultant	Time Input	Tasks, Inputs and Outputs
		✓ *Oversee and ensure the implementation of the project's M&E plan, including periodic appraisal of the Project's Theory of Change and Results Framework with reference to actual and potential project progress and results;* ✓ *Oversee/develop/coordinate the implementation of the stakeholder engagement plan;* ✓ *Oversee and guide the design of surveys/ assessments commissioned for monitoring and evaluating project results;* ✓ *Facilitate mid-term and terminal evaluations of the project; including management responses;* ✓ *Facilitate annual reviews of the project and produce analytical reports from these annual reviews, including learning and other knowledge management products;* ✓ *Support project site M&E and learning missions;* ✓ *Visit project sites as and when required to appraise project progress on the ground and validate written progress reports.* • *Support the PM with managing the Stakeholder Engagement Plan and Social and Environmental Safeguards, including:* ✓ *Monitor progress in development/implementation of the project ESMP/ESMF ensuring that UNDPs SES policy is fully met and the reporting requirements are fulfilled;* ✓ *Oversee/develop/coordinate implementation of all safeguard related plans;* ✓ *Ensure social and environmental grievances are managed effectively and transparently;* ✓ *Review the SESP annually, and update and revise corresponding risk log; mitigation/management plans as necessary;* ✓ *Ensure full disclosure with concerned stakeholders;* ✓ *Ensure environmental and social risks are identified, avoided, mitigated and managed throughout project implementation;*
Project Associate / Administrative Assistant *$1,200/month*	*36 weeks / over 3 years*	*The Project Associate / Administrative Assistant will have enhanced responsibilities combining support to the PM in the day-to-day implementation of the project and a number of administrative responsibilities including financial management of the project and its monitoring & evaluation.* *<u>Duties and Responsibilities</u>* *Under the guidance and supervision of the Project Manager, the Project Assistant will carry out the following tasks:* • *Assist the Project Manager in day-to-day management and oversight of project activities;* • *Assist the Project Manager and Project Coordinator in matters related to M&E and knowledge resources management;* • *Assist in the preparation of progress reports;* • *Ensure all project documentation (progress reports, consulting and other technical reports, minutes of meetings, etc.) are properly maintained in hard and electronic copies in an efficient and readily accessible filing system, for when required by PB, TAC, UNDP, project consultants and other PMU staff;* • *Provide PMU-related administrative and logistical assistance.*

Consultant	Time Input	Tasks, Inputs and Outputs
		• *Keep records of project funds and expenditures, and ensure all project-related financial documentation are well maintained and readily available when required by the Project Manager;* • *Review project expenditures and ensure that project funds are used in compliance with the Project Document and GoI financial rules and procedures;* • *Validate and certify FACE forms before submission to UNDP;* • *Provide necessary financial information as and when required for project management decisions;* • *Provide necessary financial information during project audit(s);* • *Review annual budgets and project expenditure reports, and notify the Project Manager if there are any discrepancies or issues;* • *Consolidate financial progress reports submitted by the responsible parties for implementation of project activities;* • *Liaise and follow up with the responsible parties for implementation of project activities in matters related to project funds and financial progress reports.* Technical duties and responsibilities: *Provide technical support for the implementation and monitoring of the project, focusing on the completion of the tasks listed below. This support is estimated at 15% of working time under component 1, 2 and 3:* • *Assist the project manager in drawing up the terms of reference for the expertise to be mobilized, and in reviewing their technical deliverables;* • *Assist the project manager in the data collection process to ensure project reporting;* • *Contribute to the development of knowledge management tools* • *Assist the Project Manager in the design and identification of new initiatives this includes, but is not limited to, the mobilization and development of partnerships with stakeholders in the project's field(s) of intervention, the elaboration of substantive documents and participation in meetings to this effect.*
For Technical Assistance		
Outcome 1 / Outcome 2 / Outcome 3		
Company Services Contracting		
Description of Approach: *The UNDP CO privileges the use of company services contracting for the provision of bundled technical assistance (all Outcome 1 and Outcome 3 outputs), laboratory services (Output 1.3 – CTMCCV) equipment (Output 2.3) and works (Output 1.6).* *The services required are described in bundles that match the budget allocation and budget notes in Section IX of the Project Document. They also match the Procurement Plan described in Annex 11.* *It will be up to the bidding company to seek the profiles required either nationally or internationally depending on market response.*		
Services Company A	*282 days*	*The company will provide the following expertise:*

Consultant	Time Input	Tasks, Inputs and Outputs
(please see budget notes)	*@USD1,000 per day (inclusive of all costs)*	• *Institutional and Regulatory Expert (36 days) – Outputs 1.1 and 1.2* • *Strategic Value Chain Expert (37 days) – Output 1.3* • *Building Energy Efficiency Expert (40 days) – Output 1.4* • *Quality Standards Expert (25 days + 20 days) – Output 1.5 and Output 2.2* • *Environmental & Social Risk Expert (40 days + 30 days + 9 days) – Output 1.6, Output 2.3 and Ouput 3.4* • *Public Lighting Energy Efficiency Expert (45 days) – Outout 2.1* *Details of delivery are described below for each expertise:* *<u>Institutional and Regulatory Expert</u>* *The expert will carry out all the activities related to Outputs 1.1 and 1.2:* • *Carry out review of existing procedures for allocating building licenses and permits;* • *Develop an Operational Manual for improved licensing process that will allow the municipality to enforce the RTNB* • *Train municipal staff on the use of the Operational Manual* • *Support the setting up and operationalisation of the cross-institutional mechanism for enforcing energy builidng code* • *Accompany the municipality staff to carry on-the-ground follow ups of builiding construction to enforce RTNB* • *Support the process of updating the RTNB, including provision for a racheting mechanism for periodic updates to reflect market dynamics and best available technologies* *Inputs: GEF financing resources + support of TWG for Component 1* *Outputs: (1) operational manual; (2) training of municipal staff on use of manual; (3) operationalisation of cross-institutional mechanism for enforcing energy building code; (3) review of the existing energy building code* *<u>Strategic Value Chain Expert</u>* *The expert will carry out Acticity 1.3.1:* • *Identify and map out all stakeholders involved in the market chain for locally-fabricated bricks* • *Carry out a baseline assessment of the existing value chain* • *Through stakeholder coordination, develop a strateic roadmap, including policy instruments for promoting local value chain for traditional bricks, including approches to increase market demand and use in energy efficient buildings* *Inputs: GEF financing resources + support of TWG for Component 1* *Outputs: (1) stakeholder mapping report; (2) value chain baseline analysis report; (3) roadmap, including policy instruments for enhancing the value chain for locally-fabricated bricks.*

Consultant	Time Input	Tasks, Inputs and Outputs
		<u>Building Energy Efficiency Expert</u> *The expert will be requred to develop and traing stakeholder on the ise of an online for carrying out comparative socio-economic and environmental analysis of buildings using life-cycle methodology under Output 1.4:* • *Carry out stakeholders' needs gaps analysis for understand the needs and expectations of stakeholders* • *Design and test the online tool based on needs gaps analysis using a collborative approach* • *Finalise the online tool and train end users* • *Propose an exit strategy for the institutionalisation of the tool at ANME* *Inputs: MOE financing resources + support of TWG for Component 1* *Outputs: (1) stakeholders' needs gaps analysis report; (2) online tool developed and adopted by stakeholders; (3) exit startegy for institutionalisation of the tool at ANME.* *<u>Quality Standards Expert</u>* *The expertise will be required to accomplish the deliverables related to quality standards under Output 1.5 and Output 2.2. The technical assistance will be delivered to INNORPI as follows:* • *Define the technical requirements for quality standards for locally-fabricated bricks (this will make use of the results produced by CTMCCV)* • *Carry out inventory of technical standards related to intelligent public lighting systems, including the technical standards of all system components, including LED lamps* • *Provide tehnical support to INNORPI related to both quality standards* *Inputs: GEF & MOE financing resources + support of TWG for Components 1 & 2* *Outputs: (1) quality standards for lcoally-fabricated bricks; (2) quality standards for intelligent public lighting system, including components.* *<u>Environmental and Social Risk Expert</u>* *The expertise for targeted social and envionmental risks assessments are to be carried out as per the requirements of the SESP and ESMF. The targeted assessments are required for Output 1.3, Output 1.6, Output 2.3 and Output 3.4. For Outputs 1.6 and 2.3, the targted assessments will result in site-specific ESMP. For Outputs 1.3 and 3.4, strategic assessments of potential environmental and social risks associated with the Strategic and Replication Plans will be carried out. The assessments are:* • *Targeted assessments for the municipal buklding at Keblia to developed a site-specific ESMP, including the management of construction waste and working conditions* • *Targeted assessments for the substitution of public lighting lamps to developed ESMP that will also cover the environmentally sound collection, storage and disposal of electronic and electrical wastes* • *Strategic assessments of potential environmental and social risks associated with the Strategic and Replication Plans will be carried out*

Consultant	Time Input	Tasks, Inputs and Outputs
		Inputs: GEF & MOE financing resources + support of TWG for Components 1 & 2 & 3 *Outputs: (1) ESMP for Keblia municipal building; (2) ESMP for public lamp substitution; (3) strategic assessments of potential social and environmental risks incorporated in the Strategic Plan for enhacing the value chain of locally-fabricated bricks and the project's Replication Plan.* *<u>Public Lighting Energy Efficiency Expert</u>* *This expertise is needed to implement Output 2.1for strengthening the capaicty of municipality staff to implement efficient lighting projects. The following are needed:* • *Identify capacity development needs of municipality staff regarding the deisgn and implementation of efficient public lighting* • *Develop training programme for municipality staff based on needs assessments* • *Provide training to municipality staff on the design, standards specification and implementation of intelligent lighting systems* *Inputs: GEF financing resources + support of TWG for Components 2* *Outputs: (1) training programme and materials developed based on needs assessments; (2) municipality staff trained on the design and implementation of public lighting intelligent systems; (3) technical specifications for the procurement of LED lamps to be procured under Output 2.3.*
Services Company - CTMCCV (please see budget note 7)	*60 samples @USD 200 per sample*	*Under Output 1.3, the public laboratory CTMCCV will be funded to carry out characterization of the physical and thermal properties of locally-fabricated bricks. These tests are needed to ensure quality standards of the bricks and will include, among others: tensile strength, compressive strength (under different conditions of humidity and temperature), thermal and insulating properties, water impremeability tests, adhesion tests* *Inputs: MOE financing resources + support of TWG for Components 2* *Outputs: (1) physical and thermal properties of bricks fabricated in Kairouan.*
Services Company - Works (please see budget note 5)	*USD 314,458*	*GEF investments will be used to enhanced the energy efficiency of the Municipal Building at Keblia that will yield global environmental benefits (Annex 13 and Section IV of the Project Document), as well as the operationalization of the procedure for enforcing the application of the building energy code (RTNB). It is expected that the Services Company – Works will be recruited within the broader call for procurement that the Municipality of Kairouan will launch for the enhancement of the municipal bulding at Keblia. The GEF investments will contribute to increasing the energy label from Class 5 to Class 3, and the technical requirements will be developed by the municipality using the results of CLIP Tunisia models (Figures 6 to 8 in the Project Document).*
Services Company –	*USD 105,457*	*GEF investments of USD 105,457 will be used to replace 574 conventional street lights with energy efficient LEDS with remote*

Consultant	Time Input	Tasks, Inputs and Outputs
Equipment (please see budget note 12)		*sensing function. The average unit cost of each LED lamp (and casing) is USD 184. The technical standards for the procerement will be developed under Output 2.1.*
		Outcome 3
Knowledge and Data Management Expert *Rate: $500/day*	*107 days / Year 1-3*	• *Carrying out annual lessons learned investigations for whole project, including conducting field/project site surveys for tracking the indicators given in the M&E Plan and the project Results Framework (Output 3.3)* • *Supporting the international consultant carrying out independent Terminal Evaluation of the project under Output 3.1* • *Supporting the development of a Replication Plan for project sustainablity under Output 3.4* • *Supporting the international consultant to set up Inventory Data Management System in the Municipality of Kairouan (Output 3.2)* • *Supporting the process of municipality membership to international organisations (Output 3.2)* *Inputs: GEF and MOE financing resources + support of TWG for Component 3* *Outputs: (1) annual lessons learned reports; (2) terminal evaluation report; (3) Replication Plan (inlcuding invstment plan); (4) climate finance concept note; (5) Inventory Data Management System; (6) membership of Municipality of Kairouan to international organisations (WCCD, C40 Cities, ICLEI etc..)*
International / Regional and global contracting		
International Knowledge Management Expert *Rate: $ 900/day*	*83 days / Year 2-3*	*The International Knowledge Management Expert will carry out the activities under Output 3.3 and Output 3.4 as follows:* • *Produce knowledge products based on the lessons learned reports* • *Formulating a replication plan for scaling up the results of the project based on lessons learned and to ensure sustainability beyond project lifetime;* • *Developing an investment plan to accompany the replication plan;* • *Carry out dissemination of the Replication Plan among all project stakehodlers.* • *Liaise with PANORAMA platform for disseminating the results of the project, including the replication plan* *Inputs: GEF and MOE financing resources + support of TWG for Component 3* *Outputs: (1) Replication Plan; (2) investment plan; (3) knowledge products; (4) climate finance concept note; (5) dissemination of results of knowledge products and replication plan*
International Data Management Expert *Rate: $900/day*	*40 days / Year 1-2*	*The International Data Management Expert will carry out the activities under Output 3.2 as follows:* • *Set up the Inventory Data Management System in the Municipality of Kairouan, as well as all data collection and quality assurance & quality control processes* • *Ensure processes for municipality membership to international organisations are successfully completed* • *Support the Municipality to carry out first reporting to the WCCD using available data and indicators that will pave the way*

Consultant	Time Input	Tasks, Inputs and Outputs
		for improvements in the future using the Inventory Data Management System *Inputs: MOE financing resources + support of TWG for Component 3* *Outputs: (1) Inventory Data Management System; (2) membership of Municipality of Kairouan to international organisations (WCCD, C40 Cities, ICLEI etc..); (3) first report for the WCCD.*

Annex 10: Stakeholder Engagement Plan

Annex 11: Environmental Social Management Framework (ESMF)

Attached separately.

Annex 12: Gender Analysis and Gender Action Plan

Attached separately.

Annex 13: Procurement Plan

Separately annexed.

Annex 14: GEF focal area specific annexes

PART A - Calculations of GHG Emissions Reductions

GHG emission reductions have been calculated for: (i) the adoption of energy efficient public lighting using LEDs (Output 2.3); (ii) the adoption of energy efficiency measures in the energy efficient municipal building at Keblia (Output 1.6); and (iii) enforcement of energy building code RTNB (Outputs 1.1 and 1.2). The GEF methodology and tool for EE in buildings was used and is provided separately as part of this annex (Annex 14 - GEF_EE_GHG_Tool_v1.0_Kairouan).

A.1. Efficient public lighting

Based on GEF Secretariat comments at PIF stage, a business-as-usual (BAU) scenario has been developed together with the Municipality of Kairouan. The BAU scenario has been developed as from 2013 (earliest year for which a complete set of data was available) and projected to 2045 so that post-project market potential for efficiency gains, and, hence, GHG emission reductions could be estimated. The BAU has been developed based on the numbers and types of lamps used between 2013 and 2022 in the City of Kairouan for public lighting. The lamps used are conventional high pressure sodium (SHP) and halogen lamps (HPL), and it can be seen that in 2022 only 400 of these conventional lamps had been replaced with LEDs. Hence, there is little effort in the existing baseline for substituting the conventional lamps with LEDs, highlighting the critical role of the UNDP-GEF project to catalyse change.

Table 14.A.1. Characteristics of lamps that are used for public lighting in the city of Kairouan: 2013 – 2022.

	SHP 70w	SHP 100w	SHP 150w	SHP 250w	Total SHP	HPL 125 W	HPL 250 W	Total HPL	LED	LED	Total LED	LEDs renovated
Year									100W	80W		
2013	0	0	730	1540	**2270**	2214	2920	**5134**	0	0	0	0
2014	0	0	900	1880	**2780**	2740	2930	**5670**	0	0	0	0
2015	0	0	2975	2730	**5705**	2370	2645	**5015**	0	0	0	0
2016	100	80	4587	3010	**7777**	2075	1610	**3685**	0	0	0	0
2017	120	455	6202	3542	**10319**	1576	1109	**2685**	0	0	0	0
2018	160	835	7308	4040	**12343**	810	711	**1521**	0	0	0	0
2019	160	1045	7608	4140	**12953**	510	611	**1121**	0	0	0	0
2020	160	1045	7608	4140	**12953**	310	611	**921**	0	0	0	0
2021	160	1045	7608	3740	**12553**	310	611	**921**	400	0	400	400
2022	160	1045	7608	3740	**12553**	310	611	**921**	400	0	400	0

Direct emission reductions

The direct project interventions will support the replacement of 574 lamps as described in Table 14.A.2. Whereas this table gives the nominal power rating of conventional lamps (either SHP or HPL), a 10% auxiliary power consumption is added to calculate effective electricity consumption. Assuming 10 hours of lighting per day, the annual electricity savings is 315.8345 MWh / year. On average the savings is 0.5502 MWh per LED lamp (not taking power rating into consideration). The calculations are contained in the Excel file named "Annex 14 - PIMS 6686 – Public lighting". The calculations assume a 10% transmission and distribution losses.

Using a grid emission factor of 0.4470 tCO_2/MWh[57], the annual emission reduction is calculated at around 141 tCO_2 per year – i.e. 424 tCO_2 between 2023 and 2026. Using a technology lifetime of 18 years, the post-project direct emissions reductions is calculated at 2,118 tCO_2 (2027-2046). Hence, the total lifetime direct emissions reductions accruing from GEF investments in LED lighting is 2,542 tCO_2.

Table 14.A.2. Details of lamp substitution using GEF investments.

Location	Number of lighting points	Existing Power rating (W)	Power rating with GEF investments, LED (W)
Avenue de l'environnement et giratoire 'Zarbiya'	206	250	100
Place Sadeka	58	150	80
Cité commerciale	60	100	80
Avenue Ibn Aghlab	70	250	100
Avenue Touhami Negra	94	250	100
Avenue Sadek Lemkaddem	24	250	100
Avenue Ibn Jazzar	34	250	100
Avenue Moez Ibn Badis	28	250	100
Total	**574**	**141570**	**55040**

Indirect emission reductions (bottom-up)

A replication factor of 5 has been used for estimating bottom-up indirect emission reductions. Discussions with the Municipality of Kairouan has shown that with the catalytic effect of the UNDP-GEF project, it will be able to mobilise additional external resources for replacing around 937 units of conventional lamps with LEDs. Over a 3-year period, this amounts to 2,811 units, which is ~4.9 times the number replaced using GEF investments. Hence, a factor of 5 has been used as catalytic effect of the UNDP-GEF project. Hence, the indirect bottom-up emission reductions between 2027 and 2046 is calculated to be 12,705 tCO_2.

[57] https://www.irena.org/IRENADocuments/Statistical_Profiles/Africa/Tunisia_Africa_RE_SP.pdf - accessed 19 May 2022.

Indirect emission reductions (top-down)

There is no data regarding the market potential for energy efficiency gains from public lighting for the City of Kairouan. Hence, BAU and energy efficiency (EE) scenarios have been constructed, and the difference between the two curves would give the market potential to 2045. In order to construct the BAU scenario, the projected increase in the numbers of these different types of lamps was estimated. The increase is due to requirements for increasing the levels of public lighting in order to better serve the public. This also includes the projected increase in infrastructure, housing and commercial activities in Kairouan. The BAU scenario also takes into account the replacement of existing lamps with efficient lamps (LEDs) that the municipality plans to carry out using its capital budget. Based on the figures provided by the Municipality of Kairouan for the unit cost of USD 104 to replace one lamp on average (including changes to lamp base and casing), the number of lamps that can be substituted with LED using the city's capital budget has been estimated at 94 units per year on average.

The growth in number of lamps has been estimated at around 231 units / year. This number has been obtained from the municipality based on growth projections it carried out between 2020 and 2026. This value has been kept fixed for the entire modelled period. Hence, the total number of lamps in 2045 has been estimated to be 18,975. The projections are shown in Table 14.A.3 below.

Table 14.A.3. Projected number of lamps of different types based in the BAU case: 2013-2045.

	SHP 70w	SHP 100w	SHP 150w	SHP 250w	Total SHP	HPL 125 W	HPL 250 W	Total HPL	LED	LED	Total LED	TOTAL
2013	0	0	730	1540	2270	2214	2920	5134	0	0	0	7404
2014	0	0	900	1880	2780	2740	2930	5670	0	0	0	8450
2015	0	0	2975	2730	5705	2370	2645	5015	0	0	0	10720
2016	100	80	4587	3010	7777	2075	1610	3685	0	0	0	11462
2017	120	455	6202	3542	10319	1576	1109	2685	0	0	0	13004
2018	160	835	7308	4040	12343	810	711	1521	0	0	0	13864
2019	160	1045	7608	4140	12953	510	611	1121	0	0	0	14074
2020	160	1045	7608	4140	12953	310	611	921	0	0	0	13874
2021	160	1045	7608	3740	12553	310	611	921	400	0	400	13874
2022	160	1045	7608	3740	12553	310	611	921	400	0	400	13874
2023	160	1095	7748	3474	12477	310	517	827	744	0	744	14048
2024	160	1131	7888	3515	12694	310	423	733	838	0	838	14265
2025	160	1181	8028	3556	12925	310	329	639	932	0	932	14496

2026	160	1231	8168	3597	13156	310	235	545	1026	0	1026	14727
2027	160	1281	8308	3638	13387	310	141	451	1120	0	1120	14958
2028	160	1331	8448	3679	13618	263	47	310	1214	0	1214	15142
2029	160	1381	8588	3720	13849	122	0	122	1308	0	1308	15279
2030	160	1431	8728	3761	14080	28	0	28	1402	0	1402	15510
2031	160	1481	8868	3736	14245	0	0	0	1496	0	1496	15741
2032	160	1531	9008	3683	14382	0	0	0	1590	0	1590	15972
2033	160	1581	9148	3630	14519	0	0	0	1684	0	1684	16203
2034	160	1631	9288	3577	14656	0	0	0	1778	0	1778	16434
2035	160	1681	9428	3524	14793	0	0	0	1872	0	1872	16665
2036	160	1731	9568	3471	14930	0	0	0	1966	0	1966	16896
2037	160	1781	9708	3418	15067	0	0	0	2060	0	2060	17127
2038	160	1831	9848	3365	15204	0	0	0	2154	0	2154	17358
2039	160	1881	9988	3312	15341	0	0	0	2248	0	2248	17589
2040	160	1931	10128	3259	15478	0	0	0	2342	0	2342	17820
2041	160	1981	10268	3206	15615	0	0	0	2436	0	2436	18051
2042	160	2031	10408	3153	15752	0	0	0	2530	0	2530	18282
2043	160	2081	10548	3100	15889	0	0	0	2624	0	2624	18513
2044	160	2131	10688	3047	16026	0	0	0	2718	0	2718	18744
2045	160	2181	10828	2994	16163	0	0	0	2812	0	2812	18975

In the EE scenario, the replacement rate for LED substitution is 937 units per year. The replacement of lamps prioritises the substitution of high power rated SHP and HPL lamps first. The replacement schedule is shown in Table 14.A.4.

Table 14.A.4. Projected number of lamps of different types based in the EE scenario: 2013-2045.

Year	SHP 70w	SHP 100w	SHP 150w	SHP 250w	Total SHP	HPL 125 W	HPL 250 W	Total HPL	LED	LED	Total LED
									100W	80W	

2013	0	0	730	1540	**2270**	2214	2920	**5134**	0	0	0
2014	0	0	900	1880	**2780**	2740	2930	**5670**	0	0	0
2015	0	0	2975	2730	**5705**	2370	2645	**5015**	0	0	0
2016	100	80	4587	3010	**7777**	2075	1610	**3685**	0	0	0
2017	120	455	6202	3542	**10319**	1576	1109	**2685**	0	0	0
2018	160	835	7308	4040	**12343**	810	711	**1521**	0	0	0
2019	160	1045	7608	4140	**12953**	510	611	**1121**	0	0	0
2020	160	1045	7608	4140	**12953**	310	611	**921**	0	0	0
2021	160	1045	7608	3740	**12553**	310	611	**921**	400	0	400
2022	160	1045	7608	3740	**12553**	310	611	**921**	400	0	400
2023	160	1045	7608	3440	**12253**	10	211	**221**	744	800	1544
2024	160	1027	7350	3284	**11821**	10	70	**80**	1250	918	2168
2025	200	1300	7047	2904	**11451**	0	0	**0**	1400	1218	2618
2026	200	1200	6500	2000	**9900**	0	0	**0**	1900	3218	5118
2027	200	1125	6131	1565	**9021**	0	0	**0**	2895	3293	6188
2028	200	1050	5762	1130	**8142**	0	0	**0**	3890	3368	7258
2029	200	975	5393	695	**7263**	0	0	**0**	4885	3443	8328
2030	200	900	5024	260	**6384**	0	0	**0**	5880	3518	9398
2031	200	825	4655	0	**5680**	0	0	**0**	6700	3593	10293
2032	200	750	4286	0	**5236**	0	0	**0**	7260	3668	10928
2033	200	675	3917	0	**4792**	0	0	**0**	7820	3743	11563
2034	200	600	3548	0	**4348**	0	0	**0**	8380	3818	12198
2035	200	525	3179	0	**3904**	0	0	**0**	8940	3893	12833
2036	200	450	2810	0	**3460**	0	0	**0**	9500	3968	13468
2037	200	375	2441	0	**3016**	0	0	**0**	10060	4043	14103
2038	200	300	2072	0	**2572**	0	0	**0**	10620	4118	14738
2039	200	225	1703	0	**2128**	0	0	**0**	11180	4193	15373
2040	200	150	1334	0	**1684**	0	0	**0**	11740	4268	16008

2041	125	75	965	0	**1165**	0	0	**0**	12300	4418	16718
2042	50	0	596	0	**646**	0	0	**0**	12860	4568	17428
2043	0	0	227	0	**227**	0	0	**0**	13420	4618	18038
2044	0	0	0	0	**0**	0	0	**0**	13838	4618	18456
2045	0	0	0	0	**0**	0	0	**0**	13838	4618	18456

The total electricity consumption was determined for 10 hrs lighting per day (all lamps) and with 10% auxiliary power added for SHP and HPL lamps. The results are shown in Figure 14.A.2. The GHG emissions were calculated by applying a grid emission factor of 0.447 tCO_2/MWh and the results are shown in Figure 14.A.1.

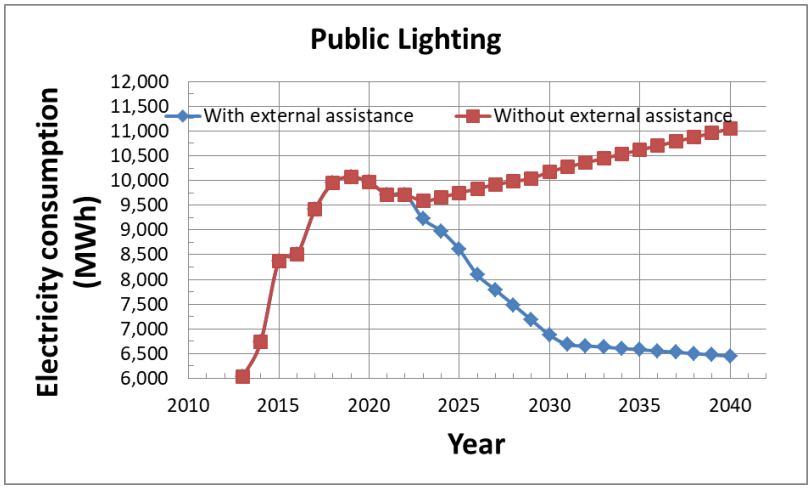


Figure 14.A.1. Public lighting electricity consumption in BAU (without external assistance) and EE (with external assistance) scenarios.

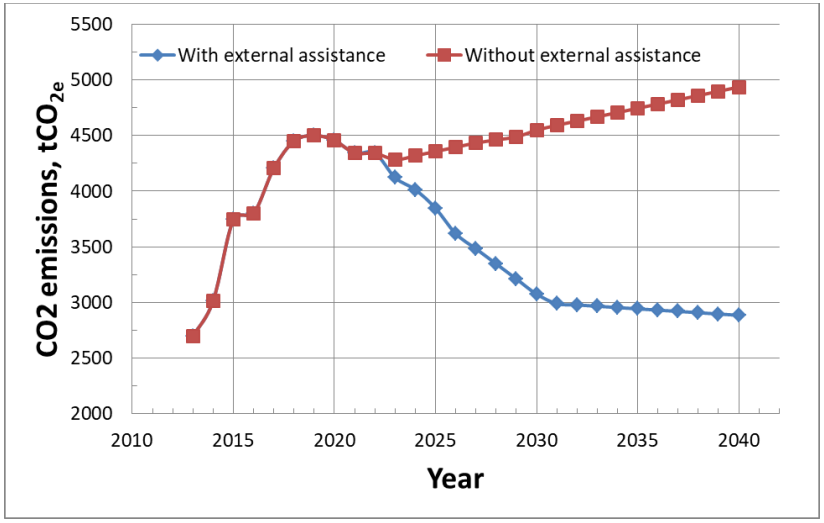


Figure 14.A.2. Public lighting CO_2 emissions in BAU (without external assistance) and EE (with external assistance) scenarios.

The cumulative emission reductions to 2045 are estimated to be 35,829 tCO_2. Applying a causality factor of 60% gives top-down emission reductions of 21,429 tCO_2.

A.2. Efficient Keblia municipal building

Direct emission reductions

The calculations are based on the CLIP Tunisia modeling results shown in Figure 6 and 7 in the discussions of Output 1.6 in the Project Document. Energy efficiency interventions in the construction of G3 and G4 will produce efficiency gains of 33 kWh/m^2. The area of one floor is 207 m^2. Hence, energy efficiency gains will be for a total of 414 m^2 in 2024, giving annual electricity savings of 13.7 MWh/year. In 2025, the G+2 existing building will also be enhanced to Class 3 ECOBat energy label. Hence, the total surface area for the G+4 municipal building will be 1035 m^2 resulting in 34.2 MWh electricity savings annually. The direct emission reductions during the project lifetime are estimated at 37 tCO2. Assuming a lifetime of 25 years for the building, the total direct GHG emission reductions are calculated as 342 tCO_2.

Indirect emission reductions (bottom-up)

The bottom-up indirect emission reductions has been determined conservatively using the number and size of municipal buildings that are planned to be built in the future. Because of the inherent uncertainties in long-term planning, it was possible to get information on municipal building construction to 2026. The total area of the proposed new and renovated buildings is five times that of the total surface area of the municipal building at Keblia. To be conservative, the post-project replication factor has been conservatively kept at 5. Hence, the bottom-up indirect emission reduction has been estimated at 1,844 tCO_2.

A.3. Enforcement of energy building code (RTNB)

In the baseline, the RTNB is not enforced. The results of Outputs 1.1 and 1.2 will lead to the application and enforcement of the RTNB, which will be updated using a racheting mechanism (Output 1.2). The RTNB is not applicable to all types of buildings. Data was collected from the Municipality to identify the buildings that are subject to the RTNB in order to know the total surface areas. The data for 2010 to 2019 are given in the worksheet named 'Bldg' in the Excel file 'Annex 14 - PIMS 6686 – Public light' and reproduced in Table 14.A.5. It can be seen that the area of commercial buildings and collective residential buildings (apartments) that are subject to the RTNB (EPEBN in the table) does not show a clear trend between 2010 and 2019. To estimate the projected increase in area for modelling purposes, the trend between 2015 and 2019 has been used – i.e. an increase of around 2.37% per annum.

Table 14.A.5. Area of buildings that are subject to RTNB, 2010-2019.

	Habitation collectif, m2	Hab Col critere EPEBN, m2	Bureau, m2	Bureau EPEBN, m2	Hat_Bur,m2	Hab+Bur EPEBN, m2
2010	97,440	12,960	33600	39,600	**131,040**	**52,560**
2015	101920	12,960	5880	7,920	**107,800**	**20,880**
2019	104160	12,960	8400	9,900	**112,560**	**22,860**

Accordingly to the existing energy building code, the minimum requirement in terms of energy consumption is equivalent to Class 5 of the ECOBat label – i.e. middle of the Class 5 band or 110 kWh/m^2. It is know that the RTNB is not enforced in the baseline, and discussions with the technical staff of the municipality, Tunisian Green Building Council, and the Association of Architects has revealed that energy use in existing buildings would be in the Class 7 band. Adopting a conservative approach in order not to overestimate GHG emission reductions, the lower end of the band is used in calculations – i.e. baseline electricity consumption is 150 kWh/m^2. With institutional capacity building under Output 1.1 and Output 1.2, it is expected that the municipality will be able to increase enforcement by 40% (of total area of building subject to RTNB) in the year 2024; 65% in 2025; and fully as from 2026.

So at the end of 2026, the direct annual GHG emission reduction is estimated at 878 tCO_2 per year, and total emission reductions are expected to be 1,391 tCO_2 during the project lifetime. Some 15,944 tCO_2 emission reductions will continue to accrue after the project lifetime. Hence, the direct project emission reductions are estimated at 17,336 tCO_2. Direct post-project emission reductions are estimated at 36,796 tCO_2.

PART B – Parallel initiatives

1. Programme 'Alliance des Communes pour la Transition Energétique – ACTE' (2017 – 2022 ; Euro 3 million ; Swiss Government): ACTE was launched in May 2015, aims to strengthen the capacity of Tunisian municipalities to contribute to the national energy transition, through energy efficiency and the use of renewable energies at the level of heritage and municipal territory. The scope of the ACTE programme at the local level covers all Tunisian municipalities and covers six areas:
    - Buildings and urban planning to support the municipality in its role as regulator and planner of the territory to promote sustainable territorial and urban planning and resilient to climate change;

- Municipal buildings and facilities to support the municipality in its role as a consumer and provider of "model" public services;
- Energy diversification to support the municipality's ability to promote energy efficiency and the use of renewable energies;
- Mobility and transport to support the municipality's ability to optimise travel on its territory and promote low-impact mobility;
- Internal organisation, monitoring and evaluation to strengthen the municipality's capacity to set up a system of management and internal governance within the municipality, including a system for monitoring and evaluating its local energy policy; and
- Cooperation and communication for the strengthening of the municipality's capacity to mobilize the support of its public, private and civil society partners.

2. Programme « Accompagnement du marché tunisien de l'éclairage dans la transition vers des technologies efficaces » (2018 – present ; USD 2.67 million ; GEF) : This project, launched by UNEP as part of the "En.lighten" initiative, aims to promote the transition of the Tunisian market to energy-efficient lighting technologies. To do this, it will intervene in the following four areas:
   - The implementation of minimum energy performance standards to ensure the efficiency of replacement bulbs and the energy savings they achieve;
   - The establishment of policies and support mechanisms to limit the supply of inefficient light bulbs and support the demand for more efficient alternatives;
   - The implementation of monitoring, verification and enforcement programs to deter the sale of non-compliant products; and
   - The implementation of measures for sustainable development, such as the determination of maximum limits for the mercury content of devices and the organization of a collection and disposal or recycling of used light bulbs.

   In addition, it is noted that this project aimed to transform the island of Djerba into an 'all-LED' island, which will promote green tourism in this region. Although the implementation of the project has been delayed, synergies will be sought during implementation. For instance, the minimum performance standards of the present project can be limited to outdoor lighting, while the UNDP-GEF project can focus on lower-power rated indoor lighting.

3. Programme 'd'Efficacité Énergétique dans le Bâtiment – PEEB' (2016-2022; MEEM-ADEME, FFEM-AFD and GIZ) : The PEEB program is an initiative of the French Development Agency (AFD), the French Ministry of the Environment, Energy and the Sea (MEEM) and the German Technical Cooperation (GIZ) that aims to promote the emergence of energy efficiency projects in buildings in developing and emerging countries. For Tunisia, this initiative will intervene on the following components:
   - Strengthening the policy framework through (i) the development of legislation relating to the thermal regulation of social housing, buildings belonging to the health sector, etc. (ii) the updating and upgrading of energy data in the sector and (iii) the strengthening of the system for monitoring the application of the thermal regulations in force;
   - Support and technical assistance in the design and implementation phases of identified EA projects;
   - Strengthening local capacities (policy makers & project developers) and developing guidance for planning and operating EE measures in hospitals; and
   - Elaboration de conventions de financement pour les hôpitaux, logements sociaux et le secteur privé.

4. Programme 'digitalisation du processus d'octroi du permis de construction' (2017-2021 ; USAID) : As part of its Tunisian Government's long-term decentralization support program "TADAEEM", USAID is supporting Tunisian municipalities to automate the building permit process. This will be possible through the establishment of a platform for the electronic management of files and the training of municipal executives to ensure widespread use of the platform throughout Tunisia.

5. Project related specifically to efficient public lighting: The City of Kairouan is beneficiary of a number of parallel initiatives supporting the adoption of energy efficient public lighting using LEDs. These are:
   - The PAI proximity project (2019) : 1,113 points of public lighting using LEDs in various residential areas and strategic axes of the city;
   - FNCT project : replacement of 138 HPS lamps of 250 W capacity with 70 W LEDs;
   - Municipal project: replacement of 34 high pressure sodium (HPS) lamps of 250 W capacity with 70 W LEDs ;
   - GIZ project: replacement of 250 HPS lamps with 70 W LEDs.

PART C – Estimation of the number of beneficiaries

1. The starting point for estimating the number of project beneficiaries is the entire population of the City of Kairouan – i.e. only the population living within the legal boundary of the City over which the Municipality has legal jurisdiction. The municipal building in Keblia and the eight (8) axes that will benefit from GEF investments will serve and benefit all citizens who fall under the municipal jurisdiction. Since, at least, the 8 road axes are so central to the city access and mobility, it is assumed that the justification is reasonable; it is reasonably expected that the entire population of the City will travel along any one of the ubiquitous road axes at least one a year (or at least once during the project duration of 3 years). Hence, the total of 187,000 represents the latest statistics of the population that live within the municipality's geographical jurisdiction. This number is differentiated by sex based on available statistics.
2. A more granular attempt is made to estimate the number of beneficiaries using data collected on passenger usage of the eight road axes targeted for efficient public lighting. Component 2 of the project is used because it will benefit a larger number of persons given the large geographic scope of its application. The layout of the city plan showing the eight road axes benefiting from GEF investments in efficient public lighting are marked in red in Figure 13.A.3. Data of traffic count were obtained from the Municipality of Kairouan based on latest available statistics for 2017. A new study of traffic count was being carried out at the time of finalizing the ProDoc and the data set was incomplete.
3. In order to estimate the usage of the axes it is assumed that the Axis 1 (Avenue de l'environnement) and Axis 5 (Avenue Touhami Negra) are the principal arteries that channel people in and out of the city. The host a multitude of public and private infrastructure such as schools, university campus, clinics, industrial zone, commercial buildings etc…). Hence, it is assumed that the same travellers split into the remaining 6 axes related to project interventions. This assumption is made in order to minimise double counting. The annualised daily average traffic (ADAT) was 13,859 for Axis 5 and 24,590 for Axis 1 giving a total of 38,449 motorised vehicles. Using average vehicle occupancy of 2 persons per vehicles, the number of persons using Axis 1 and Axis 5 on a daily basis (measured over a year) is estimated at 76,898.
4. This estimate does not include road users who travel on foot and by bicycle. The Municipality has suggested applying a weight of 40% to daily average travellers by vehicles to obtain an estimate of the number of users using the soft travel mode. The choice of weight is also influenced by peaks in road usage by inhabitants of Kairouan during religious festivities. Hence, the daily number of road users for Axis 1 and Axis 5 is estimated at 15,380 persons per day. Hence, the total number of commuters is estimated at 107,657 persons per day.
5. For the purpose of identifying the number of project beneficiaries, there is one uncertainty relating to the identities of persons behind the estimates – i.e. whether they are always the same persons (regular users), in which case the number of beneficiaries will be 107,657 persons; or whether there is a mix of regular and ad hoc (not regular) users. It is more realistic to assume that there is a combination of regular and ad hoc users, implying that the effective number of beneficiaries should be more than 107,657 persons over a one year cycle. This conclusion is reinforced when considering that there will also be inhabitants who will utilise the municipal building in Kairouan. Consequently, it is logical to treat the entire population as beneficiary of the project.

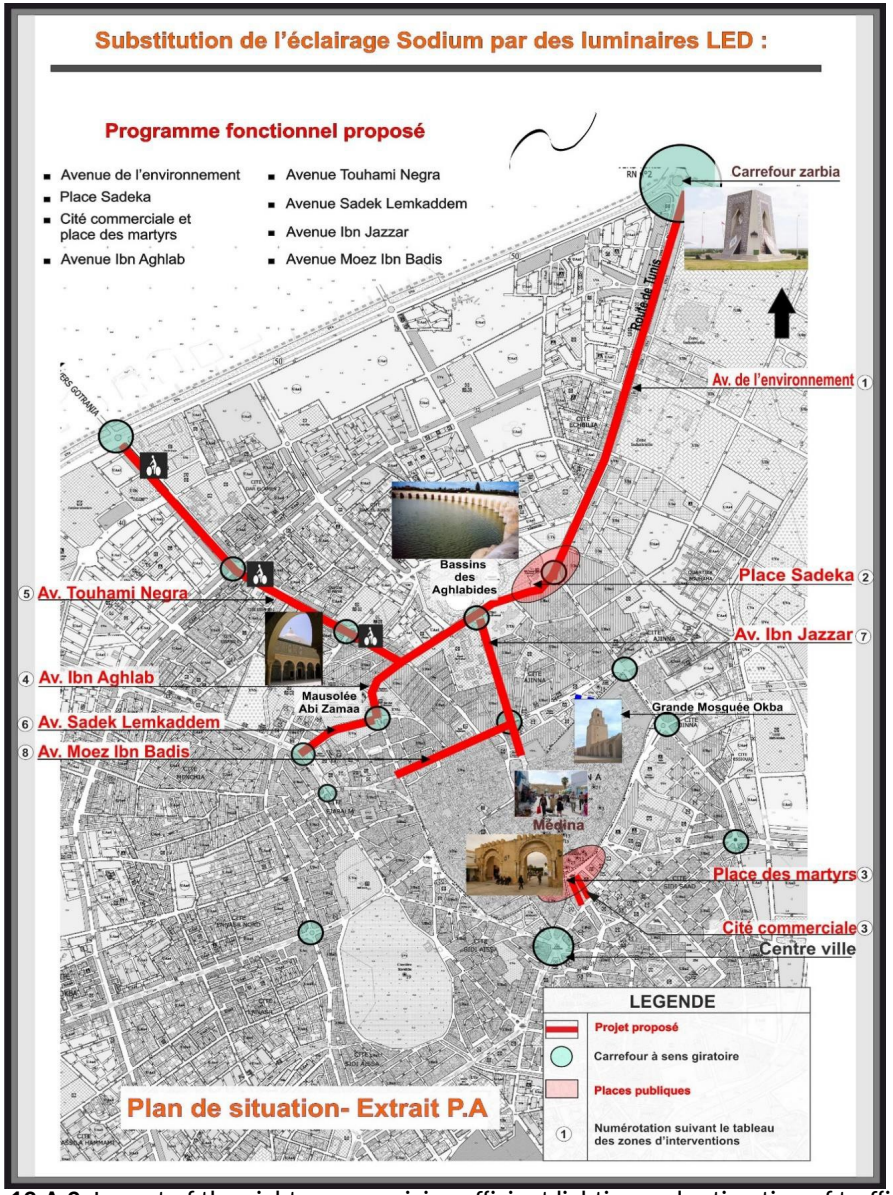


Figure 13.A.3. Layout of the eight axes receiving efficient lighting and estimation of traffic flux.

PART D – Estimation of the costs of building envelope energy efficiency

G3 and G4		2	floors						**- https://www.sm-devis.tn/tarifs-isolation-thermique-tunis/**
Table 7 in ProDoc - walls per floor									**Type d'élément à isoler Tunis** / **Prix de l'isolation au m²**
Floor	Surface insulated (m^2)	Façade	Surface area (m2)	Walls	Glass panes/windows, door	Roof			Isolation murs par l'intérieur / 55 dt / m² à 99 dt / m² / 100
G3	156	North	258	* double-walled	**tinted glass	Concrete slab 16+5			Isolation murs par l'extérieur / 82 dt / m² à 189dt / m² / 190
G4	131	East	268	* double-walled	** tinted glass	***Concrete			Isolation sols / 27 dt / m² à 58 dt / m² / 58
		South	258	* double-walled	** tinted glass				Isolation combles perdus / 22 dt / m² à 56 dt / m² / 56
Total	287	West	268	*double-walled	** tinted glass				Isolation toiture / combles aménagés par l'intérieur / 25 dt / m²à 83 dt / m² / 83
		Total	1052						Isolation toiture par l'extérieur / 92 dt / m² à 222 dt / m² / 222
Roof									
Aroof	207	m2							
Cinsol	222	DT/m2	- cost of isolation						
Croof	**45,954**	DT	- cost of insulating the roof (external)						
Walls									
Awalls	574	m2							
Cwall	100								
Cwalinsul	**57,400**	DT	**G3+G4**	**145,344**	**DT**	**- incremental cost**			
Windows									
Awin	22.1	m2							
Cwin	1,900	DT/m2	- https://www.sm-devis.tn/tarifs-menuiserie-aluminium-a-double-vitrage-tunis/						
Cwinsul	**41,990**	DT							
									Total incremental cost / **807,969 DT**
G+G1+G2		3	levels						**Total incremental cost+infla** / **889,573.9 DT**
Walls									**293,589 USD**
Awalls	3156	m2	- total of exterior walls						**Budgeted** / **314,458 USD**
Cinsol	190	DT/m2							**20,869 USD**
Cwalinsul	**599,640**	DT							
Windows									
Awin	33.15	m2	**G+G1+G2**	**662,625**	**DT**	**- incremental cost**			
Cwin	1,900	DT/m2							
Cwalinsul	**62,985**	DT/m2							

Annex 15: Additional agreements:

Composition of the cross-functional working group for the ACTE project

N°	Nom et Prénom	Fonction	Position
01	Président de la Commune ou son représentant	-	Président
02	Mestiri Mohamed Habib	Membre du Conseil municipal	Membre
03	Atallah Mohamed	Membre du Conseil municipal	Membre
04	Abdallah Hamadi	Directeur de travaux : Efficacité énergétique et déplacement urbain	Coordinateur général
05	Ghozzi Moez	Sous-directeur « Aménagement urbaine »	Membre
06	Saoudi Sahbi	Chargé de l'organisation interne et la gouvernance	Membre
07	Taktak Mohamed	Chargé de la formation	Membre
08	Amara Sahbi	Sous-directeur « Coopération internationale et communication »	Membre
09	Souidene Senda	Ingénieure Architecte chargée du permis de construction	Membre
10	Hamdani Ibtissem	Ingénieure génie civil, chargée des bâtiments communaux et le déplacement urbain	Membre
11	Gabsi Mariem	Ingénieure génie énergétique, chargée des bâtiments et de l'efficacité énergétique	Membre
12	Kalii Mohamed	Chargé de l'infrastructure et des bâtiments	Membre
13	Bouafia Mohamed Anis	Chargé de la diversification des ressources énergétiques	Membre
14	Fatnassi El Manaa	Chargé de la diversification des ressources énergétiques	Membre
15	Chouchene Mahmoud	Chargé du magasin municipal	Membre

Annex 16: GEF Core indicators

Core Indicator 6	**Greenhouse gas emission mitigated**					***(Metric tons of CO_2e)***
			Expected metric tons of CO_2e (6.1+6.2)			
			PIF stage	**Endorsement**	**MTR**	**TE**
	Expected CO2e (direct)		***8,403***	**20,219**		
	Expected CO2e (indirect)		**499,000**	**31,064**		
Indicator 6.3	**Energy saved**					
			MJ			
			Expected		**Achieved**	
			PIF stage	**Endorsement**	**MTR**	**TE**
		Direct savings	**197,813,068**	**162,836,000**		
Core Indicator 11	**Number of direct beneficiaries disaggregated by gender as co-benefit of GEF investment**					***(Number)***
			Number			
			Expected		**Achieved**	
			PIF stage	**Endorsement**	**MTR**	**TE**
		Female	***94,248***	***94,248***		
		Male	***92,752***	***92,752***		
		Total	***187,000***	***187,000***		

Annex 17: GEF 7 Taxonomy

Level 1	Level 2	Level 3	Level 4
☒ **Influencing models**			
	☐ **Transform policy and regulatory environments**		
	☒ **Strengthen institutional capacity and decision-making**		
	☐ **Convene multi-stakeholder alliances**		
	☒ **Demonstrate innovative approaches**		
	☐ **Deploy innovative financial instruments**		
☒ **Stakeholders**			
	☐ **Indigenous Peoples**		
	☒ **Private Sector**		
		☐ Capital providers	
		☐ Financial intermediaries and market facilitators	
		☐ Large corporations	
		☐ SMEs	
		☒ Individuals/Entrepreneurs	
		☐ Non-Grant Pilot	
		☐ Project Reflow	
	☒ **Beneficiaries**		
	☐ **Local Communities**		
	☐ **Civil Society**		
		☐ Community Based Organization	
		☐ Non-Governmental Organization	
		☐ Academia	
		☐ Trade Unions and Workers Unions	
	☐ **Type of Engagement**		
		☐ Information Dissemination	
		☐ Partnership	
		☐ Consultation	
		☐ Participation	
	☒ **Communications**		

		☒Awareness Raising	
		☐Education	
		☐Public Campaigns	
		☐Behavior Change	
☒**Capacity, Knowledge and Research**			
	☐**Enabling Activities**		
	☒**Capacity Development**		
	☒**Knowledge Generation and Exchange**		
	☐**Targeted Research**		
	☒**Learning**		
		☐Theory of Change	
		☒Adaptive Management	
		☐Indicators to Measure Change	
	☐**Innovation**		
	☒**Knowledge and Learning**		
		☒Knowledge Management	
		☒Innovation	
		☒Capacity Development	
		☒Learning	
	☐**Stakeholder Engagement Plan**		
☐**Gender Equality**			
	☒**Gender Mainstreaming**		
		☐Beneficiaries	
		☐Women groups	
		☒Sex-disaggregated indicators	
		☐Gender-sensitive indicators	
	☒**Gender results areas**		
		☐Access and control over natural resources	
		☐Participation and leadership	
		☐Access to benefits and services	
		☒Capacity development	
		☐Awareness raising	
		☐Knowledge generation	
☒**Focal Areas/Theme**			

	☐ **Integrated Programs**		
		☐ Commodity Supply Chains ([58]Good Growth Partnership)	
			☐ Sustainable Commodities Production
			☐ Deforestation-free Sourcing
			☐ Financial Screening Tools
			☐ High Conservation Value Forests
			☐ High Carbon Stocks Forests
			☐ Soybean Supply Chain
			☐ Oil Palm Supply Chain
			☐ Beef Supply Chain
			☐ Smallholder Farmers
			☐ Adaptive Management
		☐ Food Security in Sub-Sahara Africa	
			☐ Resilience (climate and shocks)
			☐ Sustainable Production Systems
			☐ Agroecosystems
			☐ Land and Soil Health
			☐ Diversified Farming
			☐ Integrated Land and Water Management
			☐ Smallholder Farming
			☐ Small and Medium Enterprises
			☐ Crop Genetic Diversity
			☐ Food Value Chains
			☐ Gender Dimensions
			☐ Multi-stakeholder Platforms
		☐ Food Systems, Land Use and Restoration	
			☐ Sustainable Food Systems
			☐ Landscape Restoration
			☐ Sustainable Commodity Production
			☐ Comprehensive Land Use Planning
			☐ Integrated Landscapes
			☐ Food Value Chains

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			☐ Deforestation-free Sourcing
			☐ Smallholder Farmers
		☐ Sustainable Cities	
			☐ Integrated urban planning
			☐ Urban sustainability framework
			☐ Transport and Mobility
			☐ Buildings
			☐ Municipal waste management
			☐ Green space
			☐ Urban Biodiversity
			☐ Urban Food Systems
			☐ Energy efficiency
			☐ Municipal Financing
			☐ Global Platform for Sustainable Cities
			☐ Urban Resilience
	☐ **Biodiversity**		
		☐ Protected Areas and Landscapes	
			☐ Terrestrial Protected Areas
			☐ Coastal and Marine Protected Areas
			☐ Productive Landscapes
			☐ Productive Seascapes
			☐ Community Based Natural Resource Management
		☐ Mainstreaming	
			☐ Extractive Industries (oil, gas, mining)
			☐ Forestry (Including HCVF and REDD+)
			☐ Tourism
			☐ Agriculture & agrobiodiversity
			☐ Fisheries
			☐ Infrastructure
			☐ Certification (National Standards)
			☐ Certification (International Standards)
		☐ Species	
			☐ Illegal Wildlife Trade
			☐ Threatened Species

			☐ Wildlife for Sustainable Development
			☐ Crop Wild Relatives
			☐ Plant Genetic Resources
			☐ Animal Genetic Resources
			☐ Livestock Wild Relatives
			☐ Invasive Alien Species (IAS)
		☐ Biomes	
			☐ Mangroves
			☐ Coral Reefs
			☐ Sea Grasses
			☐ Wetlands
			☐ Rivers
			☐ Lakes
			☐ Tropical Rain Forests
			☐ Tropical Dry Forests
			☐ Temperate Forests
			☐ Grasslands
			☐ Paramo
			☐ Desert
		☐ Financial and Accounting	
			☐ Payment for Ecosystem Services
			☐ Natural Capital Assessment and Accounting
			☐ Conservation Trust Funds
			☐ Conservation Finance
		☐ Supplementary Protocol to the CBD	
			☐ Biosafety
			☐ Access to Genetic Resources Benefit Sharing
	☐ **Forests**		
		☐ Forest and Landscape Restoration	
			☐ REDD/REDD+
		☐ Forest	
			☐ Amazon
			☐ Congo
			☐ Drylands

	☐ **Land Degradation**		
		☐ Sustainable Land Management	
			☐ Restoration and Rehabilitation of Degraded Lands
			☐ Ecosystem Approach
			☐ Integrated and Cross-sectoral approach
			☐ Community-Based NRM
			☐ Sustainable Livelihoods
			☐ Income Generating Activities
			☐ Sustainable Agriculture
			☐ Sustainable Pasture Management
			☐ Sustainable Forest/Woodland Management
			☐ Improved Soil and Water Management Techniques
			☐ Sustainable Fire Management
			☐ Drought Mitigation/Early Warning
		☐ Land Degradation Neutrality	
			☐ Land Productivity
			☐ Land Cover and Land cover change
			☐ Carbon stocks above or below ground
		☐ Food Security	
	☐ **International Waters**		
		☐ Ship	
		☐ Coastal	
		☐ Freshwater	
			☐ Aquifer
			☐ River Basin
			☐ Lake Basin
		☐ Learning	
		☐ Fisheries	
		☐ Persistent toxic substances	
		☐ SIDS : Small Island Dev States	
		☐ Targeted Research	
		☐ Pollution	
			☐ Persistent toxic substances

			☐Plastics
			☐Nutrient pollution from all sectors except wastewater
			☐Nutrient pollution from Wastewater
		☐Transboundary Diagnostic Analysis and Strategic Action Plan preparation	
		☐Strategic Action Plan Implementation	
		☐Areas Beyond National Jurisdiction	
		☐Large Marine Ecosystems	
		☐Private Sector	
		☐Aquaculture	
		☐Marine Protected Area	
		☐Biomes	
			☐Mangrove
			☐Coral Reefs
			☐Seagrasses
			☐Polar Ecosystems
			☐Constructed Wetlands
	☐**Chemicals and Waste**		
		☐Mercury	
		☐Artisanal and Scale Gold Mining	
		☐Coal Fired Power Plants	
		☐Coal Fired Industrial Boilers	
		☐Cement	
		☐Non-Ferrous Metals Production	
		☐Ozone	
		☐Persistent Organic Pollutants	
		☐Unintentional Persistent Organic Pollutants	
		☐Sound Management of chemicals and Waste	
		☐Waste Management	
			☐Hazardous Waste Management
			☐Industrial Waste
			☐e-Waste
		☐Emissions	
		☐Disposal	
		☐New Persistent Organic Pollutants	

		☐ Polychlorinated Biphenyls	
		☐ Plastics	
		☐ Eco-Efficiency	
		☐ Pesticides	
		☐ DDT - Vector Management	
		☐ DDT - Other	
		☐ Industrial Emissions	
		☐ Open Burning	
		☐ Best Available Technology / Best Environmental Practices	
		☐ Green Chemistry	
	☒ **Climate Change**		
		☐ **Climate Change Adaptation**	
			☐ Climate Finance
			☐ Least Developed Countries
			☐ Small Island Developing States
			☐ Disaster Risk Management
			☐ Sea-level rise
			☐ Climate Resilience
			☐ Climate information
			☐ Ecosystem-based Adaptation
			☐ Adaptation Tech Transfer
			☐ National Adaptation Programme of Action
			☐ National Adaptation Plan
			☐ Mainstreaming Adaptation
			☐ Private Sector
			☐ Innovation
			☐ Complementarity
			☐ Community-based Adaptation
			☐ Livelihoods
		☒ **Climate Change Mitigation**	
			☐ Agriculture, Forestry, and other Land Use
			☒ Energy Efficiency
			☐ Sustainable Urban Systems and Transport

			☒Technology Transfer
			☐Renewable Energy
			☐Financing
			☐Enabling Activities
		☐**Technology Transfer**	
			☐Poznan Strategic Programme on Technology Transfer
			☐Climate Technology Centre & Network (CTCN)
			☐Endogenous technology
			☐Technology Needs Assessment
			☐Adaptation Tech Transfer
		☐**United Nations Framework on Climate Change**	
			☐Nationally Determined Contribution

Annex 18: Partners Capacity Assessment Tool and HACT assessment

Annexed separately.